

- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.



NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT. EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER, CONTRACTOR, SUB-CONTRACTORS, A SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE, AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT: SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						



CLIENT: MOTIVA ENTERPRISES LLC
PLANT: PORT ARTHUR REFINERY
JOB NO:

PIPING & INSTRUMENTATION DIAGRAM
BATTERY LIMIT DRAWING
PAMC TO BOILER COMPLEX

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW



SSP ENGINEERING & CONSULTING
13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

PROJECT NO: 25MA02

DRAWN BY: GS

DESIGN BY:

DRAWING NO:

SCALE: NONE

CHECKED: APB

ENG APVL: PDN

REV

0000-PID-0001

A

CHECK PRINT

- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.



CHECK PRINT

NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT. EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER, CONTRACTOR, SUB-CONTRACTORS, A SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE, AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT, SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						



CLIENT: MOTIVA ENTERPRISES LLC
PLANT: PORT ARTHUR REFINERY
JOB NO:

PIPING & INSTRUMENTATION DIAGRAM
BATTERY LIMIT DRAWING
AROMATICS TO BOILER COMPLEX

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW



PROJECT NO: 25MA02
SCALE: NONE
DRAWN BY: GS
CHECKED: APB
DESIGN BY:
ENG APVL: PDN
DRAWING NO:

0000-PID-0002

REV

A

DRAWING LOCATION: c:\users\jason\one\active - eng\ssps files - 25ma02 - aromatics utility scope cold eyes review\04 - design\04 - process\pids - (Border Rev1 1 (6.25).
PLOT DATE: ---, PLOT BY: jason.

- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.



CHECK PRINT

NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT, EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE, AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT: SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						



CLIENT: MOTIVA ENTERPRISES LLC
PLANT: PORT ARTHUR REFINERY
JOB NO:

PIPING & INSTRUMENTATION DIAGRAM
BATTERY LIMIT DRAWING
BOILER COMPLEX TO PAMC

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW



PROJECT NO: 25MA02
SCALE: NONE
DRAWN BY: GS
CHECKED: APB
DESIGN BY: ENG APVL: PDN
DRAWING NO:

0000-PID-0010 REV A

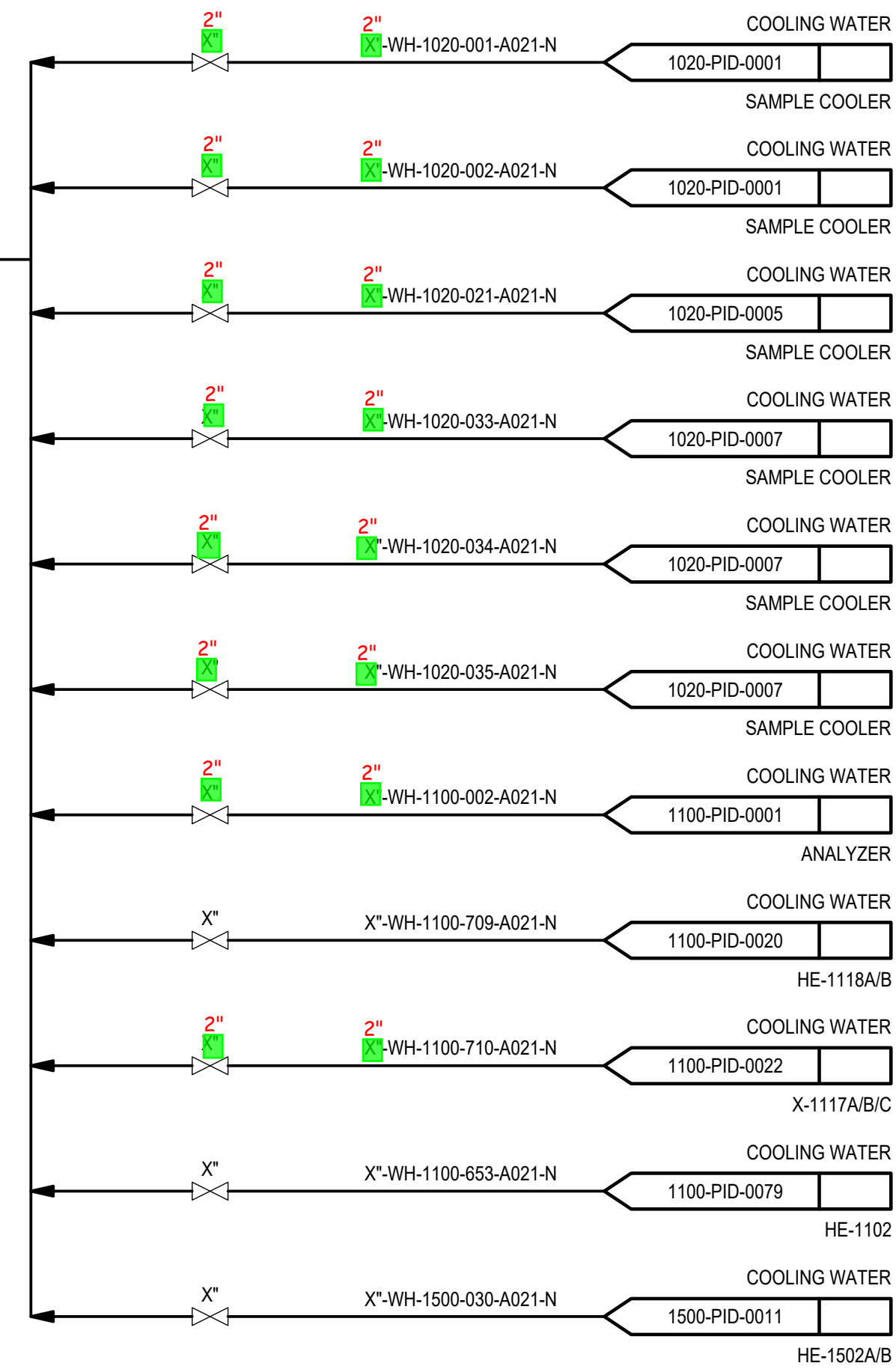
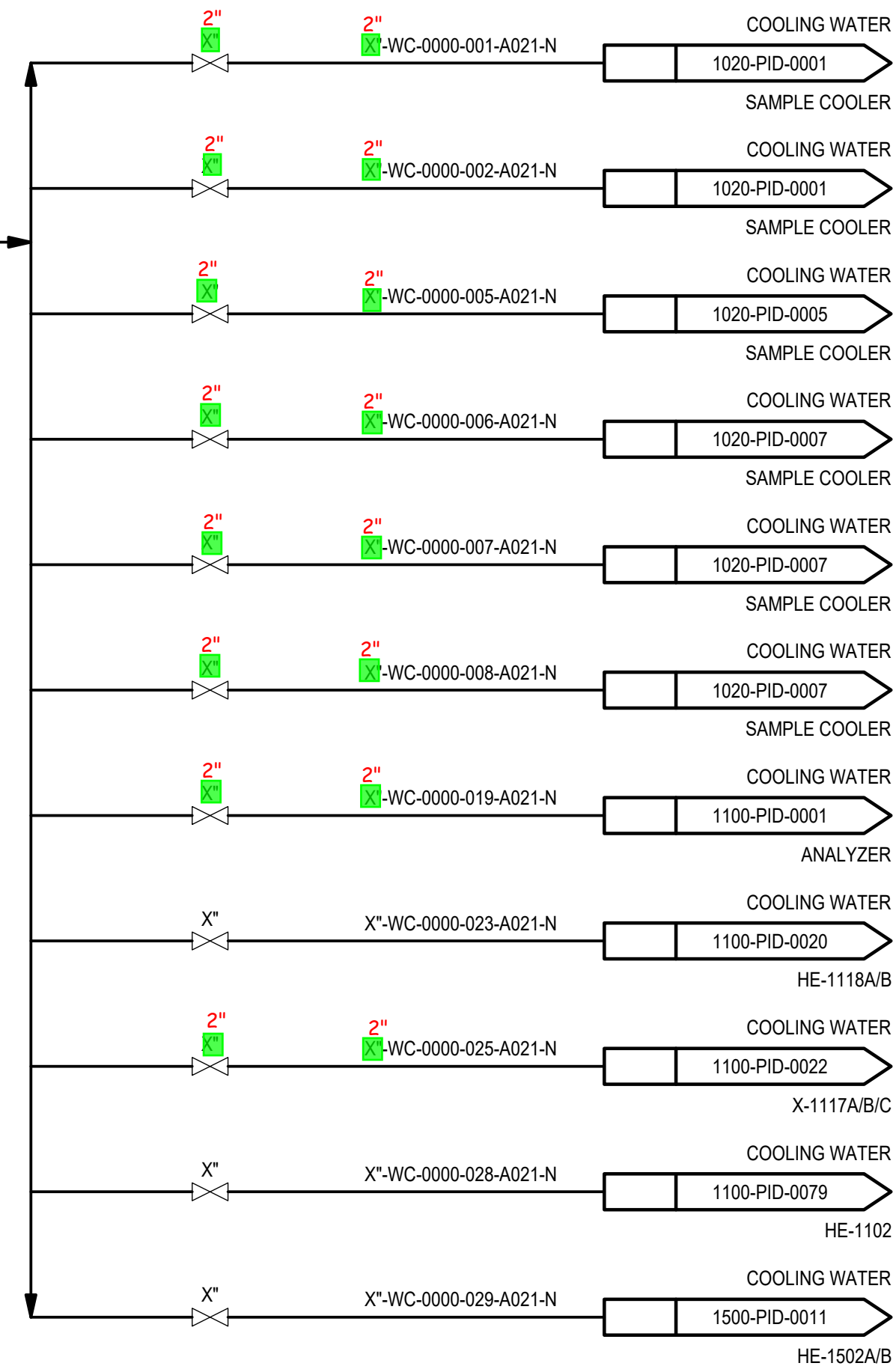
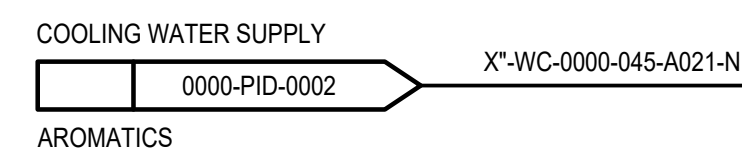
A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.

B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.

C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.



NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT. EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER, CONTRACTOR, SUB-CONTRACTORS, & SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS, DESCRIBING DESIGN INTENT. SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND CO-ORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.	REFERENCE DRAWINGS:			REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	 CLIENT: MOTIVA ENTERPRISES LLC PLANT: PORT ARTHUR REFINERY JOB NO:	PIPING & INSTRUMENTATION DIAGRAM BATTERY LIMIT DRAWING BOILER COMPLEX TO AROMATICS PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW	 SSP ENGINEERING & CONSULTING 13100 NORTHWEST FREEWAY SUITE 625, HOUSTON, TEXAS 77040	FIRM NUMBER #13619
	A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN													
											PROJECT NO: 25MA02		SCALE: NONE						
											DRAWN BY: GS		CHECKED: APB						
											DESIGN BY:		ENG APVL: PDN						
										DRAWING NO:		0000-PID-0011		REV A					

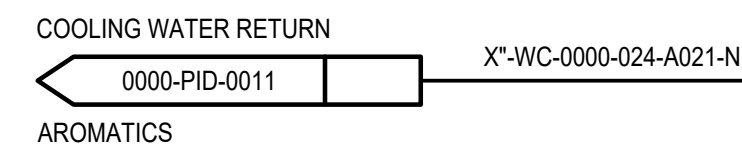


GENERAL NOTES

A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS
SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.

8. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.

C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.



CHECK PRINT

NOT PUBLISHED. ALL RISKS RESERVED BY THE ENGINEER, DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT, EXCEPT BY AGREEMENT IN WRITING AND WITHOUT THE ENGINEER'S CONSENT. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR THE USER'S RESPONSIBILITY FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE, AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT, SHALL NOT INDICATE ANY DETAIL OR SYSTEM FOR CONSTRUCTION. THE USER SHALL BE RESPONSIBLE FOR THE CONSTRUCTION OF THE PROJECT AND PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND CO-ORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR THE SELECTION OF MATERIALS AND METHODS OF CONSTRUCTION. THE USER SHALL BE RESPONSIBLE FOR THE SELECTION OF MATERIALS. ALL MATERIALS SHALL BE TRUE, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TO NEW LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, OR SEQUENCES. THE USER SHALL BE RESPONSIBLE FOR THE SELECTION OF CONSTRUCTION METHODS. ALL WORK SHALL NOT BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS

[illegible]

MOTIVA

CLIENT: MOTIVA ENTERPRISES LLC

PLANT: PORT ARTHUR REFINERY

JOB NO:

PIPING & INSTRUMENTATION DIAGRAM

COOLING WATER DISTRIBUTION

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

	SSP ENGINEERING & CONSULTING 13100 NORTHWEST FREEWAY SUITE 625 HOUSTON, TEXAS 77040		FIRM NUMBER #13619
	PROJECT NO: 25MA02		SCALE: NONE
DRAWN BY: GS		CHECKED: APB	
DESIGN BY:		ENG APVL: PDN	
DRAWING NO:		0000-PID-0020	
			REV A

DRAWING LOCATION: c:\users\jason\one\active - eng\ssps files - 25ma02 - aromatics utility scope cold eyes review\04 - design\04 - process\pids - (Border Rev1 5 16 25).
PLOT DATE: ---, ---, ---
PLOT BY: jason

- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.



CHECK PRINT

NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT. EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER, CONTRACTOR, SUB-CONTRACTORS, & SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE, AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT, SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

PIPING & INSTRUMENTATION DIAGRAM

NITROGEN DISTRIBUTION

CLIENT: MOTIVA ENTERPRISES LLC

PLANT: PORT ARTHUR REFINERY

JOB NO:

SSP ENGINEERING & CONSULTING

13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

FIRM NUMBER
#13619

PROJECT NO: 25MA02

SCALE: NONE

DRAWN BY: GS

CHECKED: APB

DESIGN BY:

ENG APVL: PDN

DRAWING NO:

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

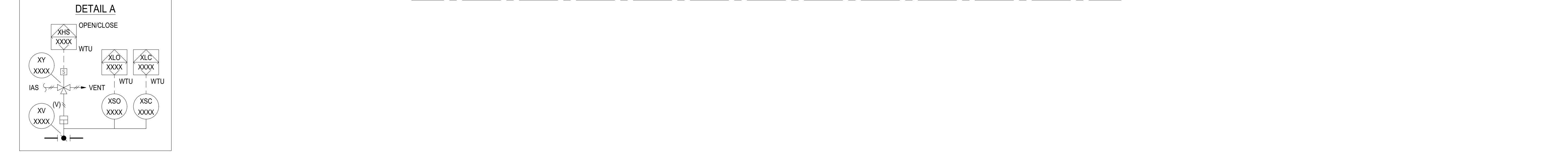
0000-PID-0030

REV

C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.



NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT, EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER, CONTRACTOR, SUB-CONTRACTORS, & SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT SHALL NOT INDICATE EXISTING DETAIL OR SYSTEM CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.										REFERENCE DRAWINGS:										REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD		PIPING & INSTRUMENTATION DIAGRAM		SSP ENGINEERING & CONSULTING 13100 NORTHWEST FREEWAY SUITE 625, HOUSTON, TEXAS 77040	FIRM NUMBER #13619																																																																																																																																																																																																																																																																																		
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN																																																																																																																																																																																																																																																																																																																	



 SSP ENGINEERING & CONSULTING 13100 NORTHVIEW FREEWAY SUITE 625 HOUSTON, TEXAS 77040		FIRM NUMBER #13619	
		CTNO: 25MA02 SCALE: NONE DRAWN BY: GS CHECKED: APB DESIGNED BY: PDN ENG APVL: PDN	
TITLING NO:		1000-PID-0001	

[illegible]

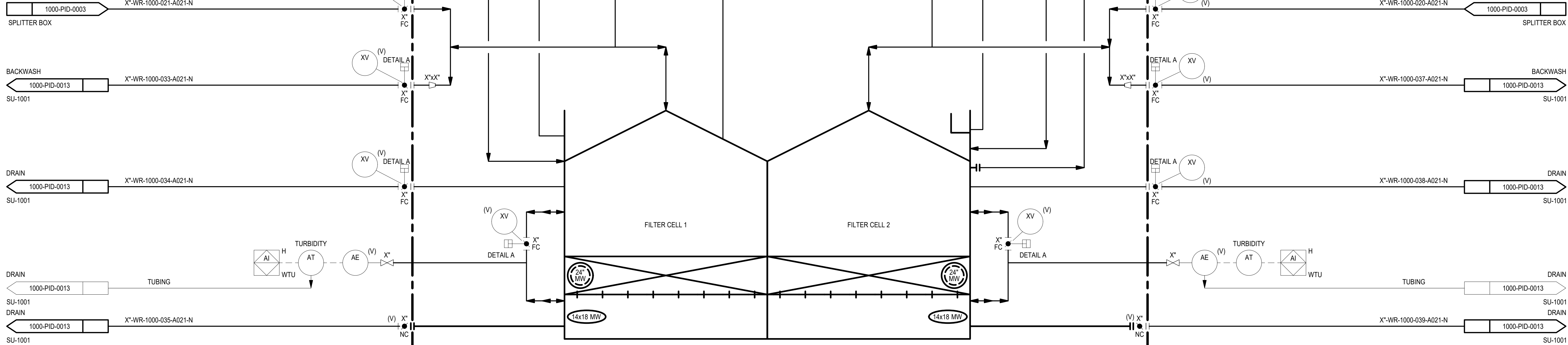
SAND FILTER
DESIGN CAPACITY :
RATED POWER :
MOC :
TRIM : VT-1000-042-A021-N
INSULATION :

A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.

B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.

1. FILTER CONTROL LOGIC BOX ORIGINATES ON P&ID 1000-PID-0003. THE BOX SHOWN ON THIS P&ID REPRESENTS THE SAME BOX.

2. FILTER CONFIGURATION IS TYPICAL. DRAWING TO BE REVISED WHEN FILTRATION SYSTEM IS SELECTED.



CHECK PRINT

ALL PUBLISHED ALL RIGHTS RESERVED BY THE ENGINEER, DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED OR REPRODUCED IN ANY MANNER WITHOUT THE WRITTEN PERMISSION OF THE ENGINEER. THE ENGINEER SHALL BE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE, AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH INFORMATION. THE ENGINEER SHALL BE RESPONSIBLE FOR THE DESIGN OF THE PROJECT, INCLUDING CONSTRUCTION, CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND CO-ORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR THE SELECTION OF MATERIALS, EQUIPMENT, AND METHODS OF CONSTRUCTION. MATERIALS SHALL BE TRUE, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND IN LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL BE RESPONSIBLE FOR THE SELECTION OF MATERIALS, EQUIPMENT, AND METHODS OF CONSTRUCTION, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS

[illegible]

CLIENT: MOTIVA ENTERPRISES LLC

PLANT: PORT ARTHUR REFINERY

JOB NO.

PIPING & INSTRUMENTATION DIAGRAM

SAND FILTER C



**SSP ENGINEERING
& CONSULTING**
13100 NORTHWEST FREEWAY
SUITE 625,
HOUSTON, TEXAS 77040

PROJECT NO: 25MA02

DRAWN BY: GS

DESIGN BY:

DRAWING NO:

SCALE: NONE

CHECKED: APR

ENG ARVI - PDN

**FIRM
NUMBER
#13619**

1000-PID-0005

A

TK-1005
REPRESSURIZATION TANK
SIZE ID x HEIGHT :
TOTAL VOLUME :
DESIGN PRESSURE :
DESIGN TEMP :
MOC :
TRIM : VT-1000-051-A021-N
INSULATION / TRACING :

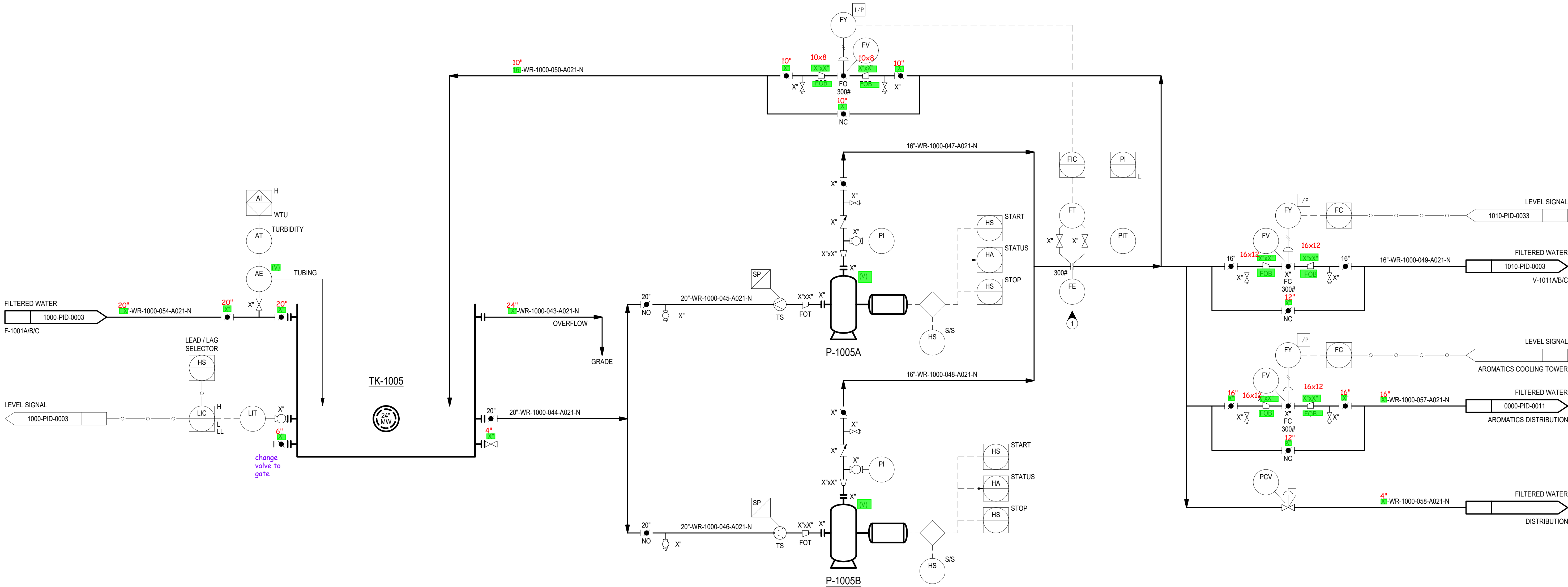
P-1005A/B
FILTERED WATER PUMPS
RATED CAPACITY :
DIFFERENTIAL PRESSURE :
RATED POWER :
MOC (CASING / INTERNAL) :
INSULATION / TRACING :
TRIM : VT-1000-052-A021-N
SEAL PLAN :

GENERAL NOTES

- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-000X.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.

NOTES

1. ORIFICE PLATE IS FLOW CONDITIONING ORIFICE, WHICH REQUIRES 2 DIAMETERS STRAIGHT RUN UPSTREAM AND DOWNSTREAM.



NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT, EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER. CONTRACTOR, SUB-CONTRACTORS, & SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE, AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT, SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	DLH	PDN						

MOTIVA		
CLIENT: MOTIVA ENTERPRISES LLC		
PLANT: PORT ARTHUR REFINERY		
JOB NO:		

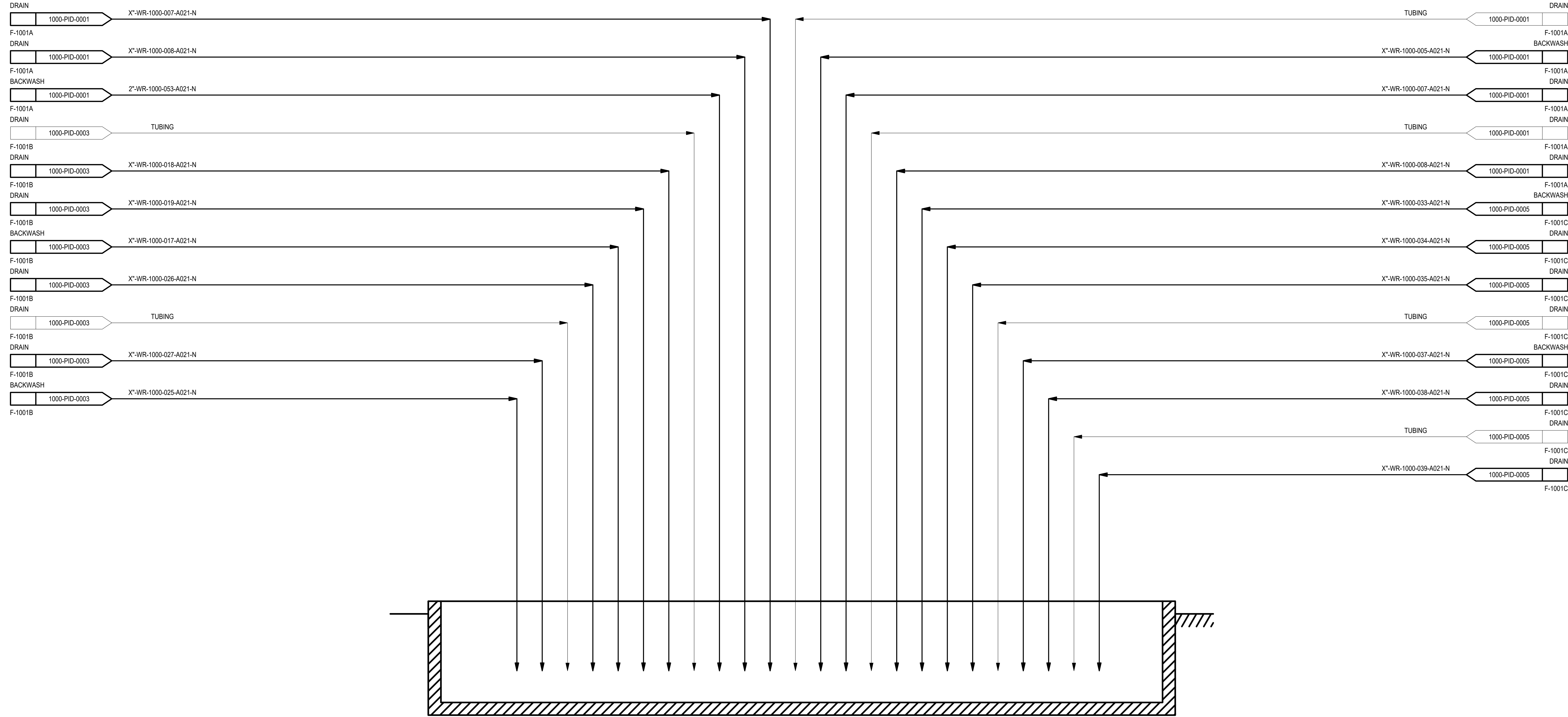
PROCESS FLOW DIAGRAM	
PIPING & INSTRUMENTATION DIAGRAM	
SUPERNATANT SUMP	
PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW	

SSP ENGINEERING & CONSULTING 13100 NORTHWEST FREEWAY SUITE 625 HOUSTON, TEXAS 77040	FIRM NUMBER #13619
PROJECT NO: 25MA02	SCALE: NONE
DRAWN BY: GS	CHECKED: APB
DESIGN BY:	ENG APVL: PDN
DRAWING NO:	1000-PID-0011

REV	A
-----	---

SU-1001
SUPERNATANT SUMP
SIZE :
WORKING CAPACITY :
INTERNALS :
MOC :
TRIM : VT-1000-056-A021-N

- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-000X.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.



CHECK PRINT

NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT. EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER, CONTRACTOR, SUB-CONTRACTORS, & SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE, AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	DLH	PDN						



CLIENT: MOTIVA ENTERPRISES LLC
PLANT: PORT ARTHUR REFINERY
JOB NO:

PROCESS FLOW DIAGRAM
PIPING & INSTRUMENTATION DIAGRAM
SUPERNATANT SUMP

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW



PROJECT NO: 25MA02
SCALE: NONE

DRAWN BY: GS
CHECKED: APB

DESIGN BY:
ENG APVL: PDN

DRAWING NO: 1000-PID-0013

FIRM
NUMBER
#13619

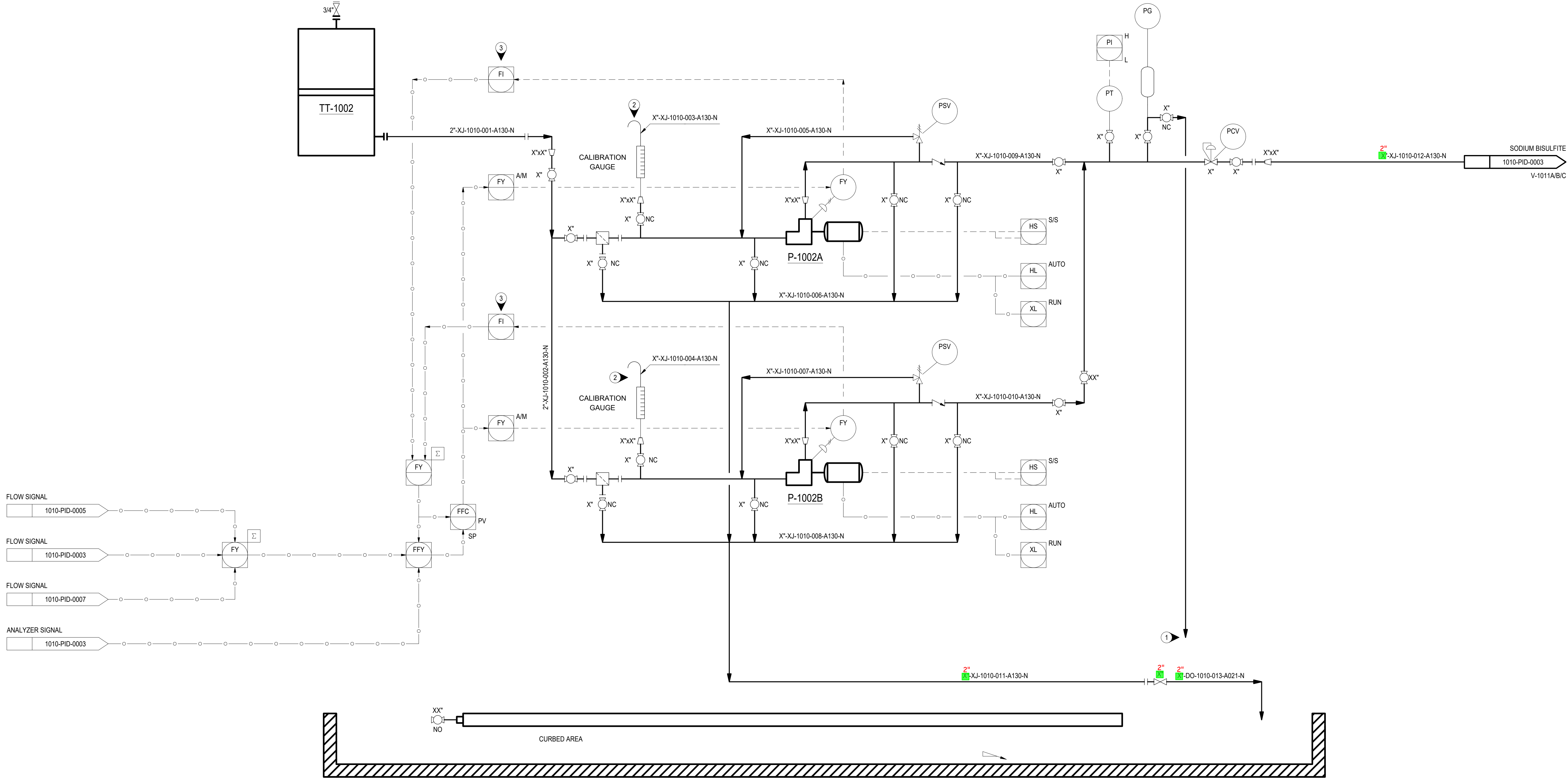
REV

A

TT-1002
SODIUM BISULFITE STACKED TOTE
SIZE ID x HEIGHT :
TOTAL VOLUME :
DESIGN PRESSURE :
DESIGN TEMP :
MOC :
TRIM : VT-1010-014-A130-N
INSULATION / TRACING :

P-1002A/B
SODIUM BISULFITE PUMPS
RATED CAPACITY :
DIFFERENTIAL PRESSURE :
RATED POWER :
MOC (CASING / INTERNAL) :
INSULATION / TRACING :
TRIM : VT-1010-015-A130-N
SEAL PLAN :

- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.
- NOTES
1. OPEN END DRAIN TO CURB AREA.
2. TOP OF VENT IS LOCATED HIGHER THAN TOP OF TANK.
3. STROKE FEEDBACK (FI-XXXX) AND STROKE CONTROL (FO-XXXX).



FLOW SIGNAL
1010-PID-0005

FLOW SIGNAL
1010-PID-0003

FLOW SIGNAL
1010-PID-0007

ANALYZER SIGNAL
1010-PID-0003

NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT, EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER. CONTRACTOR, SUB-CONTRACTORS, & SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE, AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT, SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

MOTIVA

CLIENT: MOTIVA ENTERPRISES LLC

PLANT: PORT ARTHUR REFINERY

JOB NO:

PROCESS FLOW DIAGRAM

SODIUM BISULFITE INJECTION SYSTEM

SSP ENGINEERING & CONSULTING

13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

FIRM NUMBER

#13619

PROJECT NO: 25MA02

DRAWN BY: GS

DESIGN BY:

DRAWING NO:

SCALE: NONE

CHECKED: APB

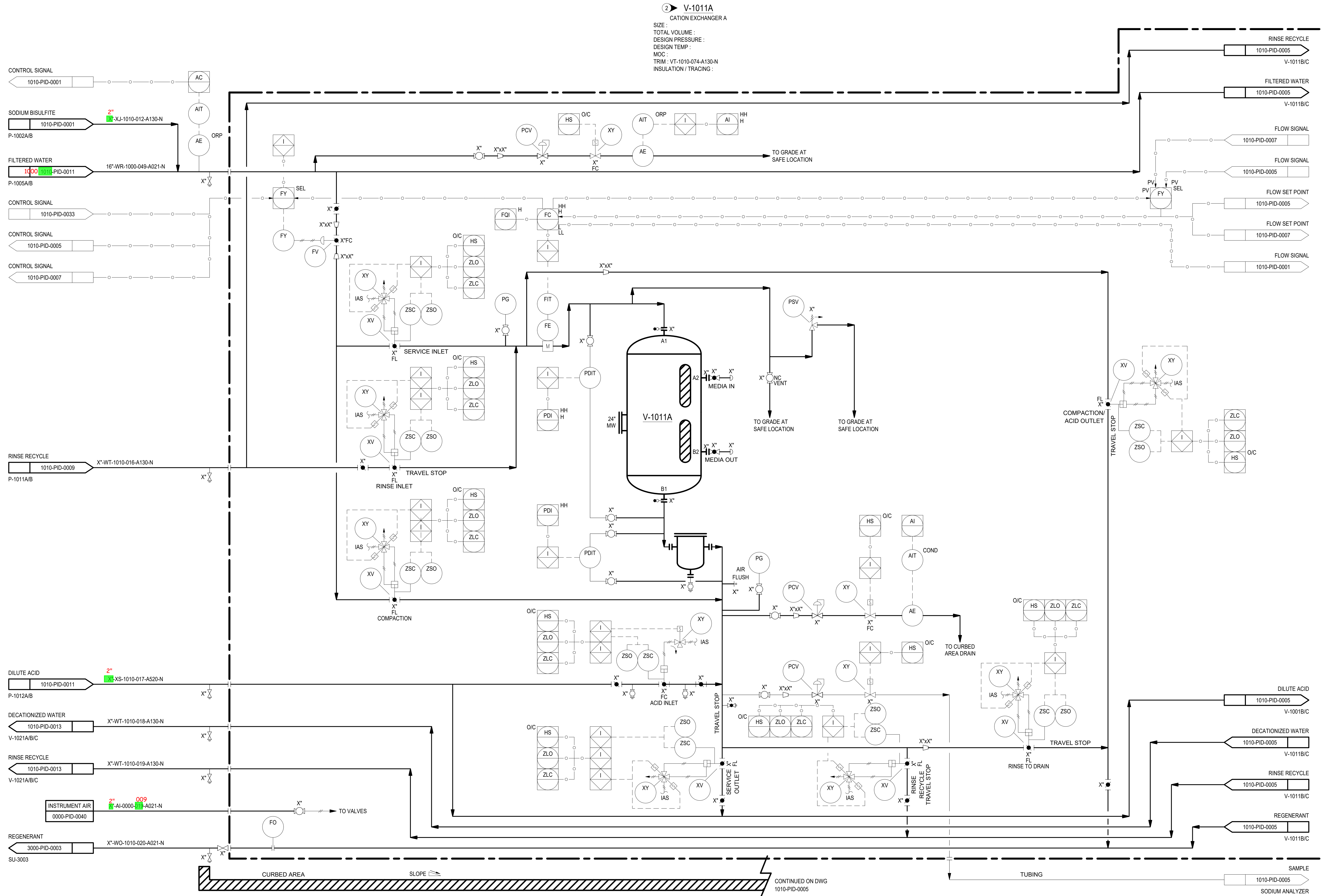
ENG APVL: PDN

1010-PID-0001

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

REV

CHECK PRINT



② V-1011A
CATION EXCHANGER A
SIZE :
TOTAL VOLUME :
DESIGN PRESSURE :
DESIGN TEMP :
MOC :
TRIM : VT-1010-074-A130-N
INSULATION / TRACING :

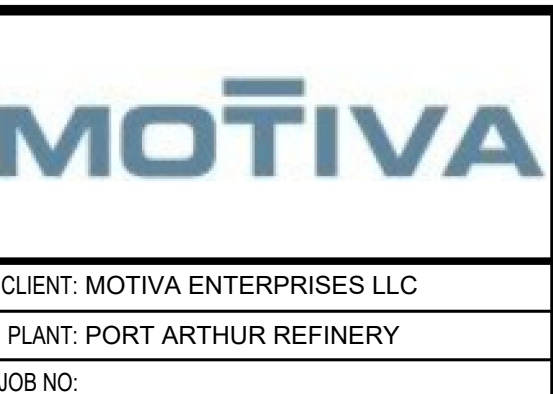
- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDS 0000-PID-0001 THRU 0000-PID-000X.
 - B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
 - C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.
- NOTES
- 1. ACID RESISTANT LINING.
 - 2. CATION EXCHANGER SHOWN IS TYPICAL EXACT CONFIGURATION WILL BE REVISED ONCE SYSTEM IS SELECTED.

DRAWING LOCATION: c:\users\jason\one-drive - eng\ssps\files - 25ma02 - aromatics utility scope old eyes review\04 - design\04 - process\04a -
PLOT DATE: ---, PLOTTED BY: jason, DRAWING: 1010-pid-0003.dwg

NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT, EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER. CONTRACTOR, SUB-CONTRACTORS, & SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:			
1010-PID-0001	1010-PID-0002	1010-PID-0003	1010-PID-0004

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

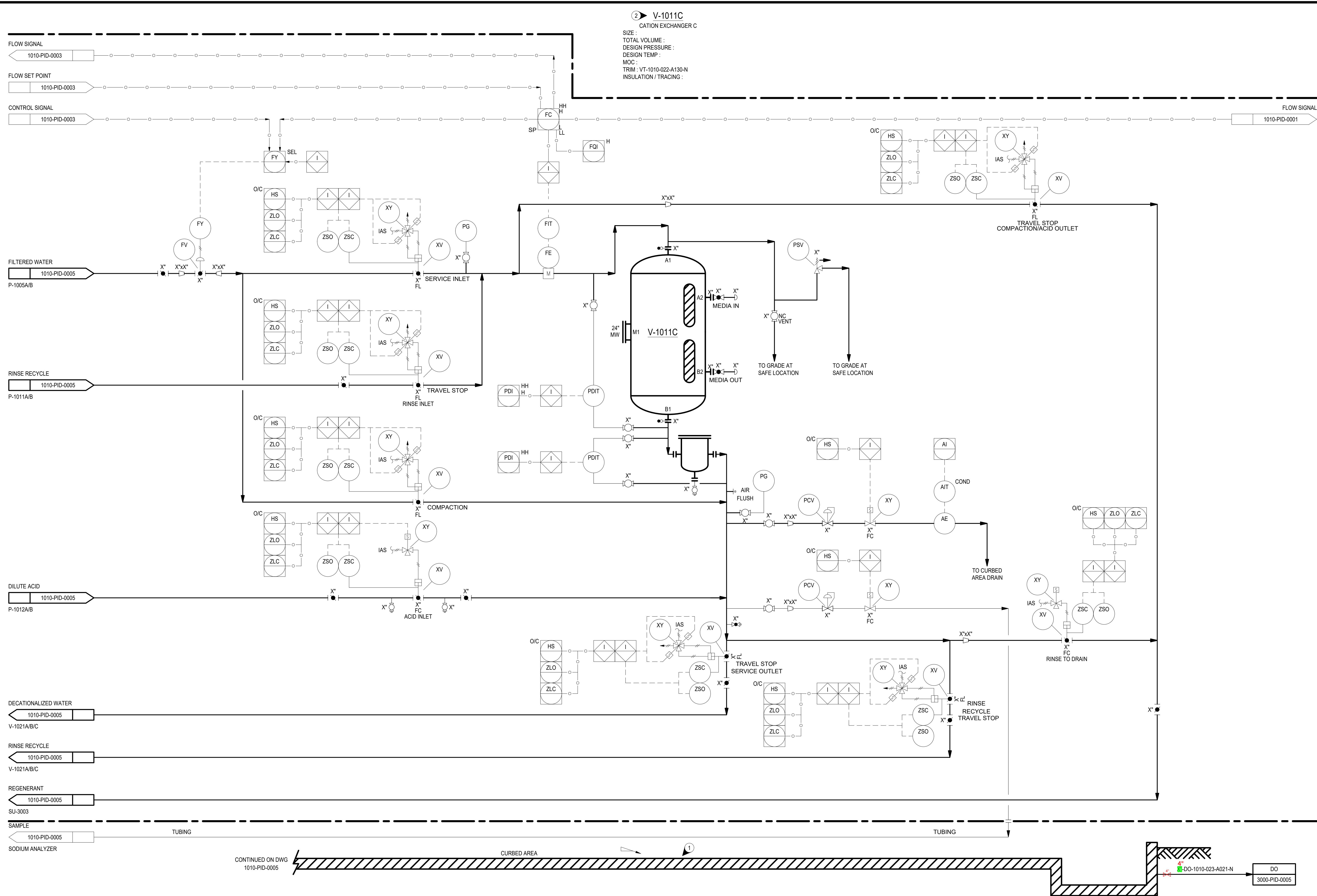


PIPING & INSTRUMENTATION DIAGRAM
DEMINERALIZED WATER SYSTEM
CATION EXCHANGER A

SSP ENGINEERING & CONSULTING 13100 NORTHWEST FREEWAY SUITE 625 HOUSTON, TEXAS 77040		FIRM NUMBER #13619
PROJECT NO: 25MA02	SCALE: NONE	
DRAWN BY: GS	CHECKED: APB	
DESIGN BY:	ENG APVL: PDN	
DRAWING NO:		

CHECK PRINT

DRAWING LOCATION: c:\users\jason\one\drive - eng\ssps\files - 25ma02 - aromatics utility scope old eyes\rev\04 - process\pids - 1010-pid-0007.dwg, PLOT DATE: ---, PLOTTED BY: jason, DRAWING LOCATION: c:\users\jason\one\drive - eng\ssps\files - 25ma02 - aromatics utility scope old eyes\rev\04 - process\pids - 1010-pid-0007.dwg, (Border Rev 1 5.18.25)



- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.
- NOTES
1. ACID RESISTANT LINING.
2. CATION EXCHANGER SHOWN IS TYPICAL. EXACT CONFIGURATION WILL BE REVISED ONCE SYSTEM IS SELECTED.

CHECK PRINT

NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT, EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE, AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT, SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

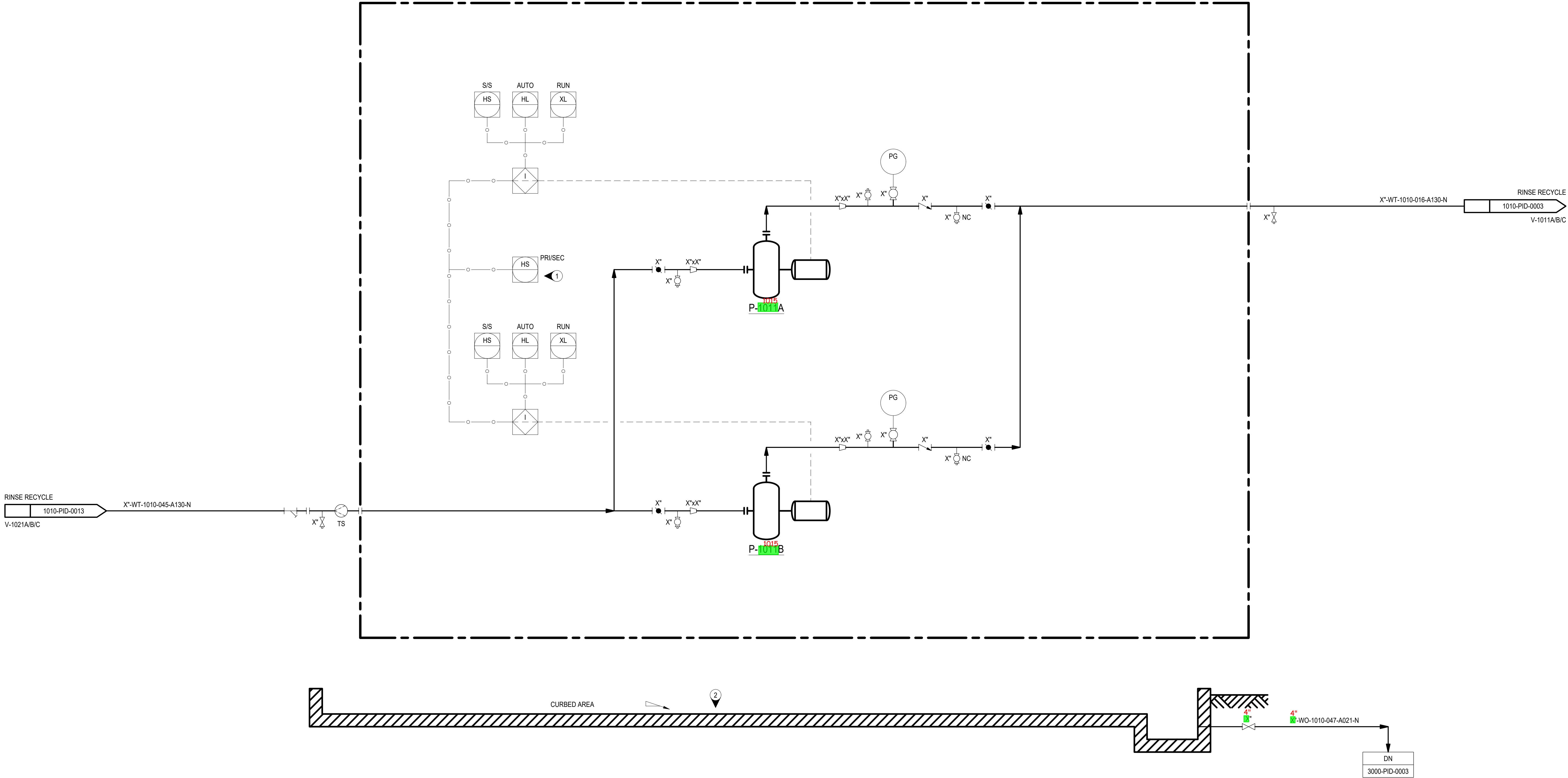
MOTIVA		
CLIENT: MOTIVA ENTERPRISES LLC		
PLANT: PORT ARTHUR REFINERY		
JOB NO:		

PIPING & INSTRUMENTATION DIAGRAM		FIRM NUMBER	
DEMINERALIZED WATER PACKAGE CATION EXCHANGER C		#13619	
PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW		REV	
DRAWING NO:		1010-PID-0007	

SSP ENGINEERING & CONSULTING 13100 NORTHWEST FREEWAY SUITE 625 HOUSTON, TEXAS 77040	
PROJECT NO: 25MA02	SCALE: NONE
DRAWN BY: GS	CHECKED: APB
DESIGN BY:	ENG APVL: PDN
DRAWING NO:	

1015
P-1011A/B
DEMIN RINSE RECYCLE PUMPS
RATED CAPACITY :
DIFFERENTIAL PRESSURE :
RATED POWER :
MOC (CASING / INTERNAL) :
INSULATION / TRACING :
TRIM : VT-1010-048-A130-N
SEAL PLAN :

- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.
- NOTES
1. RINSE RECYCLE PUMPS ARE USED BY CATION/ ANION. ONLY ONE PUMP CAN BE IN RINSE RECYCLE AT A TIME. PUMPS ALTERNATE AT EVERY START.
2. ACID/ CAUSTIC RESISTANT LINING.



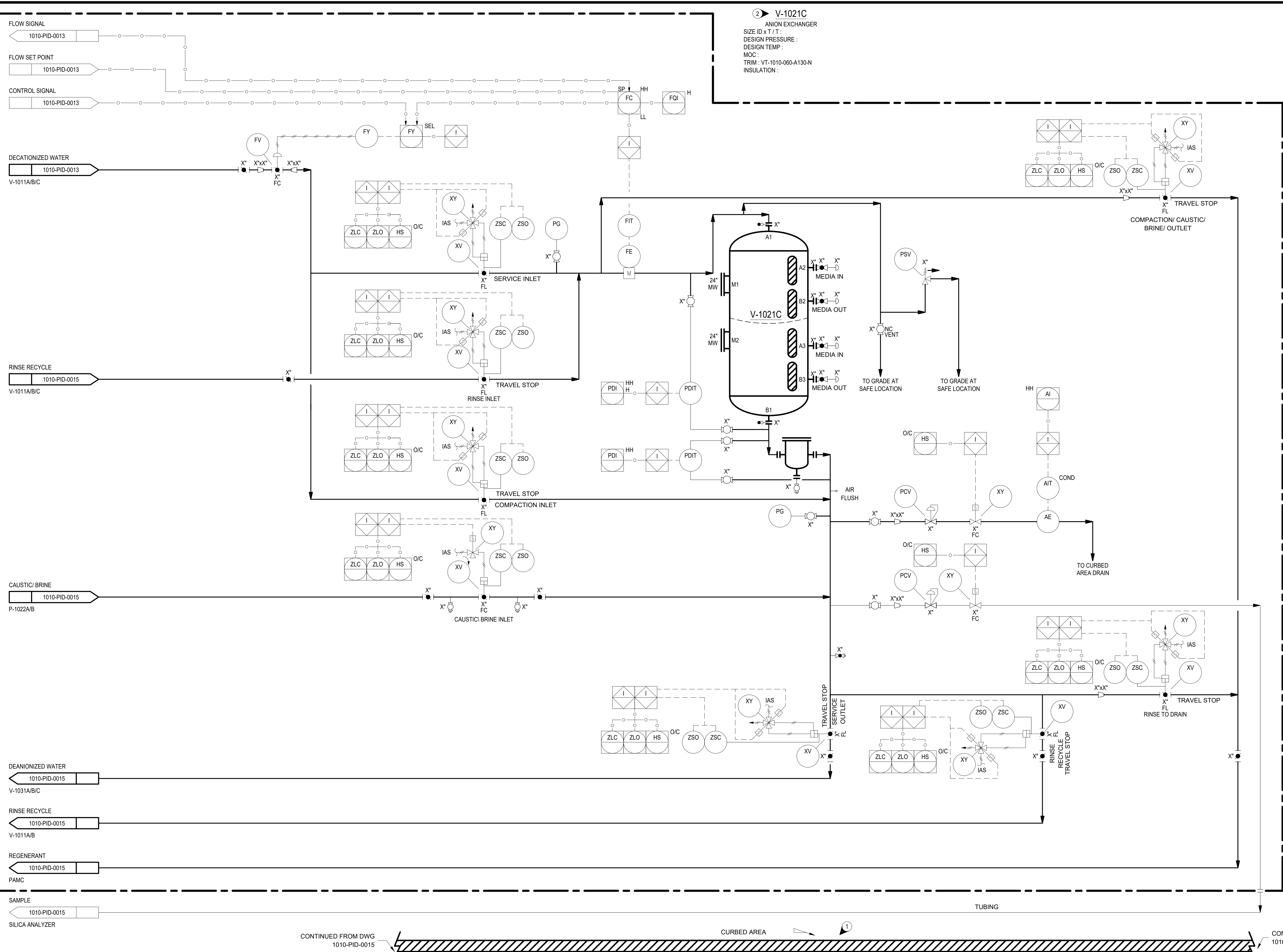
CHECK PRINT

NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT. EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER, CONTRACTOR, SUB-CONTRACTORS, & SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW UNLESS SPECIFIED OTHERWISE AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

MOTIVA			PIPING & INSTRUMENTATION DIAGRAM		SSP ENGINEERING & CONSULTING		FIRM NUMBER #13619	
CLIENT: MOTIVA ENTERPRISES LLC			DEMINERALIZED WATER PACKAGE		13100 NORTHWEST FREEWAY		SCALE: NONE	
PLANT: PORT ARTHUR REFINERY			DEMIN RINSE RECYCLE PUMPS		SUITE 625		CHECKED: APB	
JOB NO:			PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW		HOUSTON, TEXAS 77040		ENG APVL: PDN	
					DRAWING NO:		REV	
					1010-PID-0009		A	



GENERAL NOTES

- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.
- NOTES
1. ACID AND CAUSTIC RESISTANT LINING.
2. ANION EXCHANGER SHOWN IS TYPICAL. EXACT CONFIGURATION WILL BE REVISED ONCE SYSTEM IS SELECTED.

NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT. EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER, CONTRACTOR, SUB-CONTRACTORS, & SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT, SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

MOTIVA

CLIENT: MOTIVA ENTERPRISES LLC

PLANT: PORT ARTHUR REFINERY

JOB NO:

PIPING & INSTRUMENTATION DIAGRAM

DEMINERALIZED WATER PACKAGE

ANION EXCHANGER C

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

SSP ENGINEERING & CONSULTING

13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

FIRM NUMBER #13619

PROJECT NO: 25MA02

SCALE: NONE

DRAWN BY: GS

CHECKED: APB

DESIGN BY: ENG APVL: PDN

DRAWING NO: **1010-PID-0017**

REV **A**

CAUSTIC DILUTION HEATER

TYPE :
RATED DUTY :
SURFACE AREA :
SHELL DESIGN PRESS @ TEMP :
TUBE DESIGN PRESS @ TEMP :
MOC (SHELL / TUBES) :
TRIM (SHELL / TUBES) : VT-1010-061-A130-N
INSULATION (SHELL / TUBES) :

GENERAL NOTES

A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.

B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.

C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.

NOTES

1. CAUSTIC RESISTANT LINING.

1 2. CAUSTIC DILUTION HEATER CONFIGURATION IS TYPICAL.
DRAWING TO BE REVISED WHEN SYSTEM IS SELECTED.



NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER, DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT, EXCEPT BY AGREEMENT IN WRITING AND WITH THE SIGNATURE OF THE ENGINEER. THE CONTRACTOR SHALL BE RESPONSIBLE FOR OBTAINING ALL NECESSARY PERMITS AND SHALL BE RESPONSIBLE FOR OBTAINING AND CORRELATING DRAWINGS AND CONDITIONS AT THE JOB SITE, AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS AND CONDITIONS. THE CONTRACTOR SHALL BE RESPONSIBLE FOR OBTAINING ALL NECESSARY CONSTRUCTION CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND CO-ORDINATION OF ALL WORK, CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR OBTAINING ALL NECESSARY PERMITS AND SHALL BE RESPONSIBLE FOR OBTAINING ALL MATERIALS SHALL BE TRUE, UNLESS SPECIFIED OTHERWISE AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TO THE LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES, THE SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS

[illegible]**MOTIVA**

CLIENT: MOTIVA ENTERPRISES LLC

PLANT: PORT ARTHUR REFINERY

JOB NO.

PIPING & INSTRUMENTATION DIAGRAM

DEMINERALIZED WATER SYSTEM CAUSTIC DILUTION HEATER

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW



SSP ENGINEERING

13100 NORTHWEST FREEWAY
SUITE 625,
HOUSTON, TEXAS 77040

PROJECT NO: 25MA02

DRAWN BY: GS

DESIGN BY:

DRAWING NO:

SCALE: NONE

CHECKED: APB

ENG APVL: PDB

FIRM

#13619

1010-PID-0019

A

4 TK-1021
CAUSTIC DAY TANK
SIZE ID x HEIGHT :
TOTAL VOLUME :
DESIGN PRESSURE :
DESIGN TEMP :
MOC :
TRIM : VT-1010-069-A130-N
INSULATION / TRACING :

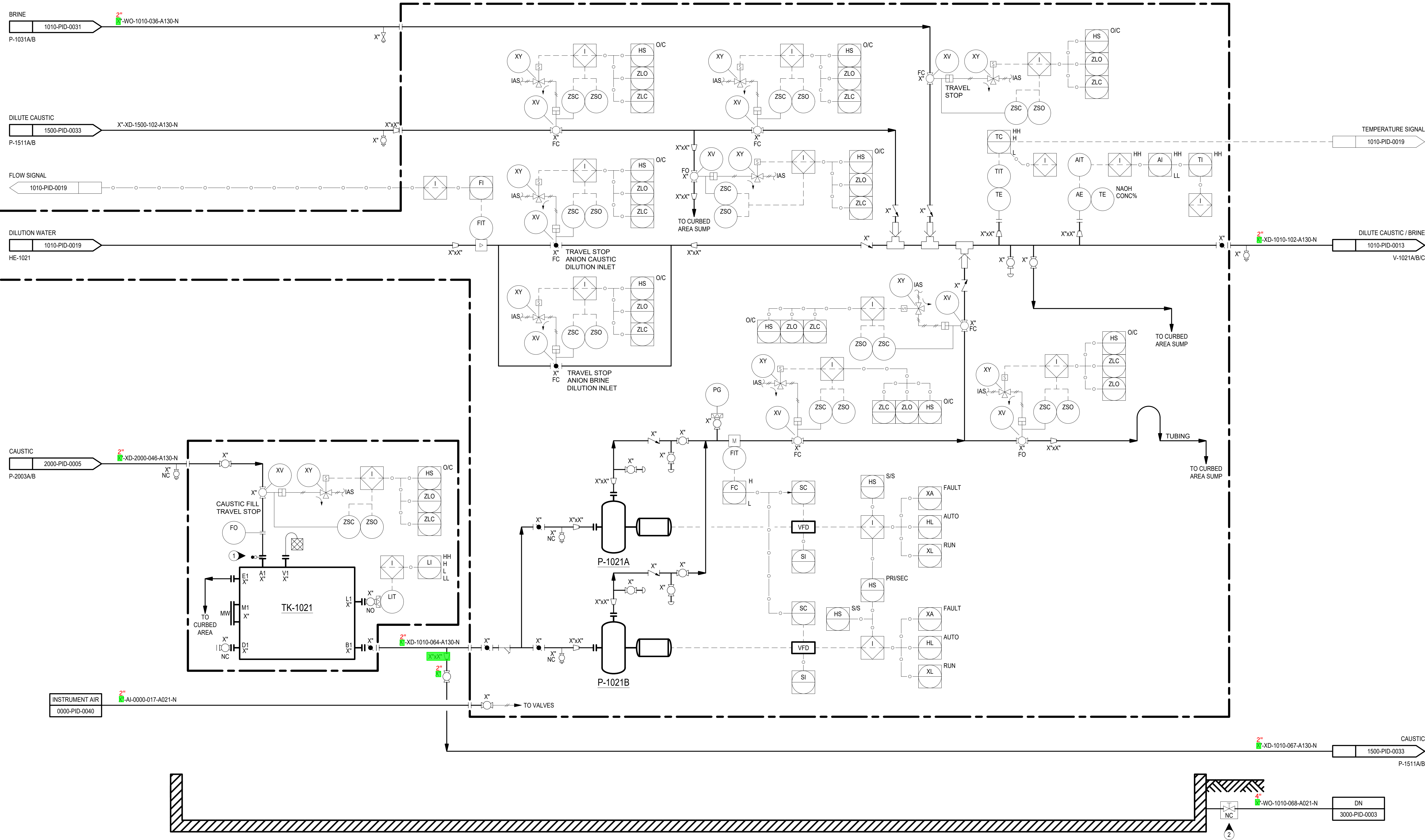
4 P-1021 A/B
CAUSTIC DOSING PUMP
RATED CAPACITY :
DIFFERENTIAL PRESSURE :
RATED POWER :
MOC (CASING / INTERNAL) :
INSULATION / TRACING :
TRIM : VT-1010-070-A130-N
SEAL PLAN :

GENERAL NOTES

- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.

NOTES

1. PROVIDED BY CONTRACTOR.
2. VALVE STEM IS ACCESSIBLE FRON GRADE.
3. CAUSTIC RESISTANCE LINING.
4. CAUSTIC DILUTION SYSTEM SHOWN IS TYPICAL. EXACT CONFIGURATION WILL BE REVISED ONCE SYSTEM IS SELECTED.



NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT. EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER, CONTRACTOR, SUB-CONTRACTORS, & SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE, AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

MOTIVA

CLIENT: MOTIVA ENTERPRISES LLC

PLANT: PORT ARTHUR REFINERY

JOB NO:

PIPING & INSTRUMENTATION DIAGRAM

DEMINERALIZED WATER SYSTEM

CAUSTIC DILUTION SYSTEM

PROJECT NO: 25MA02

DRAWN BY: GS

DESIGN BY:

DRAWING NO:

SCALE: NONE

CHECKED: APB

ENG APVL: PDN

1010-PID-0021

FIRM NUMBER #13619

REV A

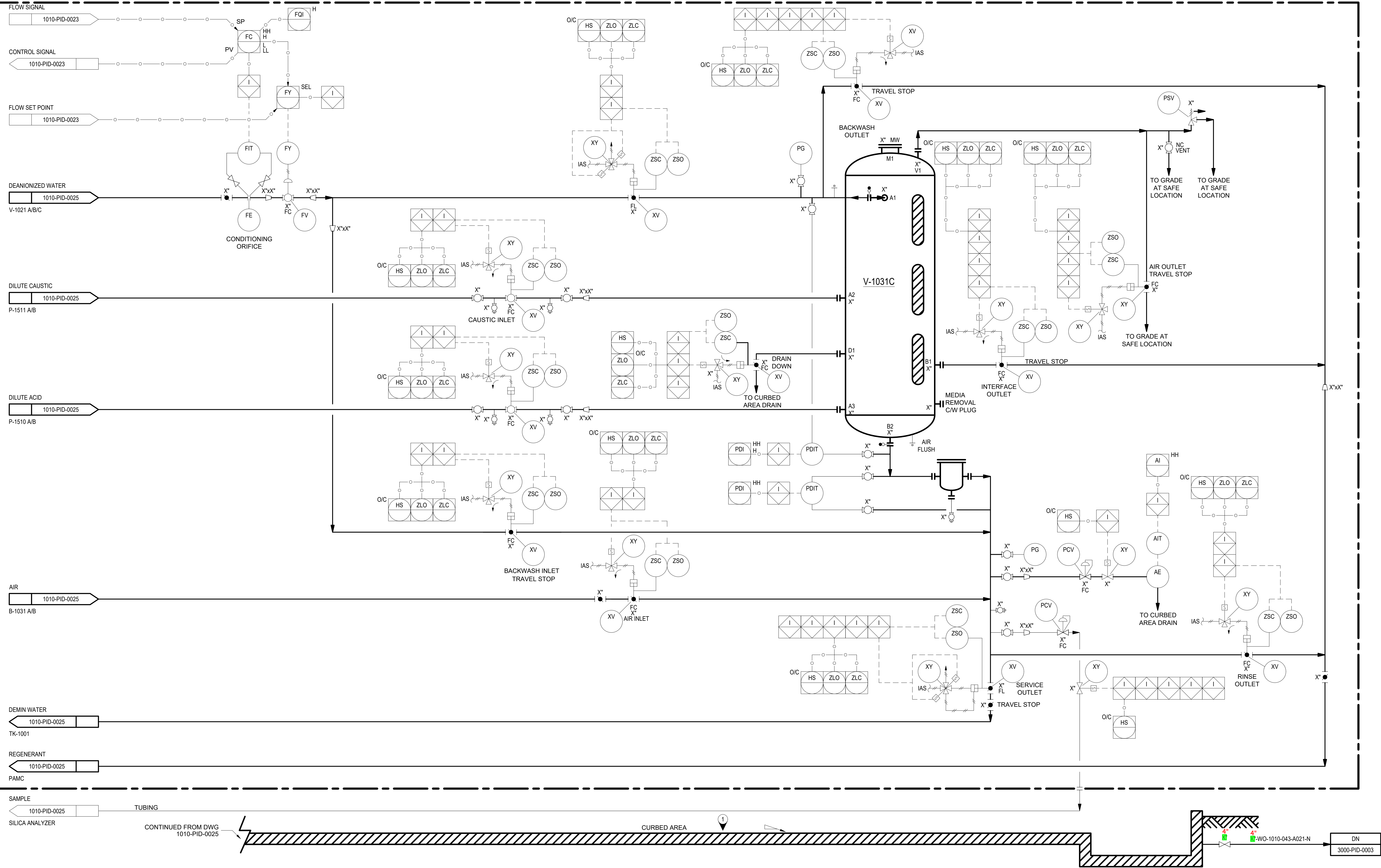
SSP ENGINEERING & CONSULTING

13100 NORTHWEST FREEWAY SUITE 625 HOUSTON, TEXAS 77040

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

② V-1031C
MIXED BED EXCHANGER C
SIZE 10 x 11 T
DESIGN PRESSURE :
DESIGN TEMP :
MOC :
TRIM : VT-1010-066-A130-N
INSULATION :

- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.
- NOTES
1. ACID RESISTANT AND CAUSTIC RESISTANT LINING.
2. MIXED BED POLISHER SHOWN IS TYPICAL. EXACT CONFIGURATION WILL BE REVISED ONCE SYSTEM IS SELECTED.



CHECK PRINT

NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT. EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER, CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

MOTIVA

CLIENT: MOTIVA ENTERPRISES LLC

PLANT: PORT ARTHUR REFINERY

JOB NO:

PIPING & INSTRUMENTATION DIAGRAM

DEMINERALIZED WATER PACKAGE

MIXED BED EXCHANGER C

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

SSP ENGINEERING & CONSULTING

13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

FIRM NUMBER #13619

PROJECT NO: 25MA02

SCALE: NONE

DRAWN BY: GS

CHECKED: APB

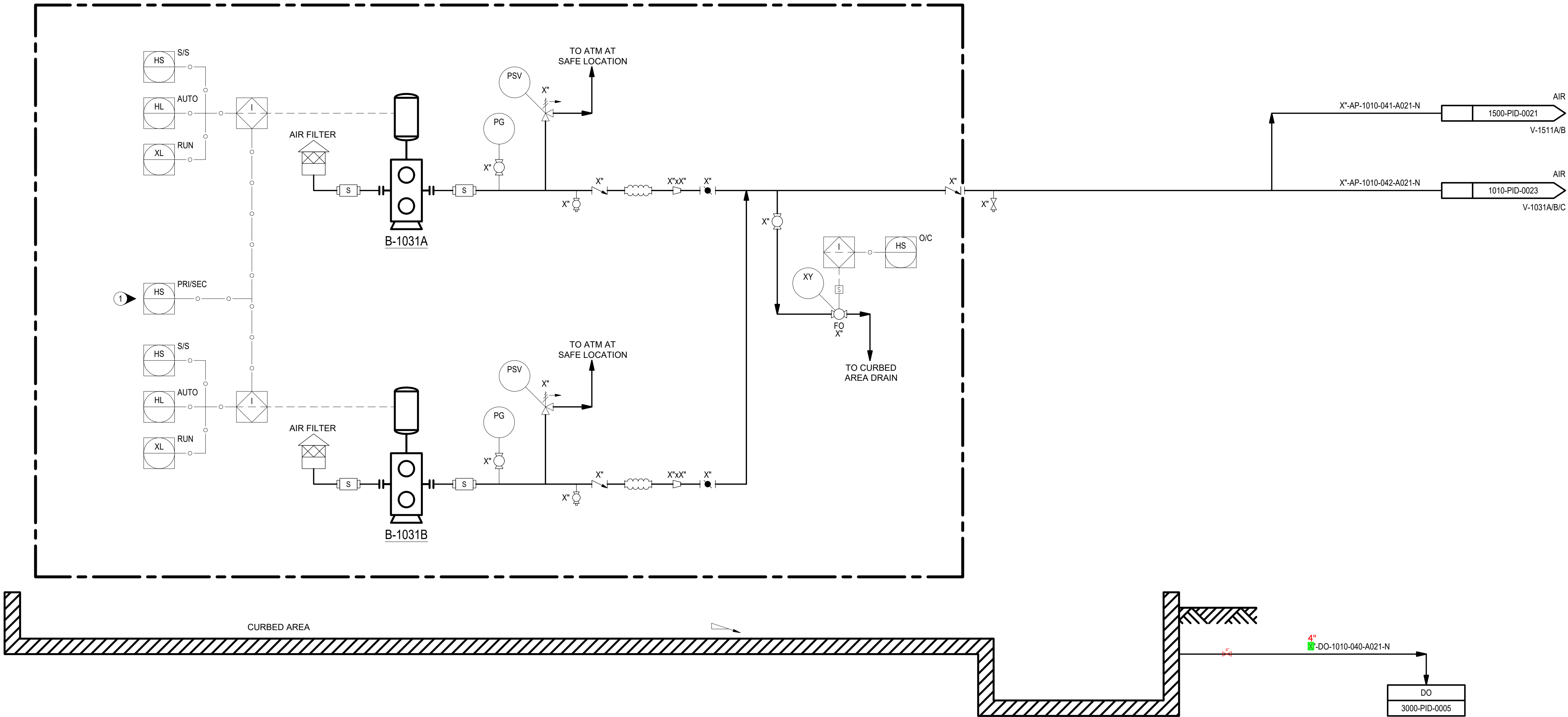
DESIGN BY: ENG APVL: PDN

DRAWING NO: 1010-PID-0027

REV A

- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.

- NOTES
1. MIXED BED SCOUR BLOWERS ALTERNATE DUTY AFTER EACH USE.
2. MIXED BED SCOUR BLOWERS CONFIGURATION IS TYPICAL. DRAWING TO BE REVISED WHEN SYSTEM IS SELECTED.



B-1031A/B
MIXED BED AIR SCOUR BLOWERS
RATED CAPACITY :
DIFFERENTIAL PRESSURE :
RATED POWER :
MOC (CASING / INTERNAL) :
INSULATION / TRACING :
TRIM : VT-1010-065-A130-N
SEAL PLAN :

CHECK PRINT

NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT. EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER, CONTRACTOR, SUB-CONTRACTORS, & SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT, SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

MOTIVA

CLIENT: MOTIVA ENTERPRISES LLC
PLANT: PORT ARTHUR REFINERY
JOB NO:

PIPING & INSTRUMENTATION DIAGRAM
DEMINERALIZED WATER PACKAGE
MIXED BED AIR SCOUR BLOWERS

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

SSP ENGINEERING & CONSULTING
13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

PROJECT NO: 25MA02 SCALE: NONE

DRAWN BY: GS CHECKED: APB

DESIGN BY: ENG APVL: PDN

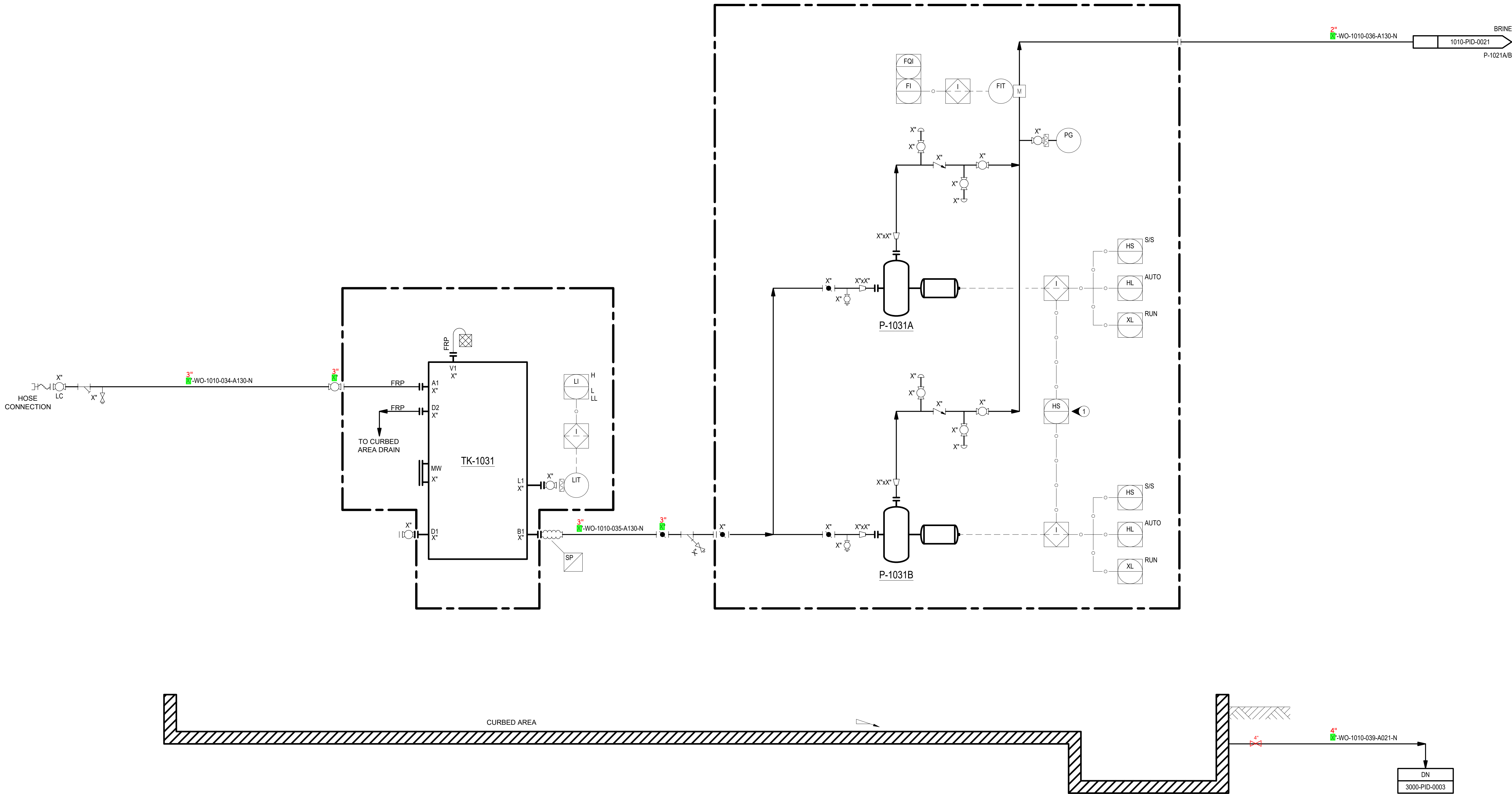
DRAWING NO: 1010-PID-0029

FIRM
NUMBER
#13619

REV

TK-1031
BRINE MEASURING TANK
SIZE ID x HEIGHT :
TOTAL VOLUME :
DESIGN PRESSURE :
DESIGN TEMP :
MOC :
TRIM : VT-1010-037-A130-N
INSULATION / TRACING :

- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.
- NOTES
1. PUMPS ALTERNATE AFTER EVERY BRINE INJECTION.
2. BRINE SYSTEM CONFIGURATION IS TYPICAL. DRAWING TO BE REVISED WHEN SYSTEM IS SELECTED.



P-1031 A/B
BRINE TRANSFER PUMPS
RATED CAPACITY :
DIFFERENTIAL PRESSURE :
RATED POWER :
MOC (CASING / INTERNAL) :
INSULATION / TRACING :
TRIM : VT-1010-038-A130-N
SEAL PLAN :

CHECK PRINT

NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT. EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER, CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE, AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:

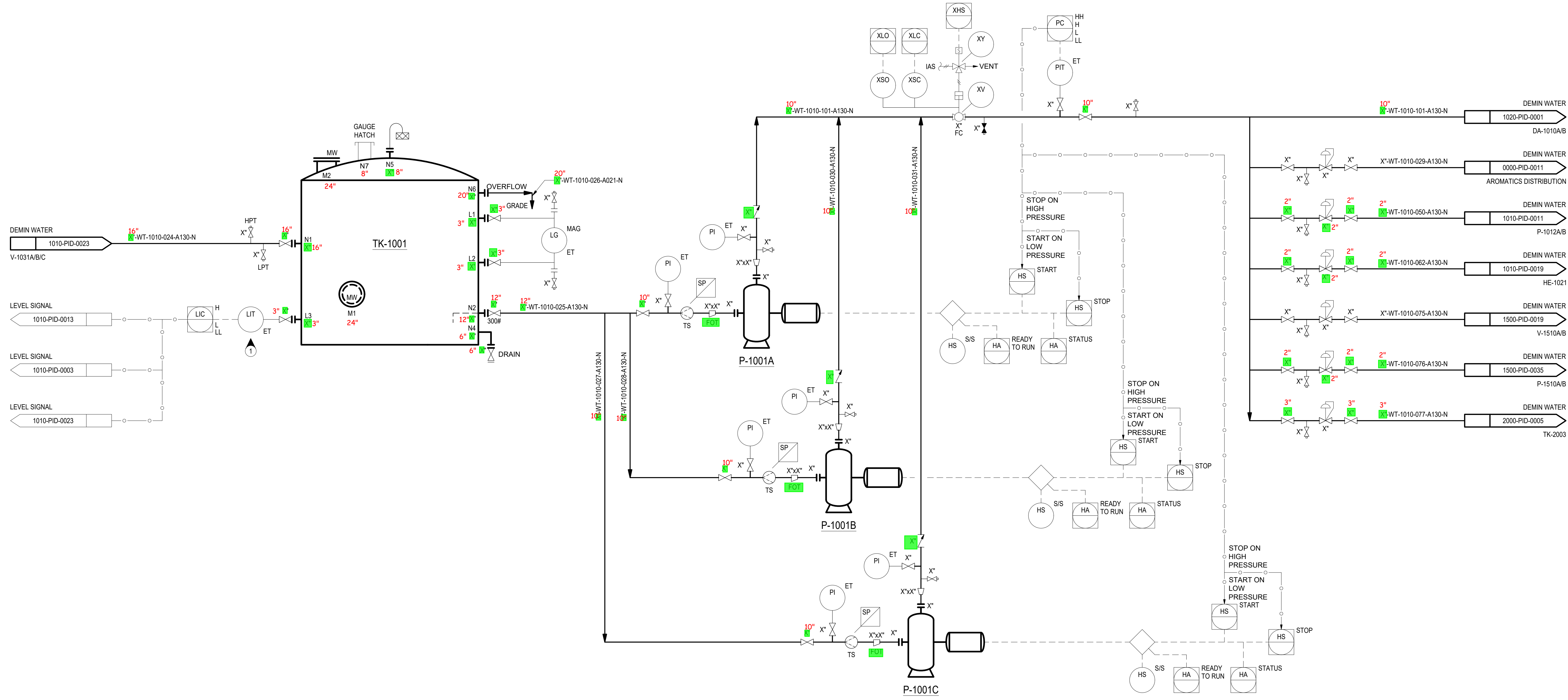
REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

MOTIVA			PIPING & INSTRUMENTATION DIAGRAM		SSP ENGINEERING & CONSULTING		FIRM NUMBER #13619	
CLIENT: MOTIVA ENTERPRISES LLC			DEMINERALIZER WATER PACKAGE		13100 NORTHWEST FREEWAY		SCALE: NONE	
PLANT: PORT ARTHUR REFINERY			BRINE SYSTEM		SUITE 625		CHECKED: APB	
JOB NO:			PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW		HOUSTON, TEXAS 77040		DESIGN BY: ENG APVL: PDN	
					DRAWING NO:		3000-PID-0003	
							1010-PID-0031	
							A	

TK-1001
DEMIN WATER TANK
SIZE ID x HEIGHT :
TOTAL VOLUME :
DESIGN PRESSURE :
DESIGN TEMP :
MOC :
TRIM : VT-1010-032-A130-N
INSULATION / TRACING :

P-1001A/B/C
DEMIN WATER PUMPS
RATED CAPACITY :
DIFFERENTIAL PRESSURE :
RATED POWER :
MOC (CASING / INTERNAL) :
INSULATION / TRACING :
TRIM : VT-1010-033-A130-N
SEAL PLAN :

- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.
- NOTES
1. THREADED BLEED RING ON WATER SERVICE.



CHECK PRINT

NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT. EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER, CONTRACTOR, SUB-CONTRACTORS, & SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE, AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT, SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

	PIPING & INSTRUMENTATION DIAGRAM			SSP ENGINEERING & CONSULTING		FIRM NUMBER #13619
	DEMIN WATER TANK			13100 NORTHWEST FREEWAY SUITE 625, HOUSTON, TEXAS 77040		
	CLIENT: MOTIVA ENTERPRISES LLC			PROJECT NO: 25MA02		
	PLANT: PORT ARTHUR REFINERY			DRAWN BY: GS		
JOB NO:	PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW		DESIGN BY:		SCALE: NONE	
			DRAWING NO:		1010-PID-0033	
					CHECKED: APB	
					ENG APVL: PDN	
					REV A	

C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.



FIRM
NUMBER
#13619

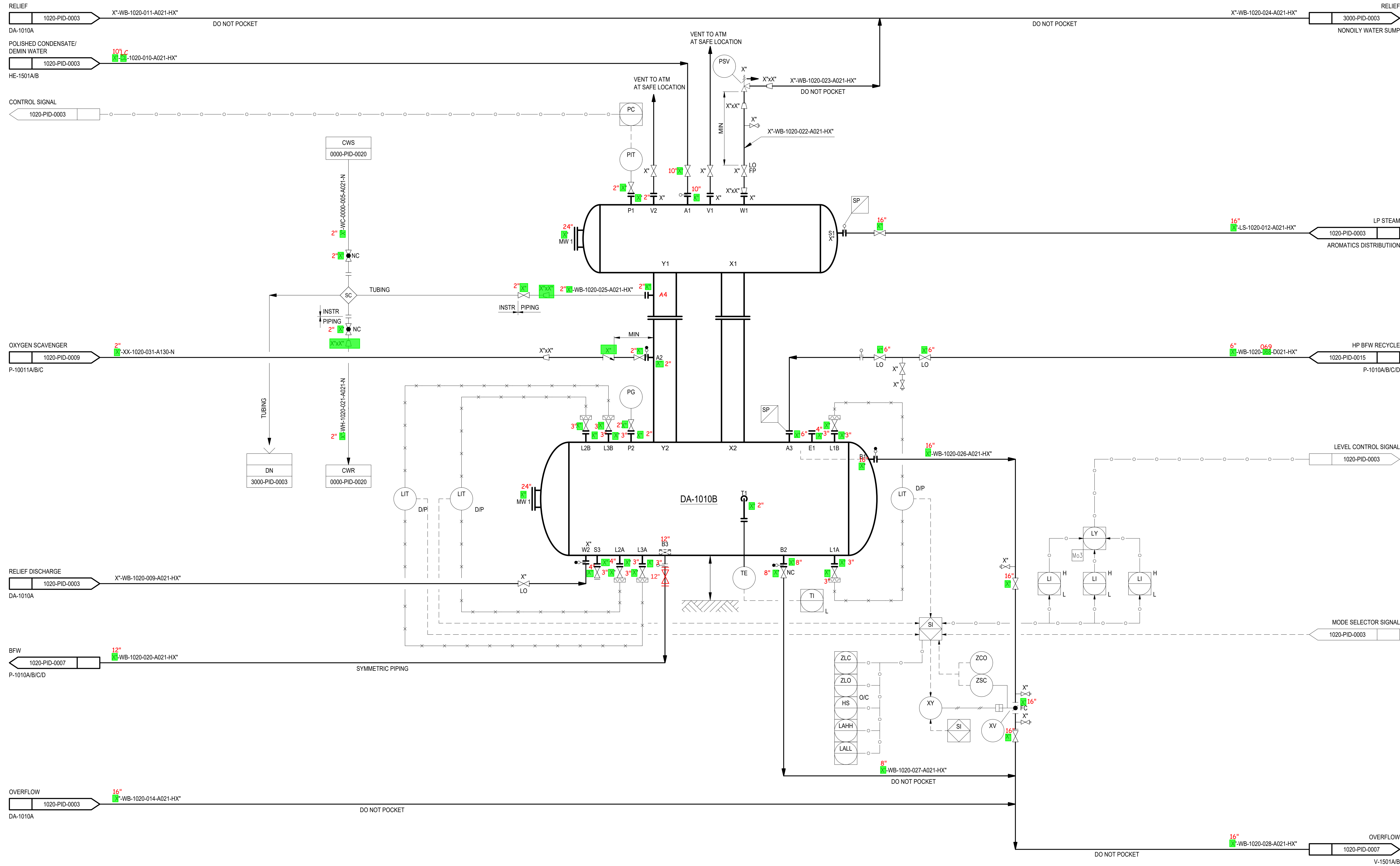
01

DA-1010B
DEAERATOR B

SIZE :
DESIGN PRESSURE :
DESIGN TEMP :
MOC :
TRIM : VT-1020-029-A021-HX*
INSULATION :

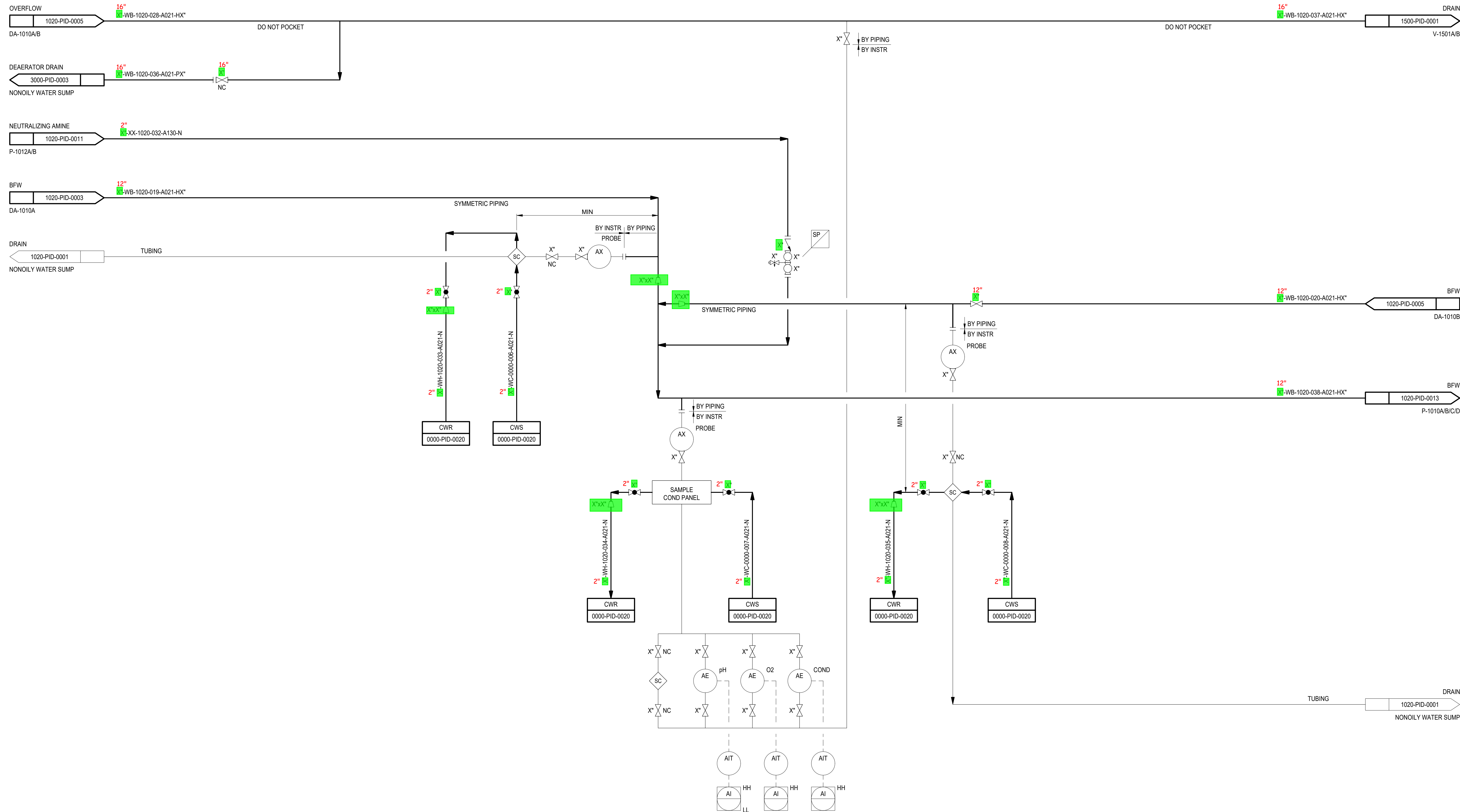
GENERAL NOTES

- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.



CHECK PRINT

- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.



NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT. EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER, CONTRACTOR, SUB-CONTRACTORS, & SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT, SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	DLH	PDN						

MOTIVA

CLIENT: MOTIVA ENTERPRISES LLC
PLANT: PORT ARTHUR REFINERY
JOB NO:

PIPING & INSTRUMENTATION DIAGRAM
STEAM UNIT
BFW SAMPLE SYSTEM

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW



SSP ENGINEERING & CONSULTING
13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

PROJECT NO: 25MA02

DRAWN BY: GS

DESIGN BY:

DRAWING NO:

SCALE: NONE

CHECKED: APB

ENG APVL: PDN

1020-PID-0007

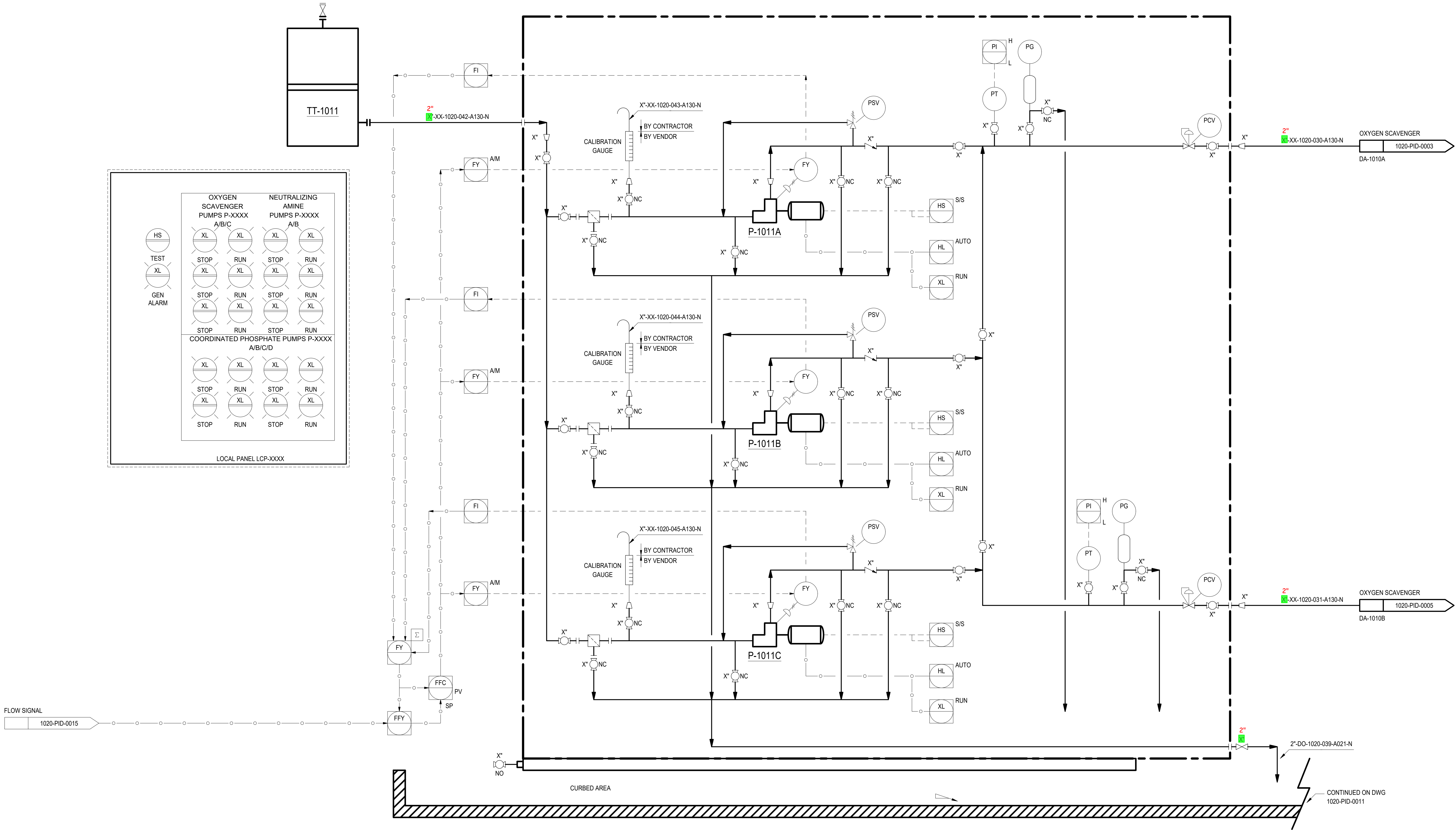
REV A

FIRM NUMBER #13619

TT-1011
OXYGEN SCAVENGER STACKED TOTE
SIZE ID x HEIGHT :
TOTAL VOLUME :
DESIGN PRESSURE :
DESIGN TEMP :
MOC :
TRIM : VT-1020-040-A130-N
INSULATION / TRACING :

P-1011A/B/C
OXYGEN SCAVENGER PUMP
RATED CAPACITY :
DIFFERENTIAL PRESSURE :
RATED POWER :
MOC (CASING / INTERNAL) :
INSULATION / TRACING :
TRIM : VT-1020-041-A130-N
SEAL PLAN :

GENERAL NOTES
A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.



NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT, EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER. CONTRACTOR, SUB-CONTRACTORS, & SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	DLH	PDN						

CLIENT: MOTIVA ENTERPRISES LLC
PLANT: PORT ARTHUR REFINERY
JOB NO:

PIPING & INSTRUMENTATION DIAGRAM
OXYGEN SCAVENGER PACKAGE

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

SSP ENGINEERING & CONSULTING
13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

FIRM NUMBER
#13619

PROJECT NO: 25MA02
DRAWN BY: GS
DESIGN BY:

SCALE: NONE
CHECKED: APB
ENG APVL: PDN

DRAWING NO:

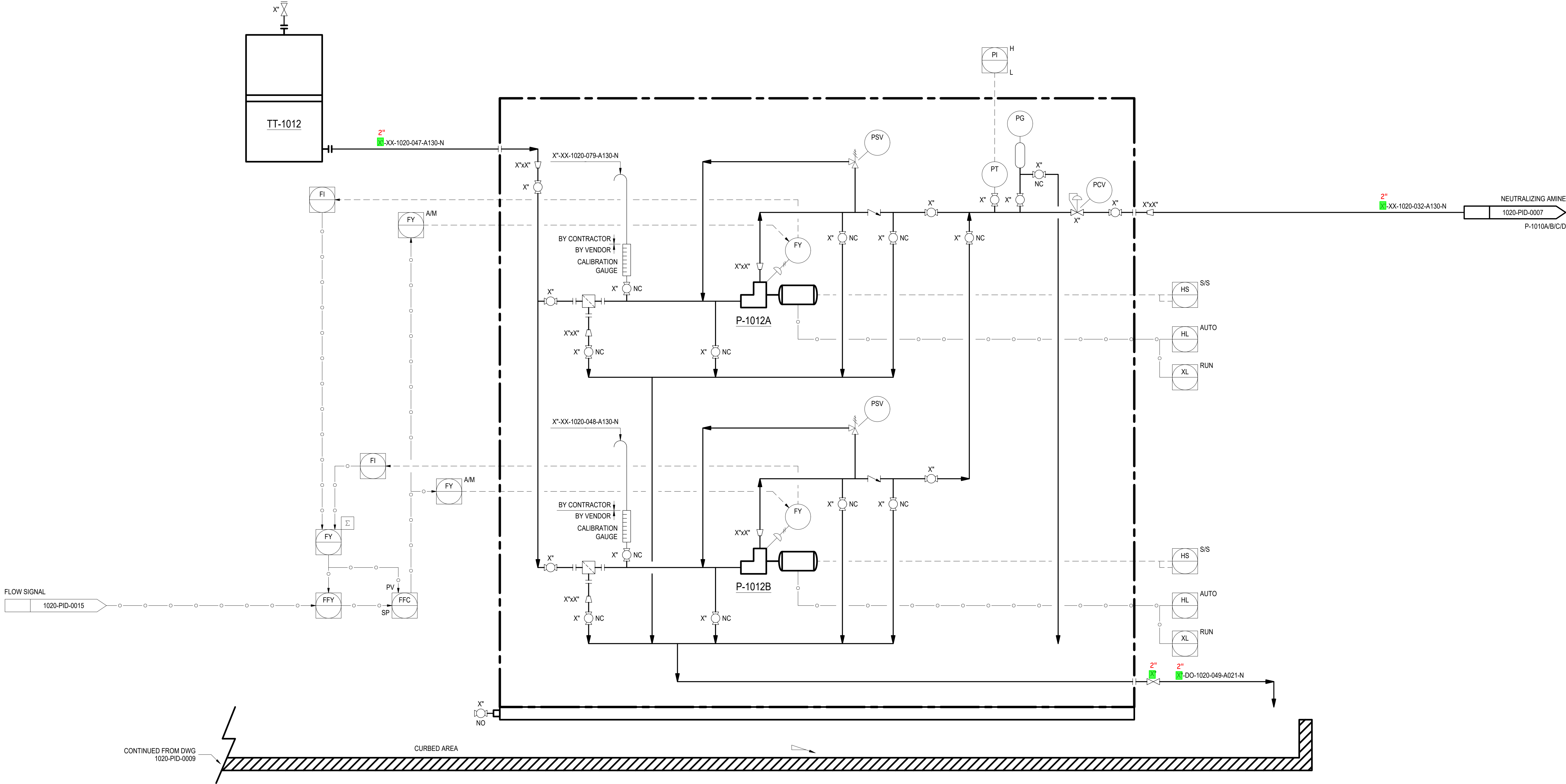
1020-PID-0009

REV
A

TT-1012
NEUTRALIZING AMINE STACKED TOTE
SIZE ID x HEIGHT:
NUMBER OF TRAYS:
RATED CAPACITY:
DESIGN PRESSURE:
DESIGN TEMP:
MOC:
TRIM : VT-1020-050-A130-N
INSULATION :

P-1012A/B
NEUTRALIZING AMINE PUMPS
RATED CAPACITY:
DIFFERENTIAL PRESSURE:
RATED POWER:
MOC (CASING / INTERNAL):
INSULATION / TRACING:
TRIM : VT-1020-051-A130-N
SEAL PLAN :

GENERAL NOTES
A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.



CHECK PRINT

NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT, EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW UNLESS SPECIFIED OTHERWISE AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	DLH	PDN						

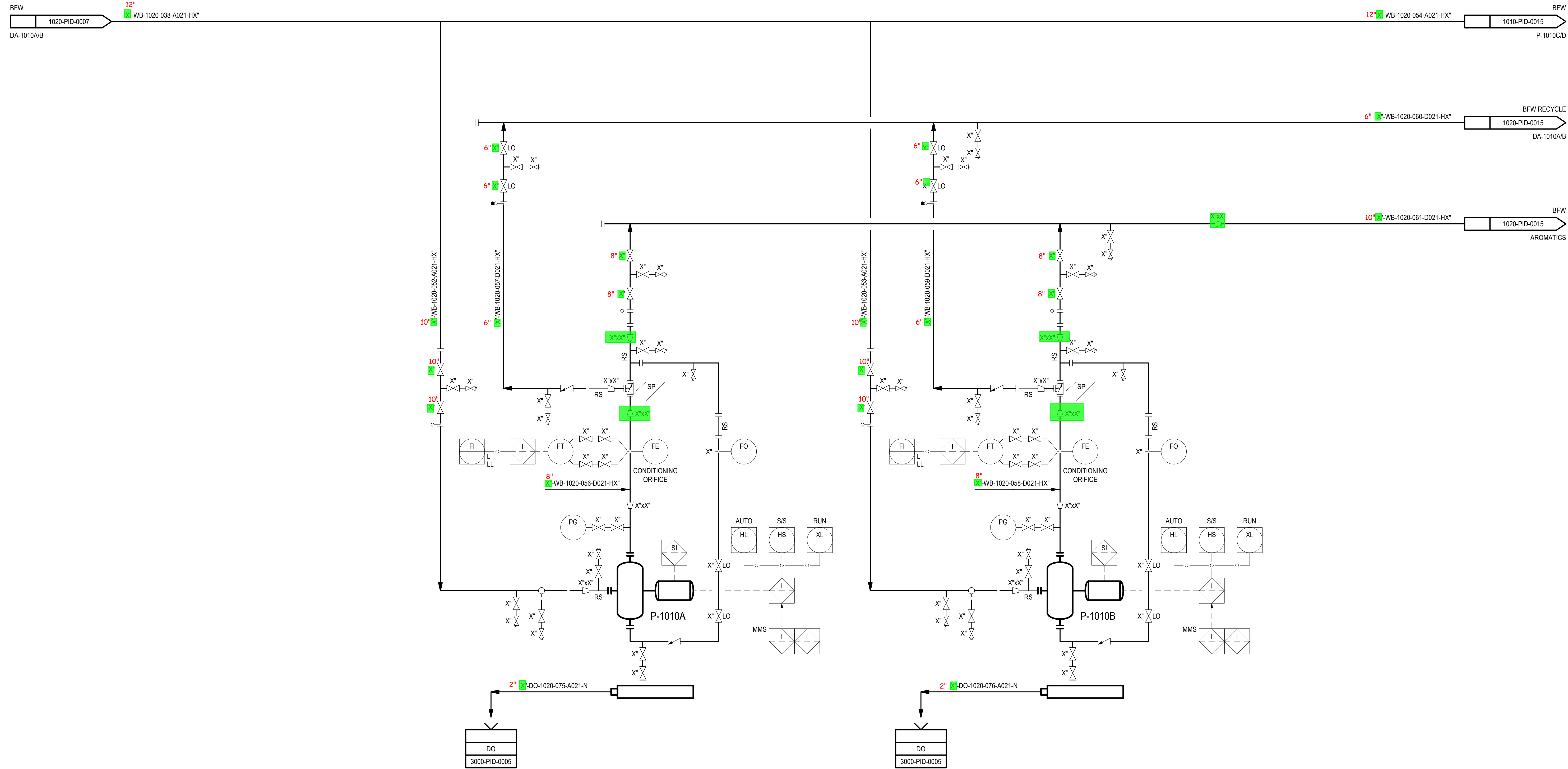
MOTIVA		
CLIENT: MOTIVA ENTERPRISES LLC		
PLANT: PORT ARTHUR REFINERY		
JOB NO:		

PIPING & INSTRUMENTATION DIAGRAM	
NEUTRALIZING AMINE PACKAGE	
PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW	

SSP ENGINEERING & CONSULTING 13100 NORTHWEST FREEWAY SUITE 625 HOUSTON, TEXAS 77040		FIRM NUMBER #13619
PROJECT NO: 25MA02	SCALE: NONE	
DRAWN BY: GS	CHECKED: APB	
DESIGN BY:	ENG APVL: PDN	
DRAWING NO:	1020-PID-0011	REV A

P-1010A/B
BFW PUMPS
RATED CAPACITY :
DIFFERENTIAL PRESSURE :
RATED POWER :
MOC (CASING / INTERNAL) :
INSULATION / TRACING :
TRIM : VT-1020-062-D021-HX
SEAL PLAN :

- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.



CHECK PRINT

NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT, EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER. CONTRACTOR, SUB-CONTRACTORS & SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT, SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	DLH	PDN						

MOTIVA

CLIENT: MOTIVA ENTERPRISES LLC
PLANT: PORT ARTHUR REFINERY
JOB NO:

PIPING & INSTRUMENTATION DIAGRAM
STEAM UNIT
BFW PUMPS A/B

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW



SSP ENGINEERING & CONSULTING
13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

PROJECT NO: 25MA02

DRAWN BY: GS

DESIGN BY:

DRAWING NO:

SCALE: NONE

CHECKED: APB

ENG APVL: PDN

REV

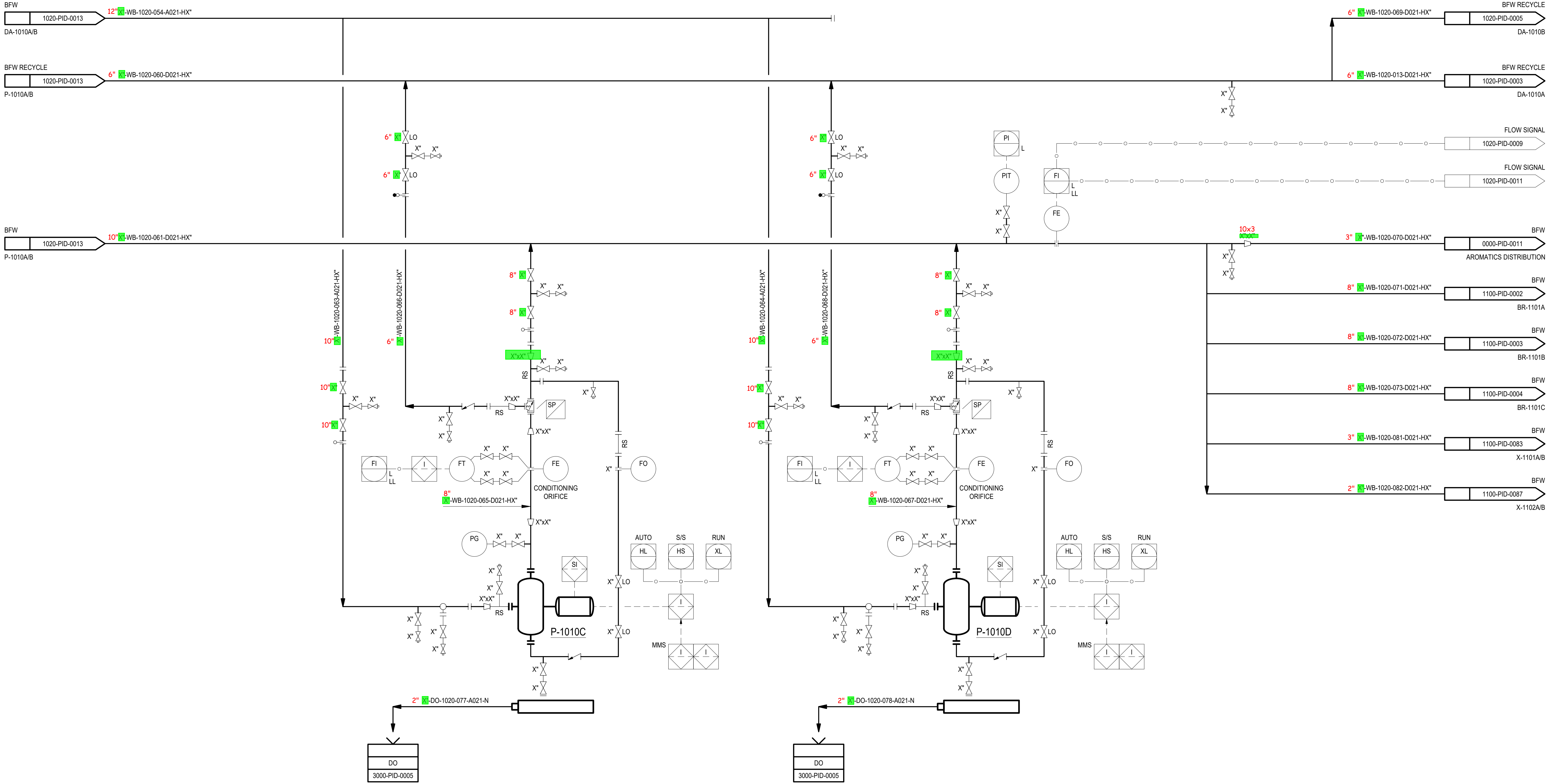
FIRM
NUMBER
#13619

1020-PID-0013

A

P-1010C/D
BFW PUMPS
RATED CAPACITY :
DIFFERENTIAL PRESSURE :
RATED POWER :
MOC (CASING / INTERNAL) :
INSULATION / TRACING :
TRIM : VT-1020-074-D021-HX"
SEAL PLAN :

- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.



CHECK PRINT

NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT, EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER, CONTRACTOR, SUB-CONTRACTORS, & SUPPLIERS. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	DLH	PDN						

CLIENT: MOTIVA ENTERPRISES LLC
PLANT: PORT ARTHUR REFINERY
JOB NO:

STEAM UNIT
BFW PUMPS C/D

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

SSP ENGINEERING & CONSULTING
13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

FIRM NUMBER
#13619

PROJECT NO: 25MA02
DRAWN BY: GS
DESIGN BY:

SCALE: NONE
CHECKED: APB
ENG APVL: PDN

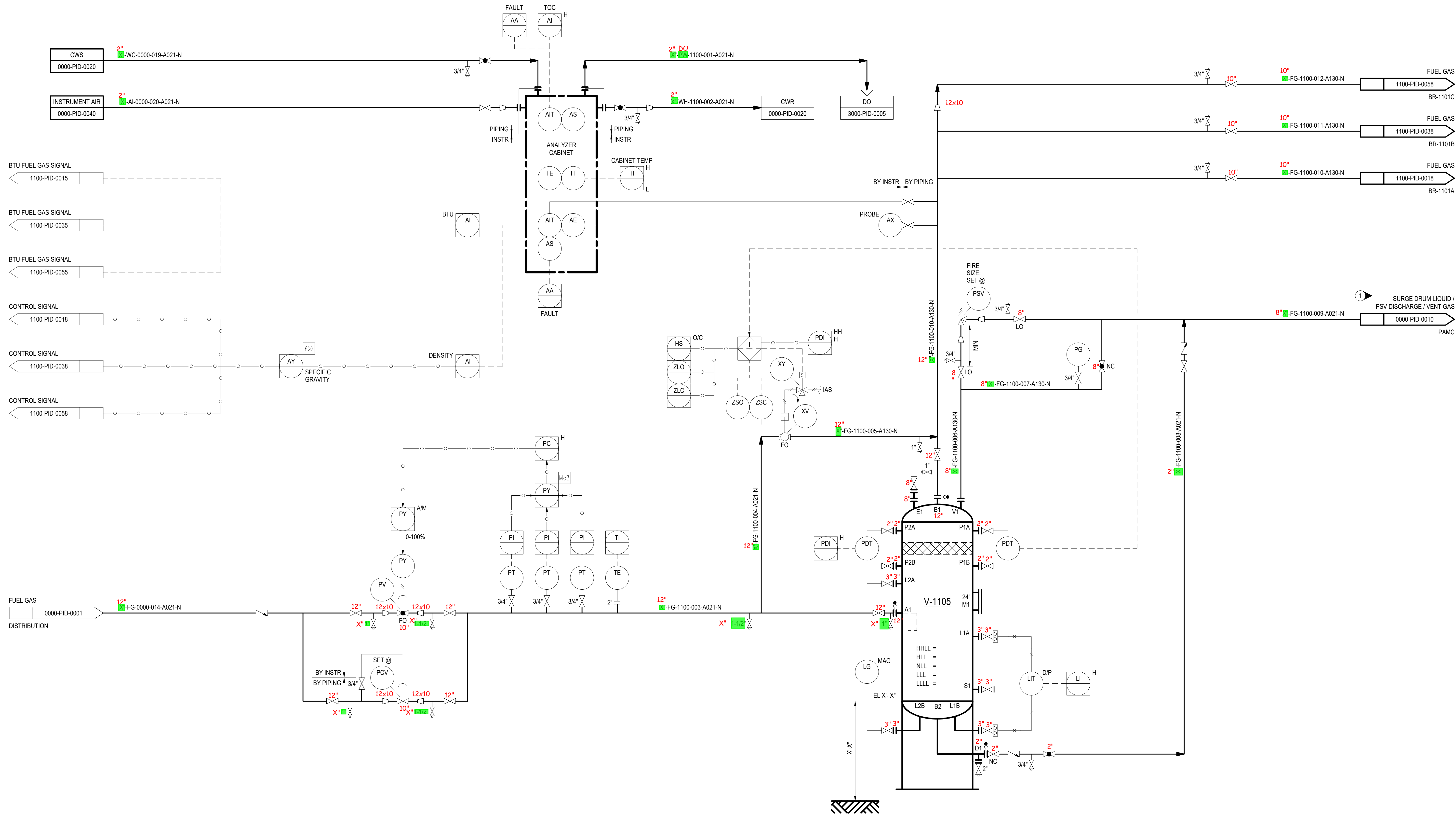
DRAWING NO:

1020-PID-0015

REV
A

V-1105
FUEL GAS COALESCER
SIZE ID x T / I :
DESIGN PRESSURE :
DESIGN TEMP :
MOC :
TRIM : VT-1100-013-A130-N
INSULATION :

- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.
- NOTES
1. COALESCER PSV WILL BE ROUTED TO AROMATICS FLARE DEPENDING ON AVAILABILITY OF TIE-IN POINT.

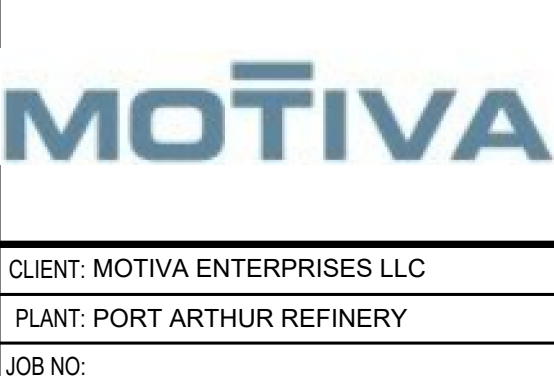


CHECK PRINT

NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT, EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER. CONTRACTOR, SUB-CONTRACTORS, A SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE, AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT, SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						



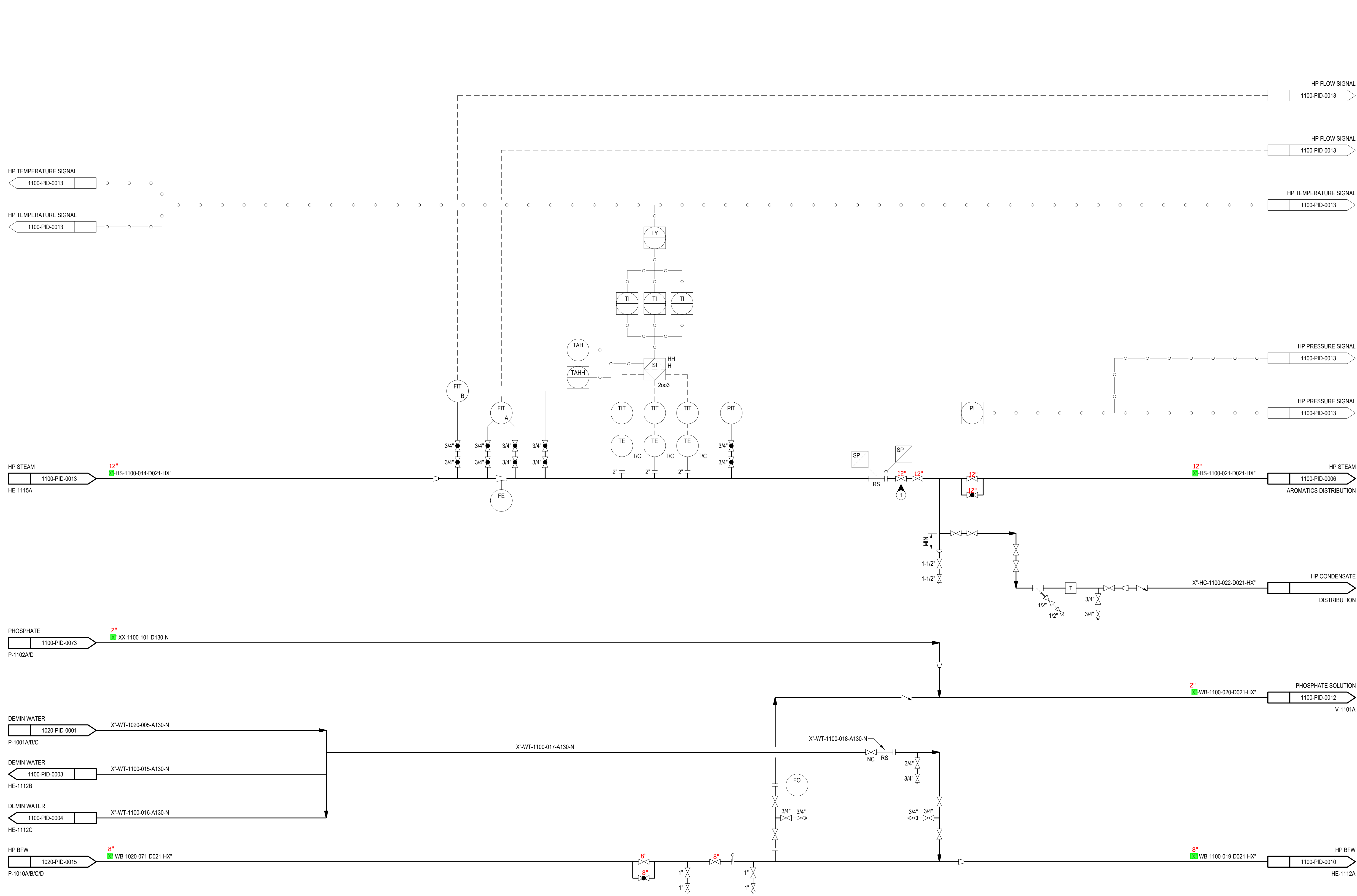
PIPING & INSTRUMENTATION DIAGRAM
STEAM UNIT
FUEL GAS COALESCER

CLIENT: MOTIVA ENTERPRISES LLC
PLANT: PORT ARTHUR REFINERY
JOB NO:

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

SSP ENGINEERING & CONSULTING 13100 NORTHWEST FREEWAY SUITE 625 HOUSTON, TEXAS 77040	FIRM NUMBER #13619
PROJECT NO: 25MA02 DRAWN BY: GS DESIGN BY: DRAWING NO:	SCALE: NONE CHECKED: APB ENG APVL: PDN 1100-PID-0001
	REV A

DRAWING: 1100-pid-0002.dwg, PLOT DATE: 12-05-25, PLOTTED BY: jason, DRAWING LOCATION: c:\users\jason\one\drive - eng\ssg\files - 25ma02 - aromatics utility scope cold eyes review\04 - design\04 - process\pids, (Border Rev1 5.16.25)



- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.
- NOTES
1. VALVE FOR STEAM BLOWS OF HP SYSTEM.

NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT, EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE, AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

CLIENT: MOTIVA ENTERPRISES LLC

PLANT: PORT ARTHUR REFINERY

JOB NO:

PIPING & INSTRUMENTATION DIAGRAM

STEAM UNIT

BFW AND HP STEAM BOILER A

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

SSP ENGINEERING & CONSULTING

13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

FIRM NUMBER
#13619

PROJECT NO: 25MA02

SCALE: NONE

DRAWN BY: GS

CHECKED: APB

DESIGN BY:

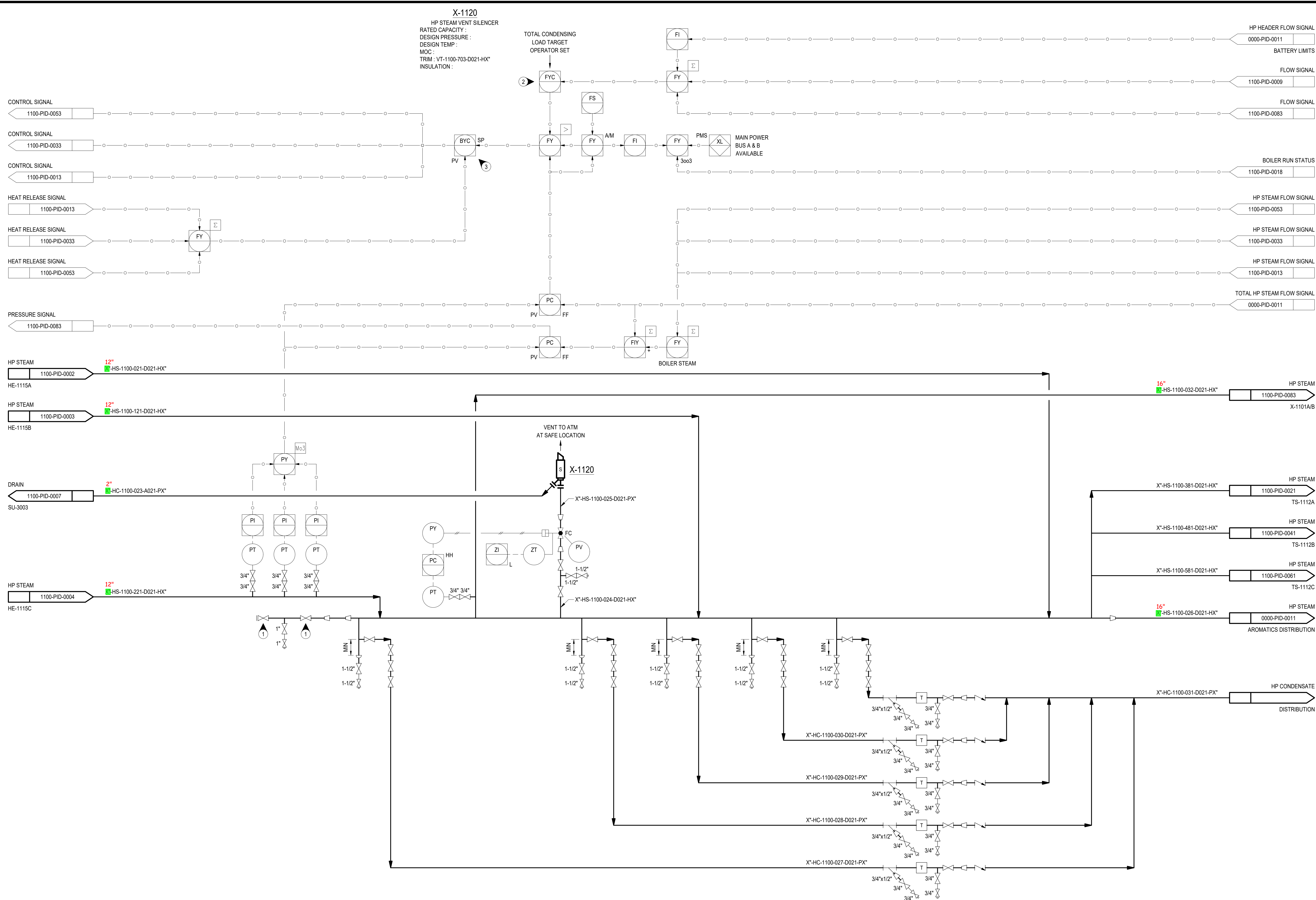
ENG APVL: PDN

DRAWING NO:

1100-PID-0002

REV
A

DRAWING: 1100-pid-0000.dwg, PLOTTED BY: Jason, PLOT DATE: ---, DRAWING LOCATION: c:\users\jason\recent - engs\pids - aromatics utility scope cold eyes review\04 - process\pids, (Border Rev 1 5 16.23)



- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
 - B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
 - C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.
- NOTES
- 1. VALVE FOR STEAM BLOWS OF HP SYSTEM.
 - 2. DE-TUNED TO PROVIDE AVERAGING CONTROL ACTION. THE PROCESS VALUE OF THIS PID CONTROLLER SHOULD BE MOVING AVERAGE PV AND NOT A HEAVILY FILTERED VALUE.
 - 3. BYC IS A LOAD ALLOCATOR (I-ONLY PID) THAT ACT AS FAST SECONDARY TO RE-DISTRIBUTE LOAD.

NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT, EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER. CONTRACTOR, SUB-CONTRACTORS, & SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE, AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT, SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:		

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

CLIENT: MOTIVA ENTERPRISES LLC

PLANT: PORT ARTHUR REFINERY

JOB NO:

PIPING & INSTRUMENTATION DIAGRAM

STEAM UNIT

HP STEAM HEADER

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

SSP ENGINEERING & CONSULTING

13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

FIRM NUMBER
#13619

PROJECT NO: 25MA02

DRAWN BY: GS

DESIGN BY:

DRAWING NO:

SCALE: NONE

CHECKED: APB

ENG APVL: PDN

1100-PID-0006

REV A

MP STEAM VENT SILENCER
RATED CAPACITY :
DESIGN PRESSURE :
DESIGN TEMP :
MOC :
TRIM : VT-1100-041-B021-PX*
INSULATION :

- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.



007

X-1140

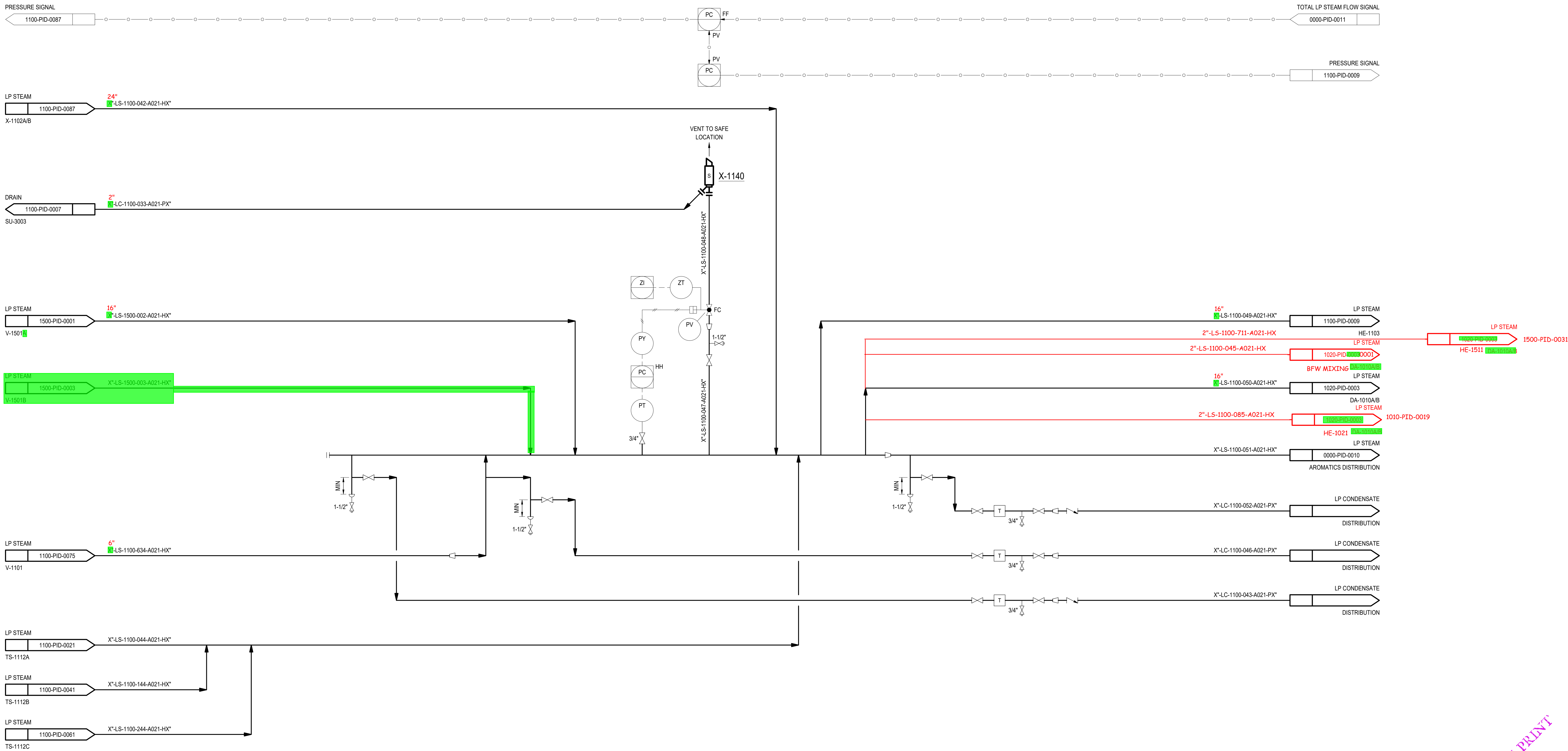
LP STEAM VENT SILENCER
RATED CAPACITY :
DESIGN PRESSURE :
DESIGN TEMP :
MOC :
TRIM : VT-1100-704-A021-HX"
INSULATION :

GENERAL NOTES

A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS
SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.

B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.

C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.



ALL PUBLISHED, ALL RIGHT RESERVED BY THE ENGINEER, DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT, EXCEPT BY AGREEMENT IN WRITING AND WITHOUT THE WRITTEN CONSENT OF THE ENGINEER. THE CONTRACTOR SHALL BE RESPONSIBLE FOR CONFIRMING AND CORRECTING DIMENSIONS AND CONDITIONS AT THE JOB SITE, AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS, DESCRIBING DESIGN INTENT. SHALL NOT INDICATE ANY DETAIL OR SYSTEM FOR CONSTRUCTION, NOR SHALL THE CONTRACTOR BE REQUIRED TO CONSTRUCT ANY DETAIL OR SYSTEM TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND MAINTENANCE. THE CONTRACTOR SHALL BE RESPONSIBLE FOR OBTAINING ALL NECESSARY CO-ORDINATION OF ALL WORK CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR OBTAINING ALL NECESSARY PERMITS AND APPROVALS. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, SQUARE, TRUE, AND TRUE IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE CONTRACTOR SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, OR SEQUENCES. THE CONTRACTOR SHALL BE RESPONSIBLE FOR OBTAINING ALL NECESSARY PERMITS AND APPROVALS. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS

[illegible]

MOTIVA

CLIENT: MOTIVA ENTERPRISES LLC

PLANT: PORT ARTHUR REFINERY

JOB NO.

PIPING & INSTRUMENTATION DIAGRAM

STEAM UNIT
LP STEAM HEADER

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW



**SSP ENGINEERING
& CONSULTING**
13100 NORTHWEST FREEWAY
SUITE 625,
HOUSTON, TEXAS 77040

NUMBER
#13619

PROJECT NO: 25MA01

DRAWN BY: GS

DESIGN BY:

DRAWING NO

SCALE: NONE

CHECKED: APB

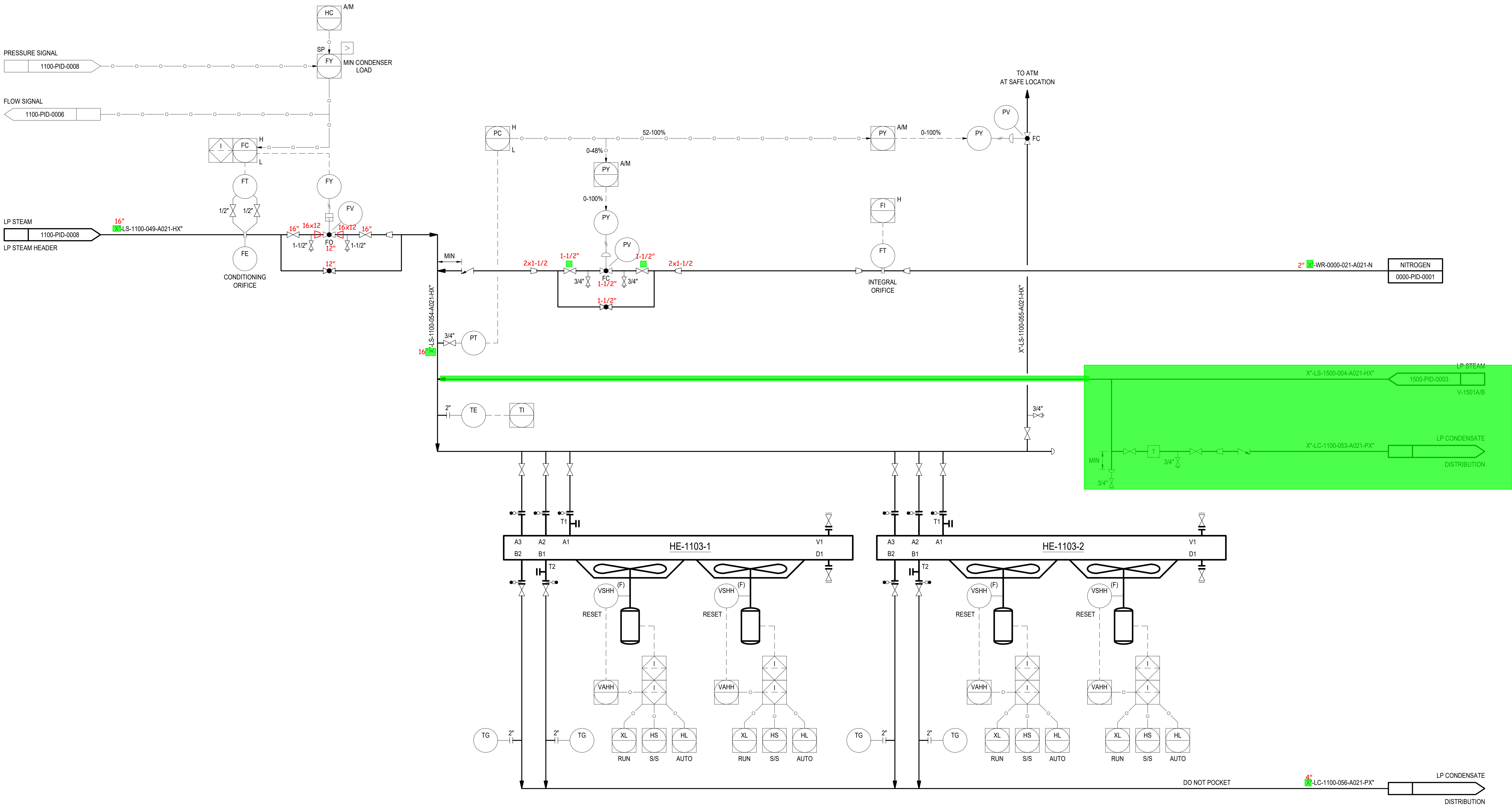
ENG APVI : PDN

1100-PID-0008

REV

HE-1103
LP STEAM VENT CONDENSER
RATED DUTY :
NUMBER OF BAYS :
DESIGN TEMP :
RATED POWER :
MOC :
TRIM : VT-1100-057-A021-PX"
INSULATION :

- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.



CHECK PRINT

NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT, EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE, AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT, SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

MOTIVA

CLIENT: MOTIVA ENTERPRISES LLC
PLANT: PORT ARTHUR REFINERY
JOB NO:

PIPING & INSTRUMENTATION DIAGRAM

STEAM UNIT
LP STEAM VENT CONDENSER

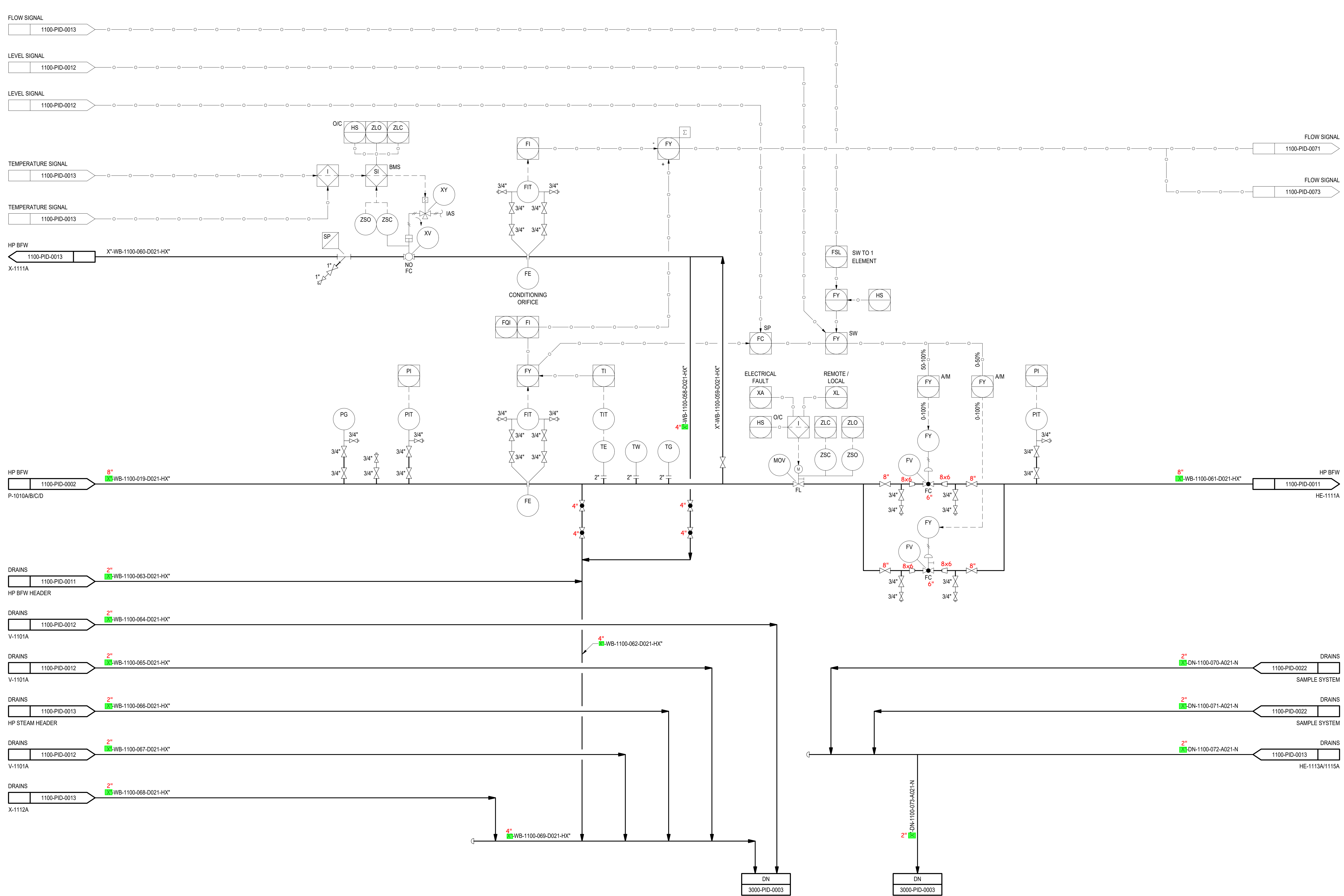
PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

SSP ENGINEERING & CONSULTING
13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

PROJECT NO: 25MA02
SCALE: NONE
DRAWN BY: GS
CHECKED: APB
DESIGN BY:
ENG APVL: PDN
DRAWING NO:

FIRM NUMBER
#13619

REV
1100-PID-0009 A



GENERAL NOTES

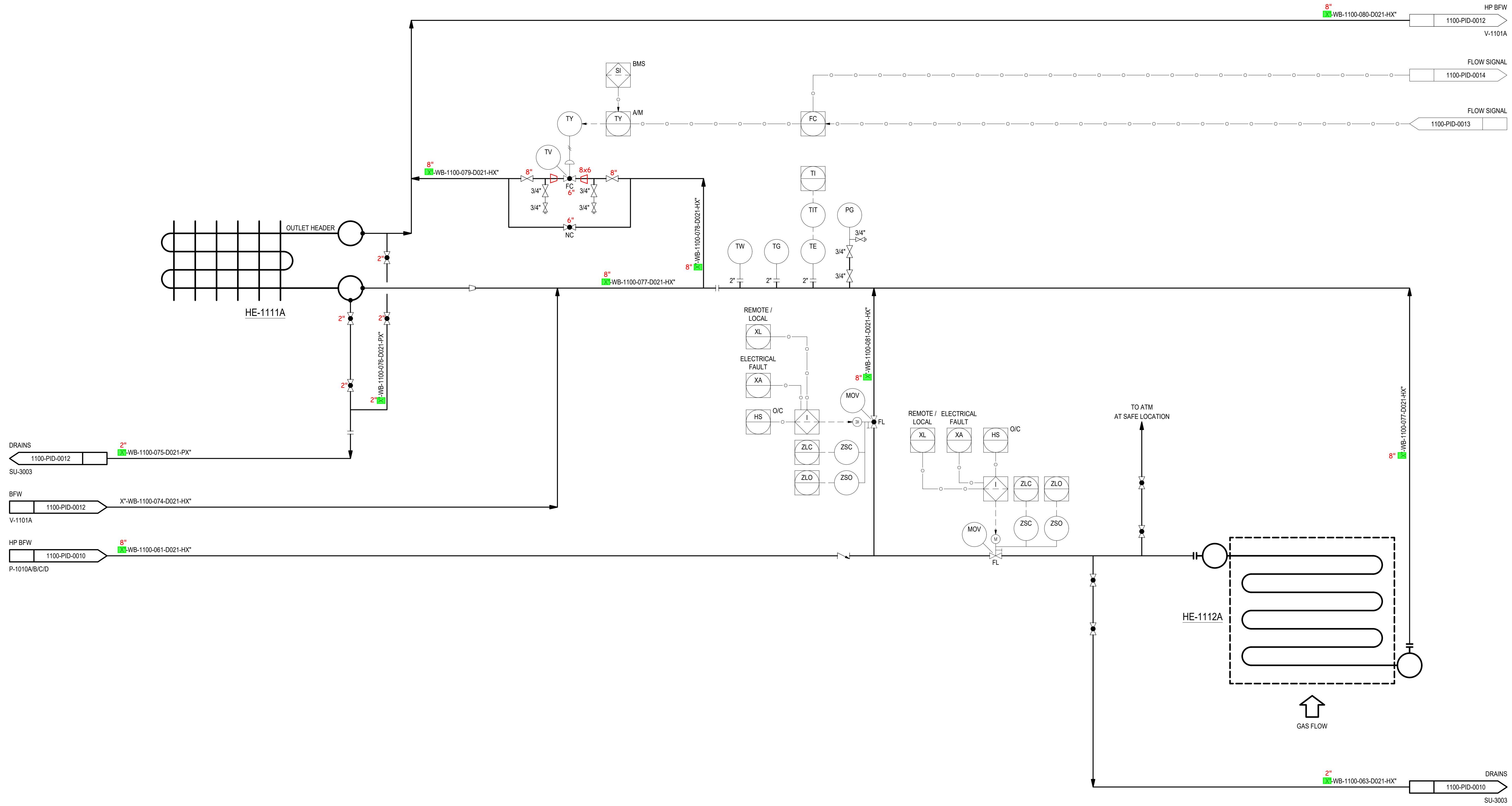
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.

NOT PUBLISHED, ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT. EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER, CONTRACTOR, SUB-CONTRACTORS, & SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE, AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND CO-ORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.										REFERENCE DRAWINGS:			<table border="1"> <thead> <tr> <th>REV #</th> <th>DATE</th> <th>DESCRIPTION</th> <th>DRWN</th> <th>CHKD</th> <th>APVD</th> </tr> </thead> <tbody> <tr> <td>A</td> <td>12-05-25</td> <td>ISSUED FOR APPROVAL</td> <td>GS</td> <td>DLH</td> <td>PDN</td> </tr> <tr> <td> </td> <td> </td> <td> </td> <td> </td> <td> </td> <td> </td> </tr> <tr> <td> </td> <td> </td> <td> </td> <td> </td> <td> </td> <td> </td> </tr> </tbody> </table>	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	A	12-05-25	ISSUED FOR APPROVAL	GS	DLH	PDN													<table border="1"> <thead> <tr> <th>REV #</th> <th>DATE</th> <th>DESCRIPTION</th> <th>DRWN</th> <th>CHKD</th> <th>APVD</th> </tr> </thead> <tbody> <tr> <td> </td> <td> </td> <td> </td> <td> </td> <td> </td> <td> </td> </tr> <tr> <td> </td> <td> </td> <td> </td> <td> </td> <td> </td> <td> </td> </tr> <tr> <td> </td> <td> </td> <td> </td> <td> </td> <td> </td> <td> </td> </tr> </tbody> </table>	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD																			 <table border="1"> <tr> <td>CLIENT: MOTIVA ENTERPRISES LLC</td> </tr> <tr> <td>PLANT: PORT ARTHUR REFINERY</td> </tr> <tr> <td>JOB NO:</td> </tr> </table>			CLIENT: MOTIVA ENTERPRISES LLC	PLANT: PORT ARTHUR REFINERY	JOB NO:	 SSP ENGINEERING & CONSULTING 13100 NORTHWEST FREEWAY SUITE 625 HOUSTON, TEXAS 77040			FIRM NUMBER #13619		
REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD																																																																					
A	12-05-25	ISSUED FOR APPROVAL	GS	DLH	PDN																																																																					
REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD																																																																					
CLIENT: MOTIVA ENTERPRISES LLC																																																																										
PLANT: PORT ARTHUR REFINERY																																																																										
JOB NO:																																																																										
PIPING & INSTRUMENTATION DIAGRAM												<table border="1"> <tr> <td colspan="3">PROJECT NO: 25MA02</td> <td colspan="3">SCALE: NONE</td> </tr> <tr> <td colspan="3">DRAWN BY: GS</td> <td colspan="3">CHECKED: APB</td> </tr> <tr> <td colspan="3">DESIGN BY:</td> <td colspan="3">ENG APVL: PDN</td> </tr> <tr> <td colspan="3">DRAWING NO:</td> <td colspan="3"> 1100-PID-0010 </td> </tr> </table>						PROJECT NO: 25MA02			SCALE: NONE			DRAWN BY: GS			CHECKED: APB			DESIGN BY:			ENG APVL: PDN			DRAWING NO:			1100-PID-0010																																			
PROJECT NO: 25MA02			SCALE: NONE																																																																							
DRAWN BY: GS			CHECKED: APB																																																																							
DESIGN BY:			ENG APVL: PDN																																																																							
DRAWING NO:			1100-PID-0010																																																																							
STEAM UNIT BOILER A HP BOILER FEED WATER HEADER						PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW																																																																				

HE-1111A
INTERNAL ECONOMIZER
RATED DUTY :
SURFACE AREA :
DESIGN PRESSURE :
DESIGN TEMP :
MOC :
TRIM : VT-1100-082-D021-HX"
INSULATION :

HE-1112A
EXTERNAL ECONOMIZER
RATED DUTY :
SURFACE AREA :
DESIGN PRESSURE :
DESIGN TEMP :
MOC :
TRIM : VT-1100-083-D021-HX"
INSULATION :

GENERAL NOTES
A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.



NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT, EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER. CONTRACTOR, SUB-CONTRACTORS, A SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE, AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT, SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	DLH	PDN						

CLIENT: MOTIVA ENTERPRISES LLC
PLANT: PORT ARTHUR REFINERY
JOB NO:

PIPING & INSTRUMENTATION DIAGRAM

STEAM UNIT
BOILER A
EXTERNAL AND INTERNAL ECONOMIZER

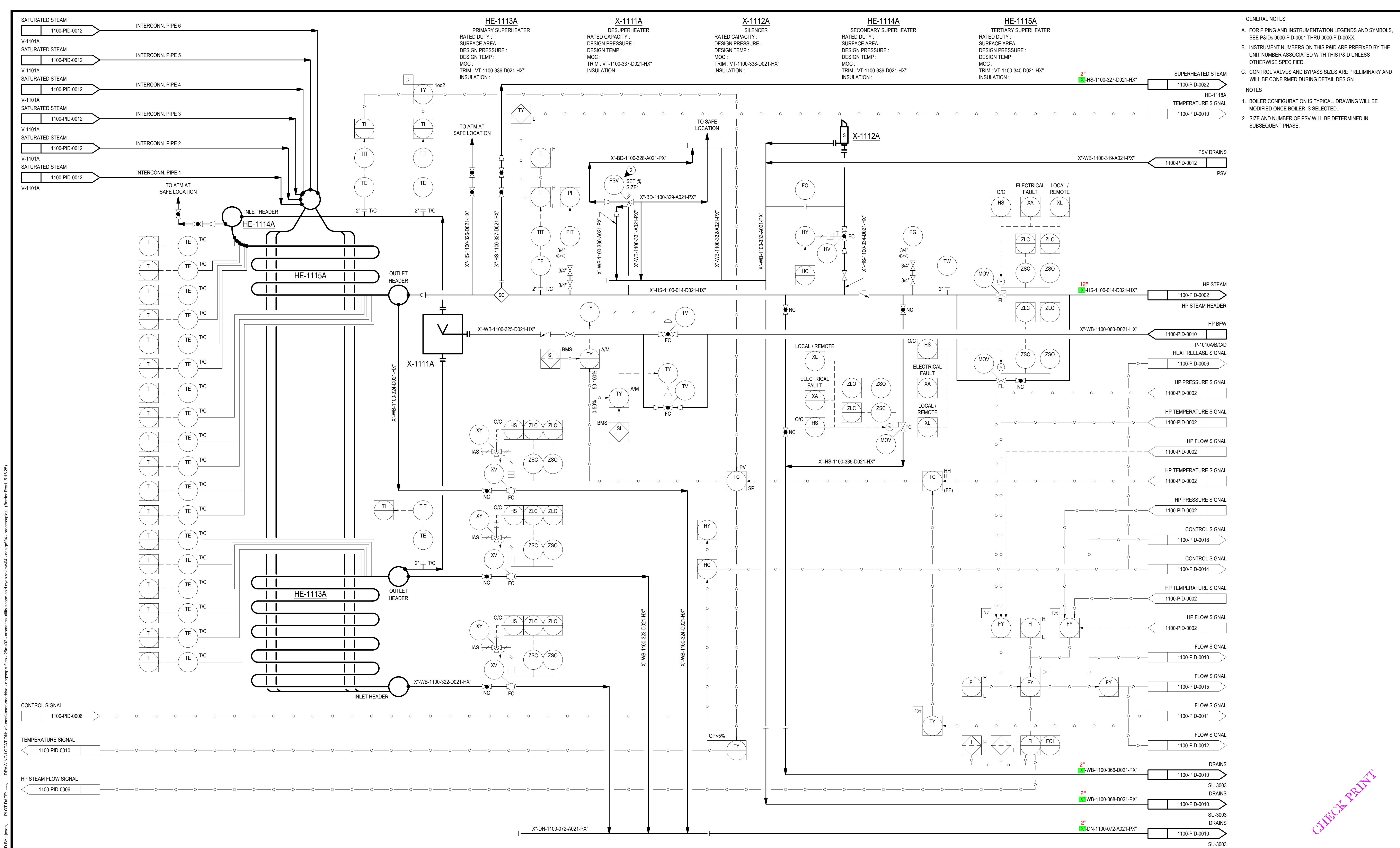
PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

SSP ENGINEERING & CONSULTING
13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

PROJECT NO: 25MA02
DRAWN BY: GS
DESIGN BY:
DRAWING NO:

SCALE: NONE
CHECKED: APB
ENG APVL: PDN

FIRM NUMBER #13619
REV A



- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDS 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.
- NOTES
1. BOILER CONFIGURATION IS TYPICAL. DRAWING WILL BE MODIFIED ONCE BOILER IS SELECTED.
2. SIZE AND NUMBER OF PSV WILL BE DETERMINED IN SUBSEQUENT PHASE.

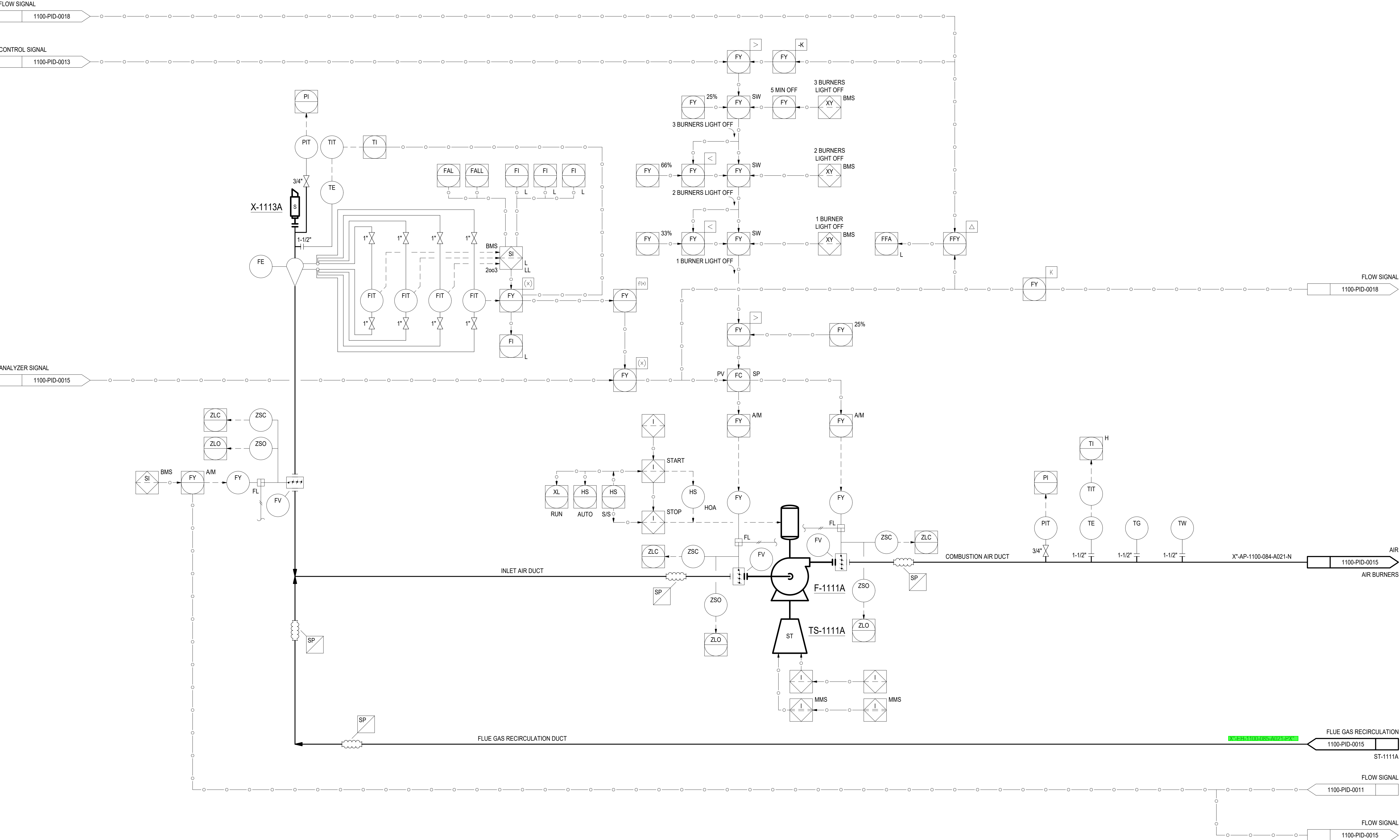
DRAWING: 1100-04-0013-000g PLOT: 1100-04-0013-000g	NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT, EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE, AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.																		
	REFERENCE DRAWINGS:																		
	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	MOTIVA CLIENT: MOTIVA ENTERPRISES LLC PLANT: PORT ARTHUR REFINERY JOB NO:						
	A	12-05-25	ISSUED FOR APPROVAL	GS	DLH	PDN													
												PIPING & INSTRUMENTATION DIAGRAM				SSP SSP ENGINEERING & CONSULTING 13100 NORTHWEST FREEWAY SUITE 625 HOUSTON, TEXAS 77040 FIRM NUMBER #13619			
												STEAM UNIT BOILER A SUPERHEATER				PROJECT NO: 25MA02 SCALE: NONE			
												DRAWN BY: GS CHECKED: APB				DESIGN BY: ENG APVL: PDN			
												DRAWING NO: 1100-PID-0013				REV A			

X-1113A
AIR SILENCER
RATED CAPACITY :
DESIGN PRESSURE :
DESIGN TEMP :
MOC :
TRIM : VT-1100-086-A021-N
INSULATION :

F-1111A
FORCED DRAFT FAN
RATED CAPACITY :
DIFFERENTIAL PRESSURE :
RATED POWER :
MOC :
TRIM : VT-1100-087-A021-N
INSULATION :

TS-1111A
STEAM TURBINE
RATED POWER :
DESIGN PRESSURE :
DESIGN TEMP :
MOC :
TRIM : VT-1100-086-D021-HX"
INSULATION :

GENERAL NOTES
A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.



CHECK PRINT

NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT, EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER. CONTRACTOR, SUB-CONTRACTORS, A SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE, AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT, SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	DLH	PDN						

MOTIVA

CLIENT: MOTIVA ENTERPRISES LLC
PLANT: PORT ARTHUR REFINERY
JOB NO:

PIPING & INSTRUMENTATION DIAGRAM

STEAM UNIT
BOILER A
FORCED DRAFT FAN

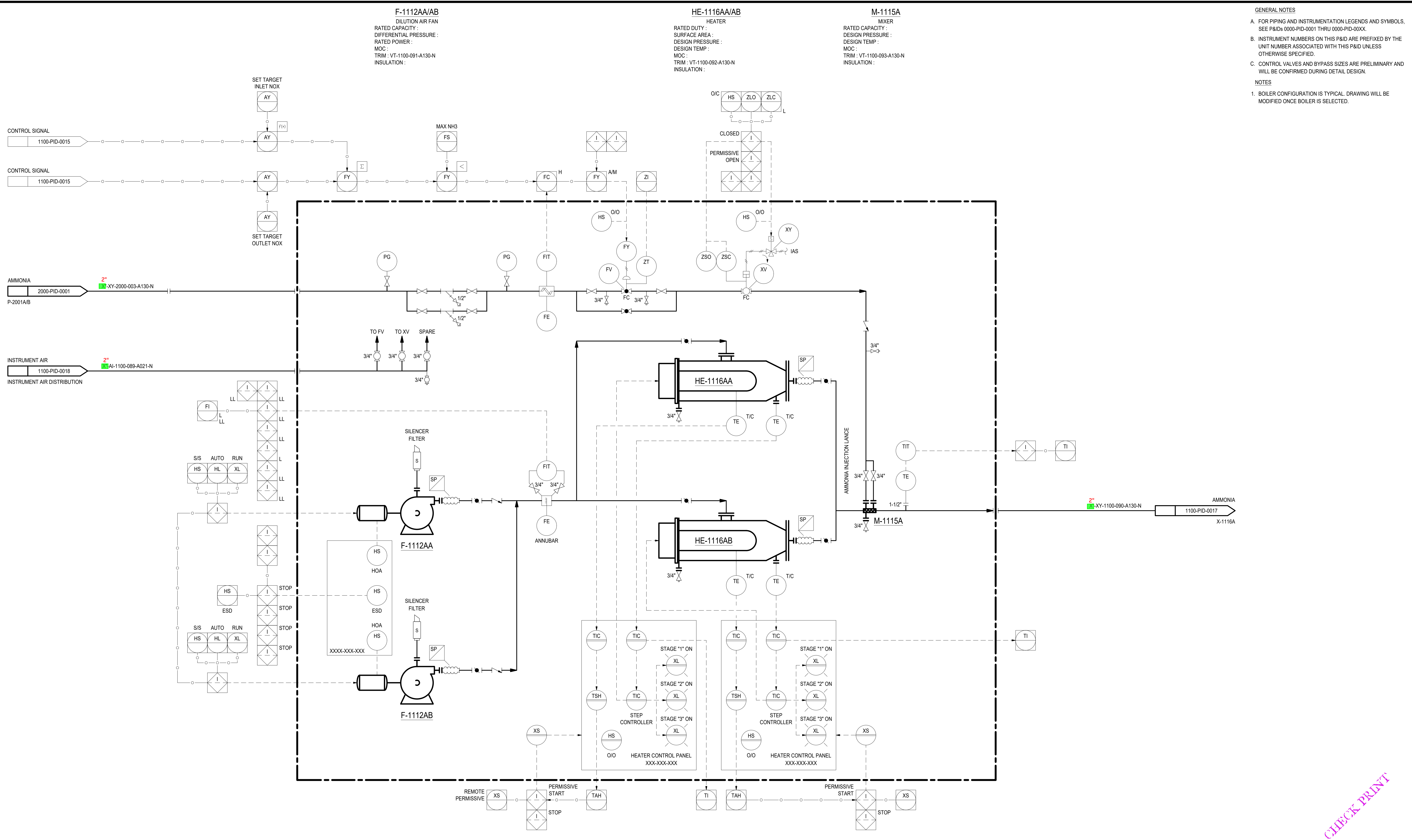
PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

SSP ENGINEERING & CONSULTING
13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

PROJECT NO: 25MA02
SCALE: NONE
DRAWN BY: GS
CHECKED: APB
DESIGN BY:
ENG APVL: PDN
DRAWING NO:

FIRM NUMBER
#13619

1100-PID-0014
REV A



- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.
- NOTES
1. BOILER CONFIGURATION IS TYPICAL. DRAWING WILL BE MODIFIED ONCE BOILER IS SELECTED.

NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT, EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE, AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT, SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:			
1100-PID-0016	1100-PID-0017	1100-PID-0018	1100-PID-0019

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	DLH	PDN						

PIPING & INSTRUMENTATION DIAGRAM

STEAM UNIT
BOILER A
AFCU

CLIENT: MOTIVA ENTERPRISES LLC
PLANT: PORT ARTHUR REFINERY
JOB NO:

SSP ENGINEERING & CONSULTING
13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

FIRM NUMBER
#13619

PROJECT NO: 25MA02
DRAWN BY: GS
DESIGN BY:
DRAWING NO:

SCALE: NONE
CHECKED: APB
ENG APVL: PDN

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

1100-PID-0016

REV
A

DRAWING LOCATION: c:\users\jason\onepointive - eng\ssps files - 25ma02 - aromatics utility scope cold eyes review\04 - design\04 - process\pids - 1100-pid-0016.dwg, PLOT DATE: 12-05-25, PLOT BY: jason

CHECK PRINT

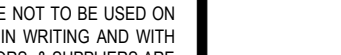
GENERAL NOTES

A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.

B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.

C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.

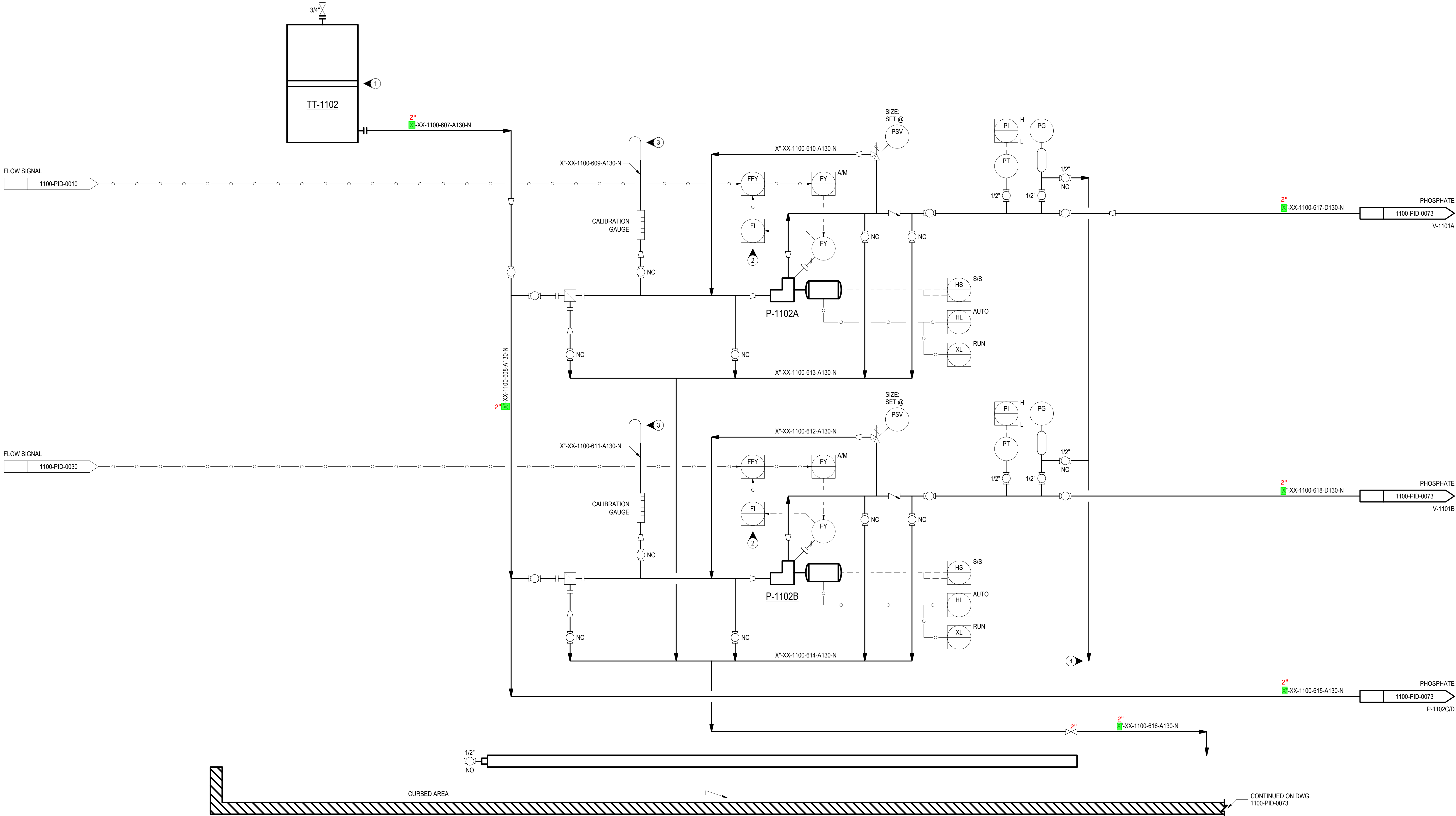


ALL PUBLISHED, ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT, EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER, CONTRACTOR, SUB-CONTRACTORS, & SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE, AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT, SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNOLOGIES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.										REFERENCE DRAWINGS:										<table><tr><th>REV #</th><th>DATE</th><th>DESCRIPTION</th><th>DRWN</th><th>CHKD</th><th>APVD</th><th>REV #</th><th>DATE</th><th>DESCRIPTION</th><th>DRWN</th><th>CHKD</th><th>APVD</th></tr><tr><td>A</td><td>12-05-25</td><td>ISSUED FOR APPROVAL</td><td>GS</td><td>DLH</td><td>PDN</td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>										REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	A	12-05-25	ISSUED FOR APPROVAL	GS	DLH	PDN							 STEAM UNIT BOILER A SCR										PIPING & INSTRUMENTATION DIAGRAM										 SSP ENGINEERING & CONSULTING 13100 NORTHWEST FREEWAY SUITE 625, HOUSTON, TEXAS 77040										FIRM NUMBER #13619									
REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD																																																																																		
A	12-05-25	ISSUED FOR APPROVAL	GS	DLH	PDN																																																																																								
CLIENT: MOTIVA ENTERPRISES LLC										PROJECT NO: 25MA02										SCALE: NONE																																																																									
DRAWN BY: GS										CHECKED: APB										DESIGN BY:										ENG APVL: PDN																																																															
PLANT: PORT ARTHUR REFINERY										PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW										DRAWING NO:										1100-PID-0017																																																															
JOB NO:																														REV A																																																															

TT-1102
PHOSPHATE STACKED TOTE
SIZE ID x HEIGHT :
TOTAL VOLUME :
DESIGN PRESSURE :
DESIGN TEMP :
MOC :
TRIM : VT-1100-631-A130-N
INSULATION / TRACING :

P-1102A/B
PHOSPHATE PUMPS
RATED CAPACITY :
DIFFERENTIAL PRESSURE :
RATED POWER :
MOC (CASING / INTERNAL) :
INSULATION / TRACING :
TRIM : VT-1100-632-D130-N
SEAL PLAN :

- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.
- NOTES
1. STACKED TOTE.
2. STROKE FEEDBACK.
3. TOP OF VENT IS LOCATED HIGHER THAN TOP OF TANK.
4. OPEN END DRAIN TO CURB AREA.
5. ALL MOTOR DRIVEN EQUIPMENT PROVIDED WITH A 3 POSITION START / RUN / STOP SWITCH. SPRING RETURN FROM START TO RUN, MAINTAINED STOP.
6. ALL CHEMICAL INJECTIONS WILL BE IN COMMON CURBED AREA WITH ONE SUMP AND ONE 2" STAND PIPE.
7. THE SELECTION SWITCH IS USED TO SELECT THE SPARE INJECTION DEVICE TO REPLACE ANY ONE OF THE THREE INJECTION DEVICES WHICH IS OFFLINE FOR MAINTENANCE.



CHECK PRINT

NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT. EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER, CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE, AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT, SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

MOTIVA		
CLIENT: MOTIVA ENTERPRISES LLC		
PLANT: PORT ARTHUR REFINERY		
JOB NO:		

PIPING & INSTRUMENTATION DIAGRAM	
STEAM UNIT	
PHOSPHATE PACKAGE	
PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW	

SSP ENGINEERING & CONSULTING		FIRM NUMBER #13619
13100 NORTHWEST FREEWAY SUITE 625 HOUSTON, TEXAS 77040		
PROJECT NO: 25MA02	SCALE: NONE	
DRAWN BY: GS	CHECKED: APB	
DESIGN BY:	ENG APVL: PDN	
DRAWING NO:	1100-PID-0071	REV A

P-1102C/D
PHOSPHATE PUMPS
 RATED CAPACITY :
 DIFFERENTIAL PRESSURE :
 RATED POWER :
 MOC (CASING / INTERNAL) :
 INSULATION / TRACING :
 TRIM : VT-1100-632-D130-N
 SEAL PLAN :

GENERAL NOTES

A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS
SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.

8. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.

C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.

NOTES

1. TOP OF VENT IS LOCATED HIGHER THAN TOP OF TANK.

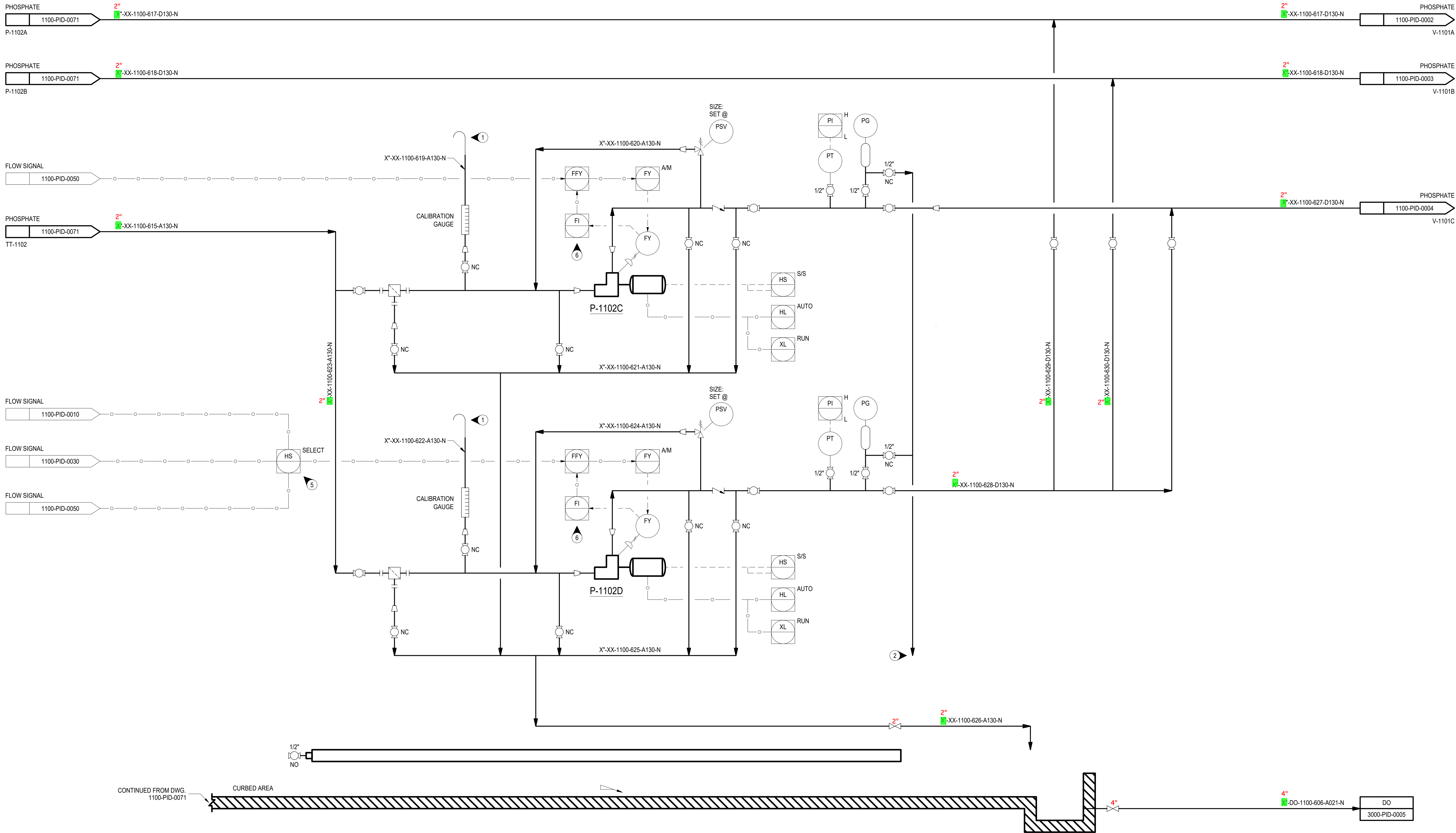
2. OPEN END DRAIN TO CURB AREA.

3. ALL MOTOR DRIVEN EQUIPMENT PROVIDED WITH A 3 POSITION START / RUN / STOP SWITCH. SPRING RETURN FROM START TO RUN, MAINTAINED STOP.

4. ALL CHEMICAL INJECTIONS WILL BE IN COMMON CURBED AREA WITH ONE SUMP AND ONE 2" STAND PIPE.

5. THE SELECTION SWITCH IS USED TO SELECT THE SPARE INJECTION DEVICE TO REPLACE ANY ONE OF THE THREE INJECTION DEVICES WHICH IS OFFLINE FOR MAINTENANCE.

3. STROKE FEEDBACK



CHECK PRINT

NOT PUBLISHED. ALL RISKS RESERVED BY THE ENGINEER, DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT, EXCEPT BY AGREEMENT IN WRITING AND WITHOUT THE ENGINEER'S CONSENT. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR THE USER'S RESPONSIBILITY FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE, AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT, SHALL NOT INDICATE ANY DETAIL OR SYSTEM FOR CONSTRUCTION. THE USER SHALL BE RESPONSIBLE FOR OBTAINING ALL NECESSARY PERMITS AND PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND CO-ORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR OBTAINING ALL NECESSARY PERMITS AND FOR THE PROPER INSTALLATION OF MATERIALS. SHALL BE TRUE, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TO NEW LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, OR SEQUENCES. THE USER SHALL BE RESPONSIBLE FOR THE PROPER INSTALLATION OF MATERIALS. WORK SHALL BE DONE IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:

[illegible]

MOTIVA

CLIENT: MOTIVA ENTERPRISES LLC

PLANT: PORT ARTHUR REFINERY

JOB NO

PIPING & INSTRUMENTATION DIAGRAM

STEAM UNIT

PHOSPHATE PACKAGE

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW



**SSP ENGINEERING
& CONSULTING**
13100 NORTHWEST FREEWAY
SUITE 625,
HOUSTON, TEXAS 77040

PROJECT NO: 25MA0:

DRAWN BY: GS

DESIGN BY

SCALE: NONE

CHECKED: APB

ENG APVI : PDN

1100-PID-0073

RE
A

V-1101
CONTINUOUS BLOWDOWN DRUM
SIZE :
TOTAL VOLUME :
DESIGN PRESSURE :
DESIGN TEMP :
MOC :
TRIM : VT-1100-639-D021-PX"
INSULATION / TRACING :

GENERAL NOTES

A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS
SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.

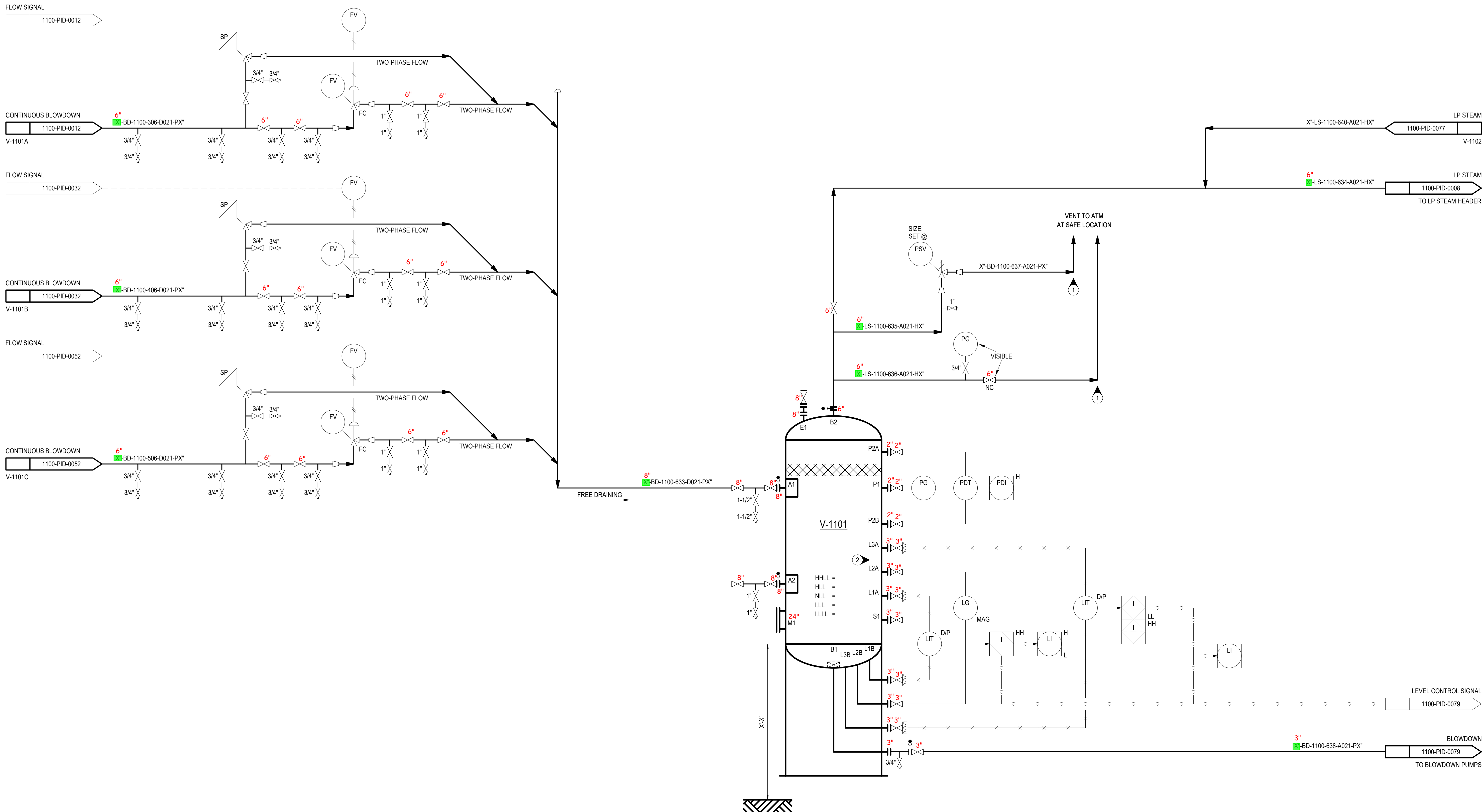
8. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.

C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.

NOTES

1. 3/8" WEEP HOLE AT LOW POINT.

2. ALL UPPER L1A, L2A AND L3A NOZZLES ARE AT SAME ELEVATION. ALL LOWER L1B, L2B AND L3B NOZZLES ARE AT SAME ELEVATION.



NOT PUBLISHED, ALL ITS RESERVED BY THE ENGINEER, DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE USED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OR FOR ANY OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT, EXCEPT BY AGREEMENT IN WRITING AND WITH THE WRITTEN CONSENT OF THE ENGINEER. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR THE USER'S RESPONSIBILITY FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE, AND SHALL NOT NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DIMENSIONS. DESCRIBING DESIGN INTENT, SHALL NOT INDICATE ANY DETAIL OR SYSTEM FOR CONSTRUCTION. THE USER SHALL BE RESPONSIBLE FOR OBTAINING ALL NECESSARY PERMITS, AND PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND MAINTENANCE. THE USER SHALL BE RESPONSIBLE FOR OBTAINING ALL NECESSARY PERMITS, AND CO-OPERATION OF ALL WORK CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE THE USER'S RESPONSIBILITY. THE USER SHALL BE RESPONSIBLE FOR OBTAINING ALL NECESSARY PERMITS, AND MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE USER SHALL BE RESPONSIBLE FOR CONSTRUCTION METHODS AND MODES, TECHNIQUES, AND PROCEDURES. THE USER SHALL BE RESPONSIBLE FOR OBTAINING ALL NECESSARY PERMITS, AND WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS

[illegible]

	
CLIENT: MOTIVA ENTERPRISES LLC	
PLANT: PORT ARTHUR REFINERY	
JOB NO:	

PIPING & INSTRUMENTATION DIAGRAM

STEAM UNIT
CONTINUOUS BLOWDOWN DRUM

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

	SSP ENGINEERING & CONSULTING 13100 NORTHWEST FREEWAY SUITE 625, HOUSTON, TEXAS 77040		FIRM NUMBER #13619
	PROJECT NO: 25MA02		SCALE: NONE
	DRAWN BY: GS		CHECKED: APB
	DESIGN BY:		ENG APVL: PDN
DRAWING NO:			1100-PID-0075 REV A

③ **V-1102**
 INTERMITTENT BLOWDOWN DRUM
 SIZE :
 TOTAL VOLUME :
 DESIGN PRESSURE :
 DESIGN TEMP :
 MOC :
 TRIM :
 INSULATION / TRACING :

GENERAL NOTES

A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.

8. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.

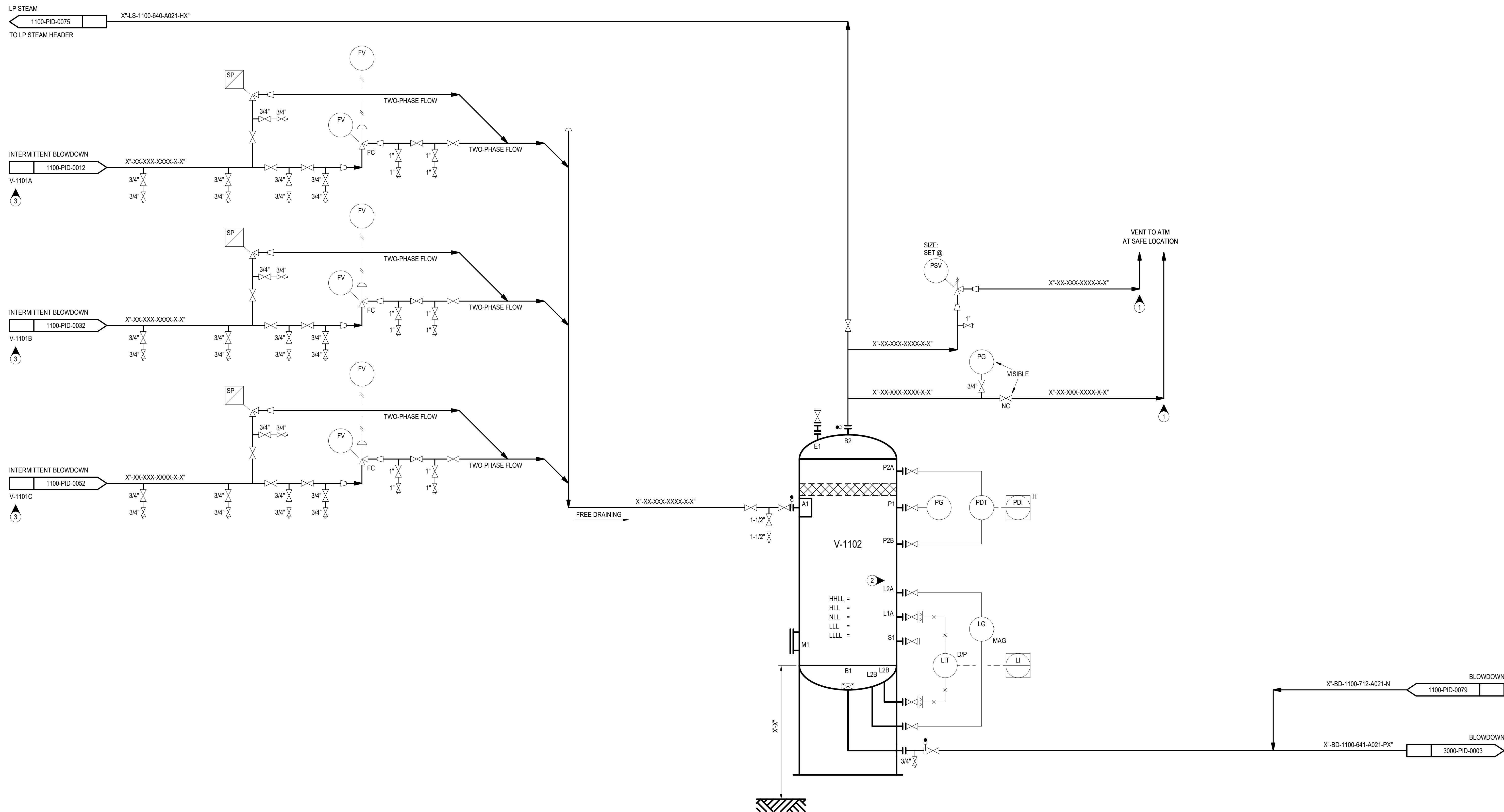
C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.

NOTES

1. 3/8" WEEP HOLE AT LOW POINT.

2. UPPER L1A AND L2A NOZZLES ARE AT SAME ELEVATION.
LOWER L1B AND L2B NOZZLES ARE AT SAME ELEVATION.

3. NEED FOR INTERMITTENT BLOWDOWN WILL BE DETERMINED WHEN BOILER IS SELECTED.

[illegible]

GENERAL NOTES

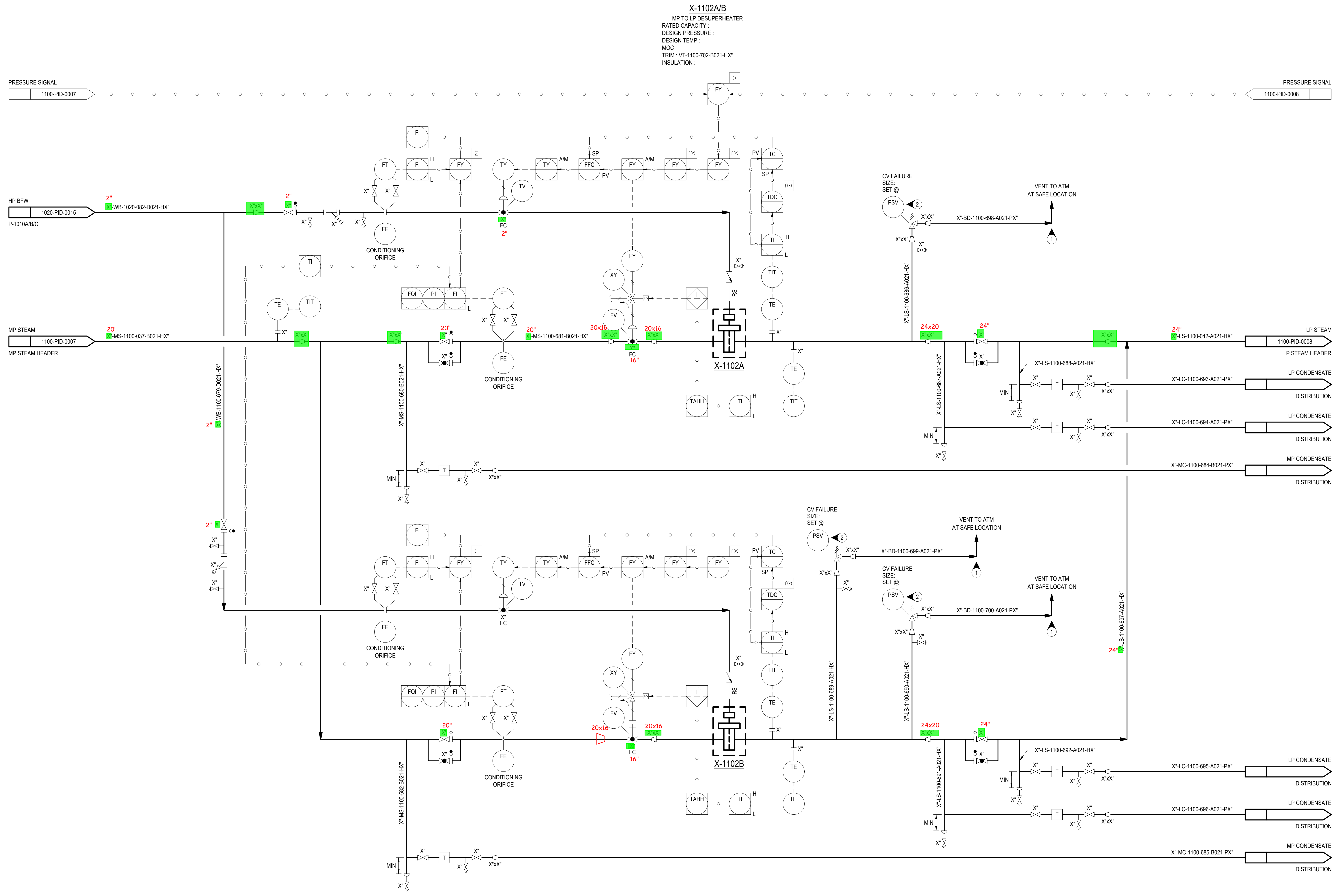
A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.

B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.

C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.



NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT, EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER, CONTRACTOR, SUB-CONTRACTORS, & SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.										REFERENCE DRAWINGS:										REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	 STEAM UNIT BLOWDOWN PUMPS AND COOLER	PIPING & INSTRUMENTATION DIAGRAM	 SSP ENGINEERING & CONSULTING 13100 NORTHWEST FREEWAY SUITE 625 HOUSTON, TEXAS 77040	FIRM NUMBER #13619																																																																																																																																																																																																																																																																																								
										A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN																																																																																																																																																																																																																																																																																																												



NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT, EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER, CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:			
REV #	DATE	DESCRIPTION	
A	12-05-25	ISSUED FOR APPROVAL	

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD

CLIENT: MOTIVA ENTERPRISES LLC	
PLANT: PORT ARTHUR REFINERY	
JOB NO:	

PROJECT NO: 25MA02		SCALE: NONE	
DRAWN BY: GS		CHECKED: APB	
DESIGN BY:		ENG APVL: PDN	
DRAWING NO:		1100-PID-0087	

FIRM NUMBER
#13619

CHECK PRINT

GENERAL NOTES

1. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.

2. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.

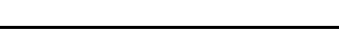
3. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.

NOTES

1. PRESSURE INDICATOR TO BE READABLE FROM BYPASS VALVE.

2. PROVIDE 3/8" WEEP HOLE AT LOW POINT.



NOT PUBLISHED, ALL RIGHTS RESERVED BY THE ENGINEER, DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT. EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER, CONTRACTOR, SUB-CONTRACTORS, & SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE, AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT, SHALL NOT INDICATE VERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, & SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, & SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADE. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.												REFERENCE DRAWINGS:											
REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	 PROCESS FLOW DIAGRAM PIPING & INSTRUMENTATION DIAGRAM CONDENSATE FLASH DRUM A											
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN							 SSP ENGINEERING & CONSULTING 13100 NORTHWEST FREEWAY SUITE 625 HOUSTON, TEXAS 77040											
												PROJECT NO: 25MA02 DRAWN BY: GS DESIGN BY: SCALE: NONE CHECKED: APB ENG APVL: PDN											
CLIENT: MOTIVA ENTERPRISES LLC												PLANT: PORT ARTHUR REFINERY JOB NO: PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW DRAWING NO:											
JOB NO:												1500-PID-0001											
												REV A											

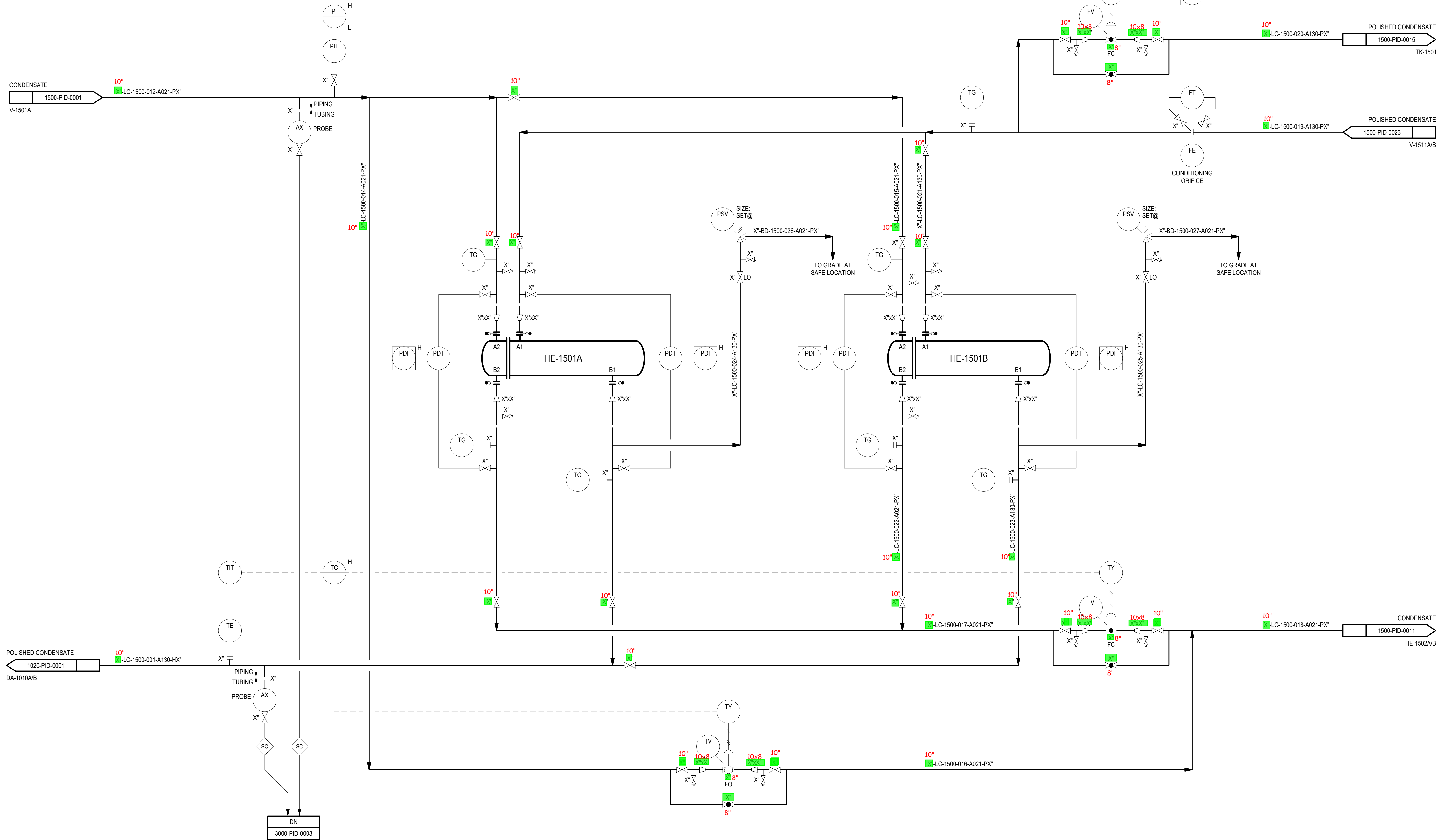
HE1501A/B

CONDENSATE EXCHANGER

TYPE :
RATED DUTY :
SURFACE AREA :
DESIGN PRESS @ TEMP (HOT SIDE) :
DESIGN PRESS @ TEMP (COLD SIDE) :
MOC :
TRIM (HOT / COLD) : VT-1500-028-A021-PX*
VT-1500-029-A130-PX*
INSULATION :

GENERAL NOTES

- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.



CHECK PRINT

NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT. EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER, CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

MOTIVA

CLIENT: MOTIVA ENTERPRISES LLC
PLANT: PORT ARTHUR REFINERY
JOB NO:

PIPING & INSTRUMENTATION DIAGRAM
CONDENSATE SYSTEM
CONDENSATE EXCHANGERS

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

SSP ENGINEERING & CONSULTING
13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

PROJECT NO: 25MA02
SCALE: NONE

DRAWN BY: GS
CHECKED: APB

DESIGN BY:
ENG APVL: PDN

DRAWING NO: 1500-PID-0007

FIRM
NUMBER
#13619

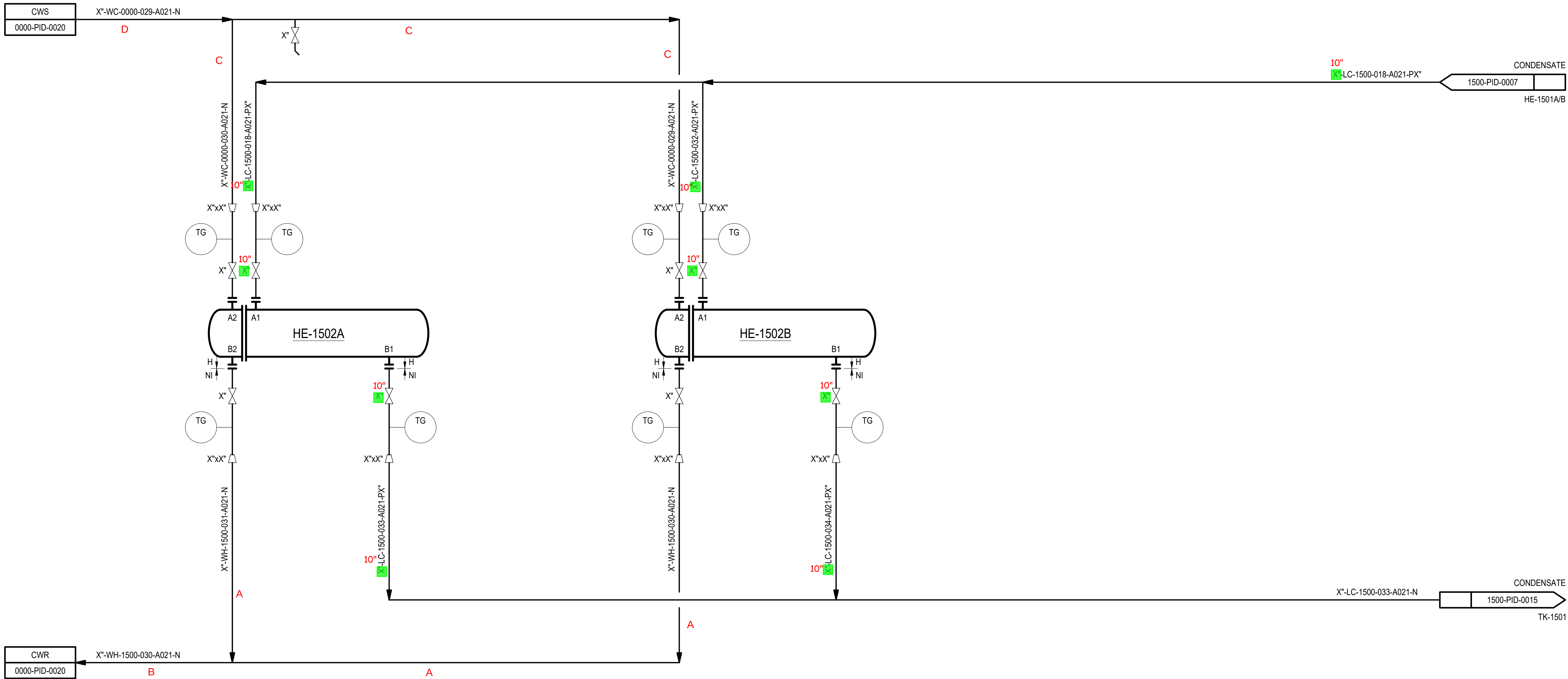
REV

A

HE-1502A/B
CONDENSATE TRIM COOLER

TYPE :
RATED DUTY :
SURFACE AREA :
SHELL DESIGN PRESS @ TEMP :
TUBE DESIGN PRESS @ TEMP :
MOC (SHELL / TUBES) :
TRIM (SHELL / TUBES) : VT-1500-035 A021-N
VT-1500-036 A021-N
INSULATION (SHELL / TUBES) :

- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.



CHECK PRINT

NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT. EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER, CONTRACTOR, SUB-CONTRACTORS, & SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW UNLESS SPECIFIED OTHERWISE AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

CLIENT: MOTIVA ENTERPRISES LLC
PLANT: PORT ARTHUR REFINERY
JOB NO:

PROCESS FLOW DIAGRAM

PIPING & INSTRUMENTATION DIAGRAM

CONDENSATE SYSTEM

CONDENSATE TRIM COOLERS

SSP ENGINEERING & CONSULTING
13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

FIRM NUMBER #13619

PROJECT NO: 25MA02
DRAWN BY: GS
DESIGN BY:

SCALE: NONE
CHECKED: APB
ENG APVL: PDN

DRAWING NO:

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

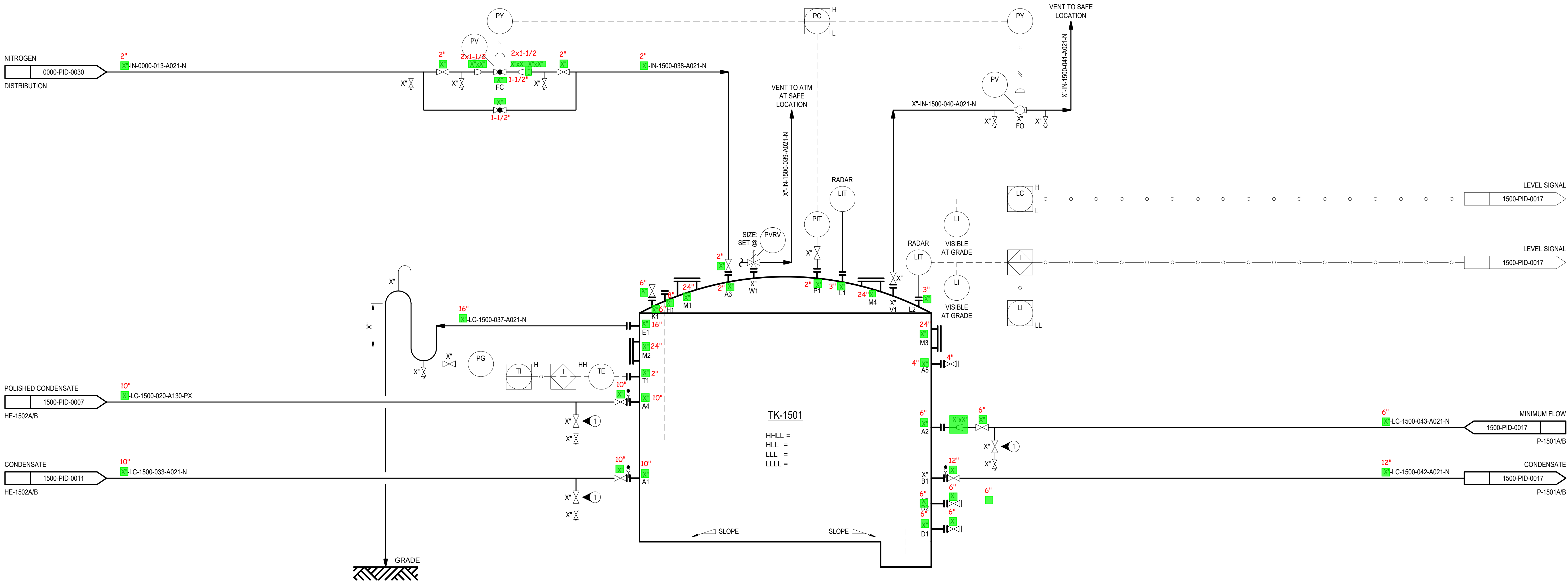
1500-PID-0011

REV **A**

TK-1501
CONDENSATE TANK
SIZE ID x HEIGHT :
TOTAL VOLUME :
DESIGN PRESSURE :
DESIGN TEMP :
MOC :
TRIM : VT-1500-044-A021-N
INSULATION / TRACING :

3. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.

1. CONNECTION TO BYPASS TANK DURING TANK INSPECTION AND MAINTENANCE.



CHECK PRINT

NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER, DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR FOR EXTENSIONS TO THIS PROJECT, EXCEPT BY AGREEMENT IN WRITING AND SIGNED BY THE ENGINEER. THE USER SHALL BE RESPONSIBLE FOR OBTAINING NECESSARY PERMITS, ERECTION INSTALLATION, AND COORDINATION OF ALL WORK CONTRACTOR, SUBCONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR VERIFYING THAT MATERIALS AND METHODS OF CONSTRUCTION MEET THE REQUIREMENTS OF THE DRAWINGS. MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE; AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, OR SEQUENCES. THE USER SHALL BE RESPONSIBLE FOR OBTAINING NECESSARY PERMITS, ERECTION INSTALLATION, AND COORDINATION OF ALL WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS

[illegible]**MOTIVA**

JOB NO.

PIPING & INSTRUMENTATION DIAGRAM

CONDENSATE SYSTEM
CONDENSATE TANK

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW



**SSP ENGINEERING
& CONSULTING**
13100 NORTHWEST FREEWAY
SUITE 625,
HOUSTON, TEXAS 77040

PROJECT NO: 25MA02

DRAWN BY: GS

DESIGNED BY

ENGINEERING CONSULTING 1000 WEST FREEWAY SUITE 625, HOUSTON, TEXAS 77057-1361 TEL: 713/865-1361 FAX: 713/865-1362 WWW: WWW.BURNSMCDERMOTT.COM	FIRM NUMBER #13619
--	---

SCALE: NONE

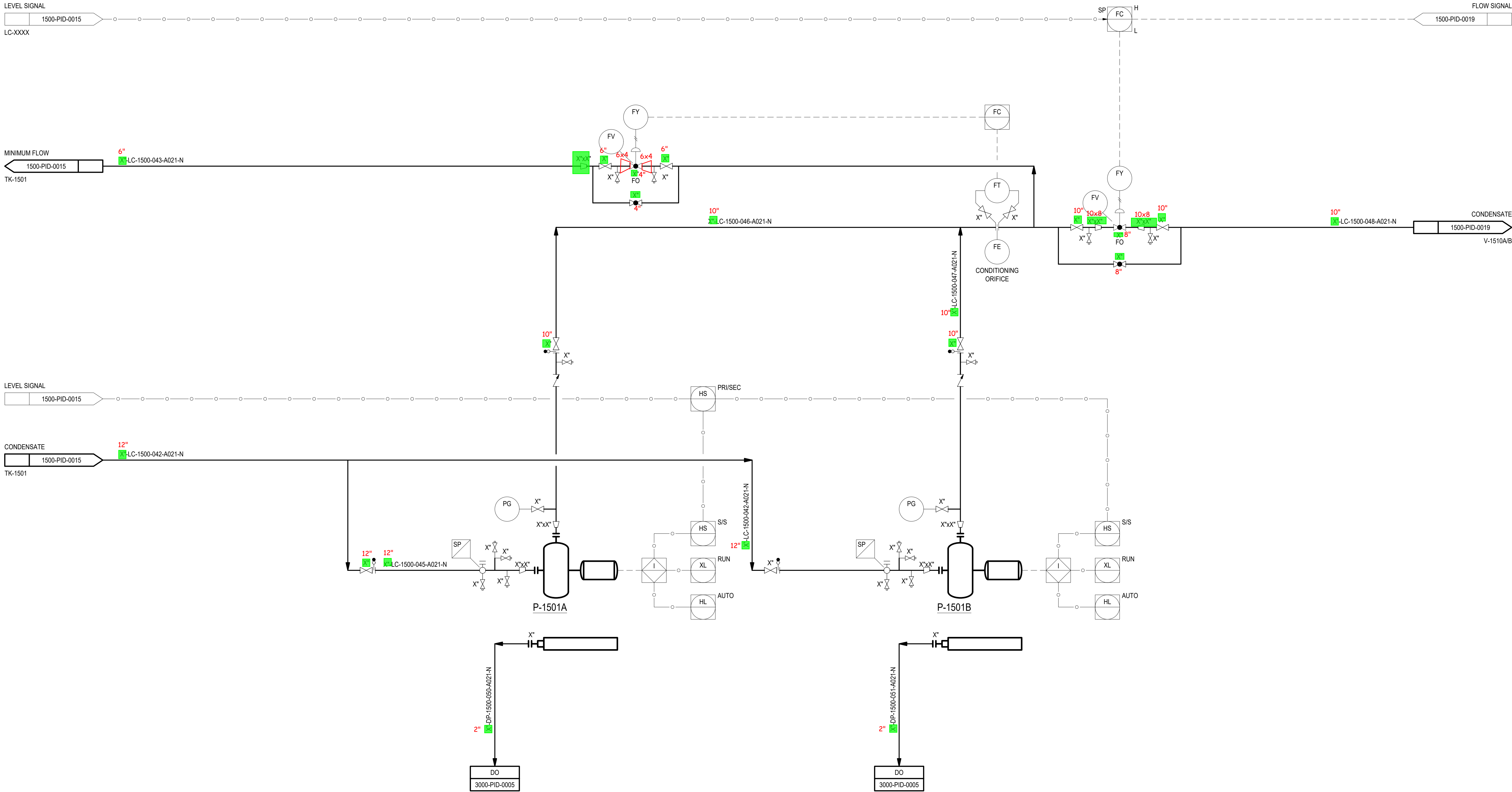
CHECKED. AFD
FILED. APR 10 1968

DRAWING NO:	REF:
-------------	------

1500-PID-0015 | A

P-1501A/B
CONDENSATE PUMPS
RATED CAPACITY :
DIFFERENTIAL PRESSURE :
RATED POWER :
MOC (CASING / INTERNAL) :
INSULATION / TRACING :
TRIM : VT-1500-048-A021-N
SEAL PLAN :

- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.



CHECK PRINT

NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT. EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER, CONTRACTOR, SUB-CONTRACTORS, & SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

MOTIVA

CLIENT: MOTIVA ENTERPRISES LLC

PLANT: PORT ARTHUR REFINERY

JOB NO:

PIPING & INSTRUMENTATION DIAGRAM

CONDENSATE SYSTEM

CONDENSATE PUMPS

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

SSP ENGINEERING & CONSULTING

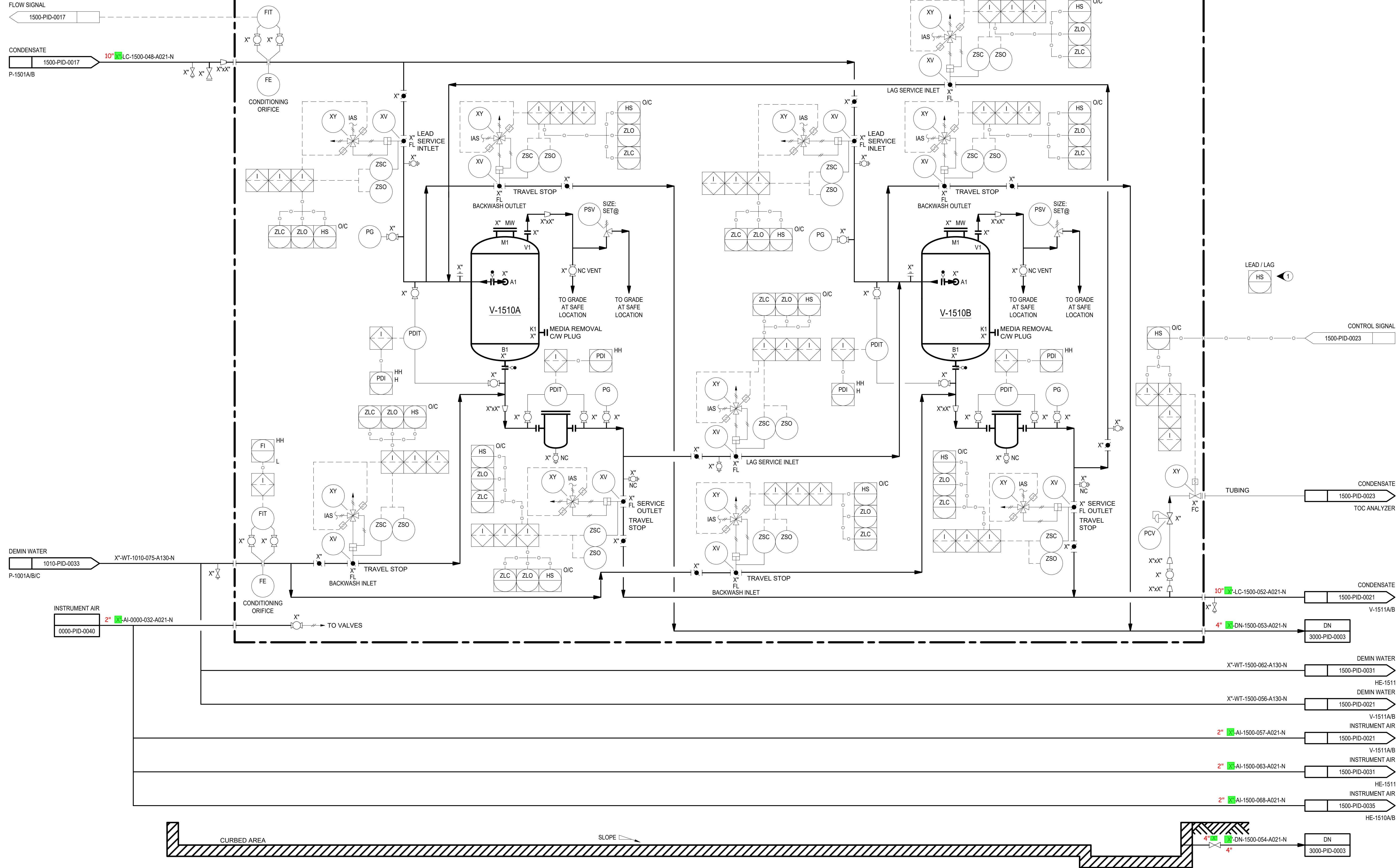
13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

FIRM NUMBER #13619

PROJECT NO: 25MA02
SCALE: NONE
DRAWN BY: GS
CHECKED: APB
DESIGN BY: ENG APVL: PDN
DRAWING NO: **1500-PID-0017**

ACTIVATED CARBON FILTERS

TRIM : VT-1500-055-A021-N
INSULATION / TRACING :



A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS,

3. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.

1. SELECTOR SWITCH FOR SELECTING BETWEEN ACF V-1510A

2. CARBON FILTERS SHOWN ARE TYPICAL. CONFIGURATION WILL BE REVISED ONCE SYSTEM IS SELECTED.

NOT PUBLISHED: ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS AS INSTRUMENTS OF SERVICE ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT. EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER, CONTRACTOR, SUB-CONTRACTORS, & SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE, AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBED DESIGN INTENT SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMIT, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.										REFERENCE DRAWINGS:			REV #			DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #			DATE	DESCRIPTION	DRWN	CHKD	APVD	
MOTIVA CONDENSATE POLISHING PACKAGE ACTIVATED CARBON FILTERS										A			12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN												
										SSP SSP ENGINEERING & CONSULTING 13100 NORTHWEST FREEWAY SUITE 625, HOUSTON, TEXAS 77040										PROJECT NO: 25MA02			SCALE: NONE						
										CLIENT: MOTIVA ENTERPRISES LLC										DRAWN BY: GS			CHECKED: APB						
										PLANT: PORT ARTHUR REFINERY										DESIGN BY:			ENG APVL: PDN						
JOB NO:										PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW			DRAWING NO:			1500-PID-0019				REV A									

V-1511A
CONDENSATE POLISHER

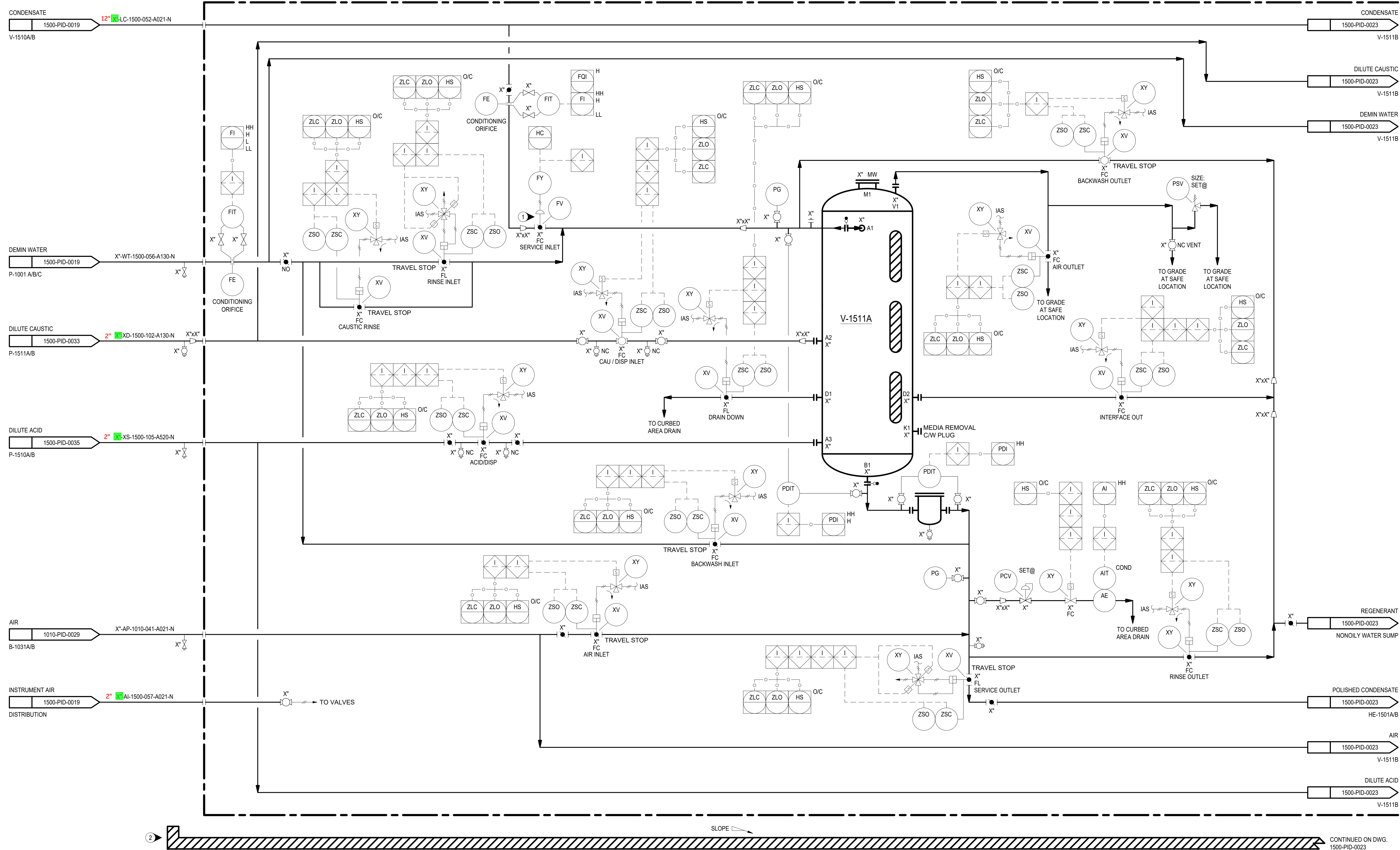
SIZE :
TOTAL VOLUME :
DESIGN PRESSURE :
DESIGN TEMP :
MOC :
TRIM : VT-1500-058-A130-N
INSULATION / TRACING :

GENERAL NOTES

- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.

NOTES

1. FV-XXXX APPLY AS AN ON/OFF VALVE AND HC-XXXX IS MANUAL SET TO FULLY OPEN OR CLOSE THE VALVE.
2. COMMON CURBED AREA FOR MIXED BED POLISHER TRAIN 1 AND 2. ACID AND CAUSTIC RESISTANT LINING.
3. CONDENSATE POLISHER SHOWN IS TYPICAL CONFIGURATION WILL BE REVISED ONCE SYSTEM IS SELECTED.



CONTINUED ON DWG.
1500-PID-0023

[illegible]

REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN

MOTIVA

CLIENT: MOTIVA ENTERPRISES LLC
PLANT: PORT ARTHUR REFINERY
JOB NO:

PIPING & INSTRUMENTATION DIAGRAM

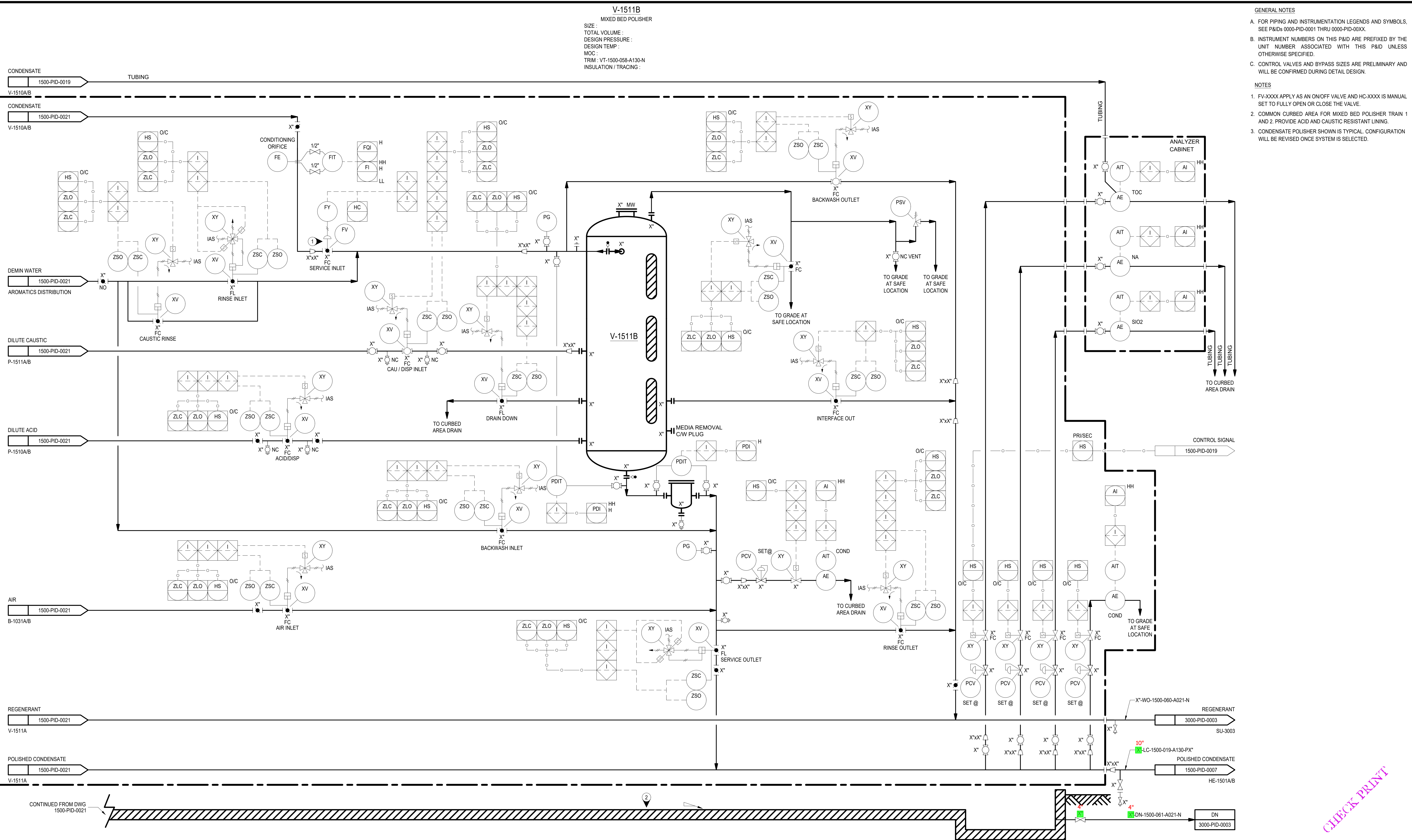
CONDENSATE POLISHING PACKAGE

CONDENSATE POLISHER A

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

	SSP ENGINEERING & CONSULTING 13100 NORTHWEST FREEWAY SUITE 625 HOUSTON, TEXAS 77040		FIRM NUMBER #13619
	PROJECT NO: 25MA02		SCALE: NONE
	DRAWN BY: GS		CHECKED: APB
	DESIGN BY:		ENG APVL: PDN
DRAWING NO:			1500-PID-0021 REV A

CHECK PRINT



- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.
- NOTES
1. FV-XXXX APPLY AS AN ON/OFF VALVE AND HC-XXXX IS MANUAL SET TO FULLY OPEN OR CLOSE THE VALVE.
2. COMMON CURBED AREA FOR MIXED BED POLISHER TRAIN 1 AND 2. PROVIDE ACID AND CAUSTIC RESISTANT LINING.
3. CONDENSATE POLISHER SHOWN IS TYPICAL. CONFIGURATION WILL BE REVISED ONCE SYSTEM IS SELECTED.

NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT, EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER. CONTRACTOR, SUB-CONTRACTORS, & SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE, AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

MOTIVA

CLIENT: MOTIVA ENTERPRISES LLC
PLANT: PORT ARTHUR REFINERY
JOB NO:

PIPING & INSTRUMENTATION DIAGRAM

CONDENSATE POLISHING PACKAGE
MIXED BED POLISHER B

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

SSP ENGINEERING & CONSULTING

13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

PROJECT NO: 25MA02
DRAWN BY: GS
DESIGN BY:
DRAWING NO:

SCALE: NONE
CHECKED: APB
ENG APVL: PDN

FIRM NUMBER
#13619

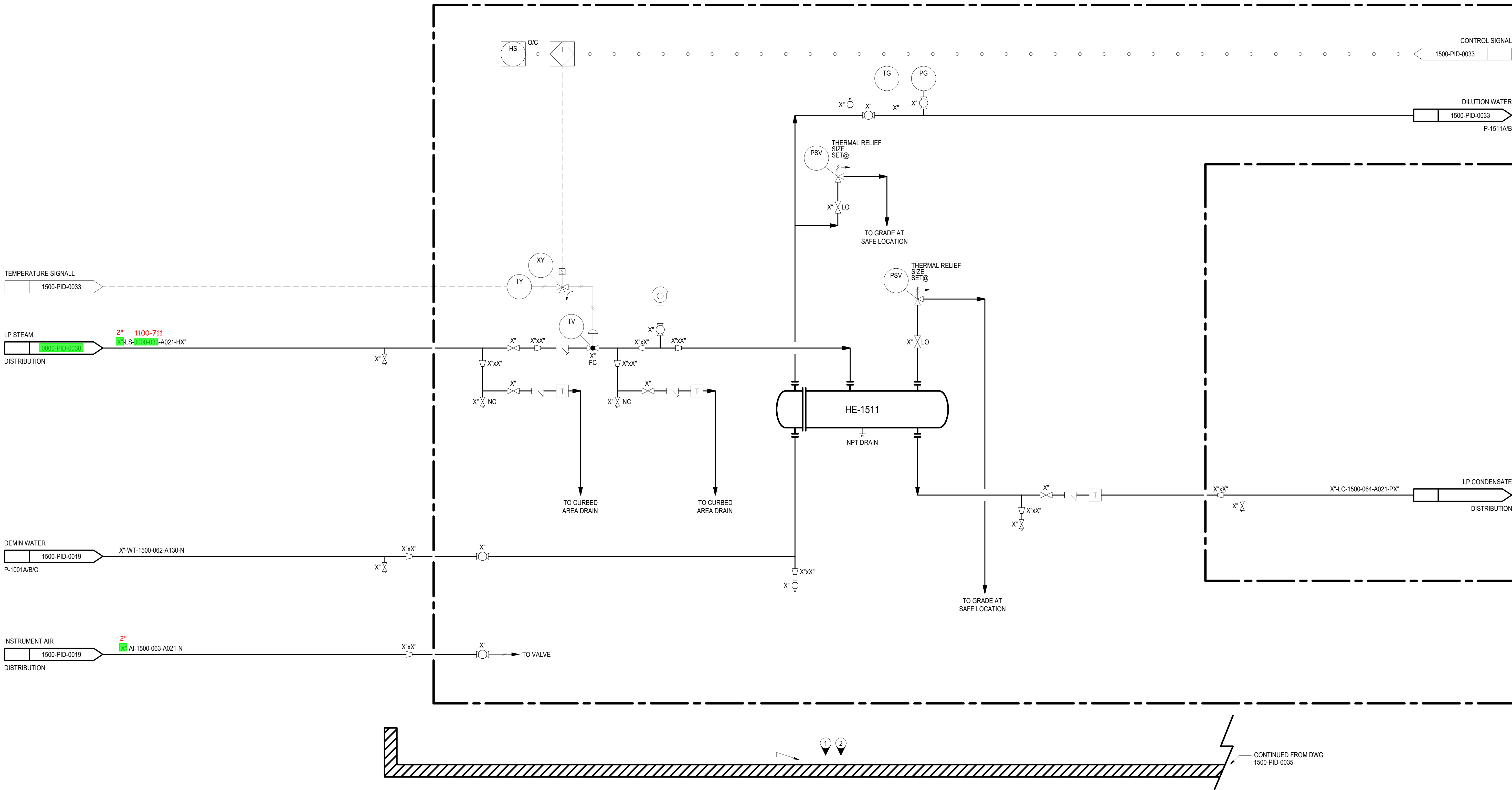
1500-PID-0023

REV
A

HE-1511
CONDENSATE CAUSTIC DILUTION HEATER
TYPE:
RATED DUTY :
SURFACE AREA :
SHELL DESIGN PRESS @ TEMP :
TUBE DESIGN PRESS @ TEMP :
MOC (SHELL / TUBES) :
TRIM (SHELL / TUBES) : VT-1500-065 A021-HX*
VT-1500-066 A130-PX*
INSULATION (SHELL / TUBES) :

- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.

- NOTES
1. COMMON CURBED AREA FOR DEMIN CAUSTIC DILUTION SKID 1 (DRAWING XXXX-PID-XXXX), AND CONDENSATE POLISHING PACKAGE CAUSTIC DILUTION SKID.
2. CAUSTIC RESISTANT LINING.



CHECK PRINT

NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT, EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE, AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT, SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

MOTIVA		
CLIENT: MOTIVA ENTERPRISES LLC		
PLANT: PORT ARTHUR REFINERY		
JOB NO:		

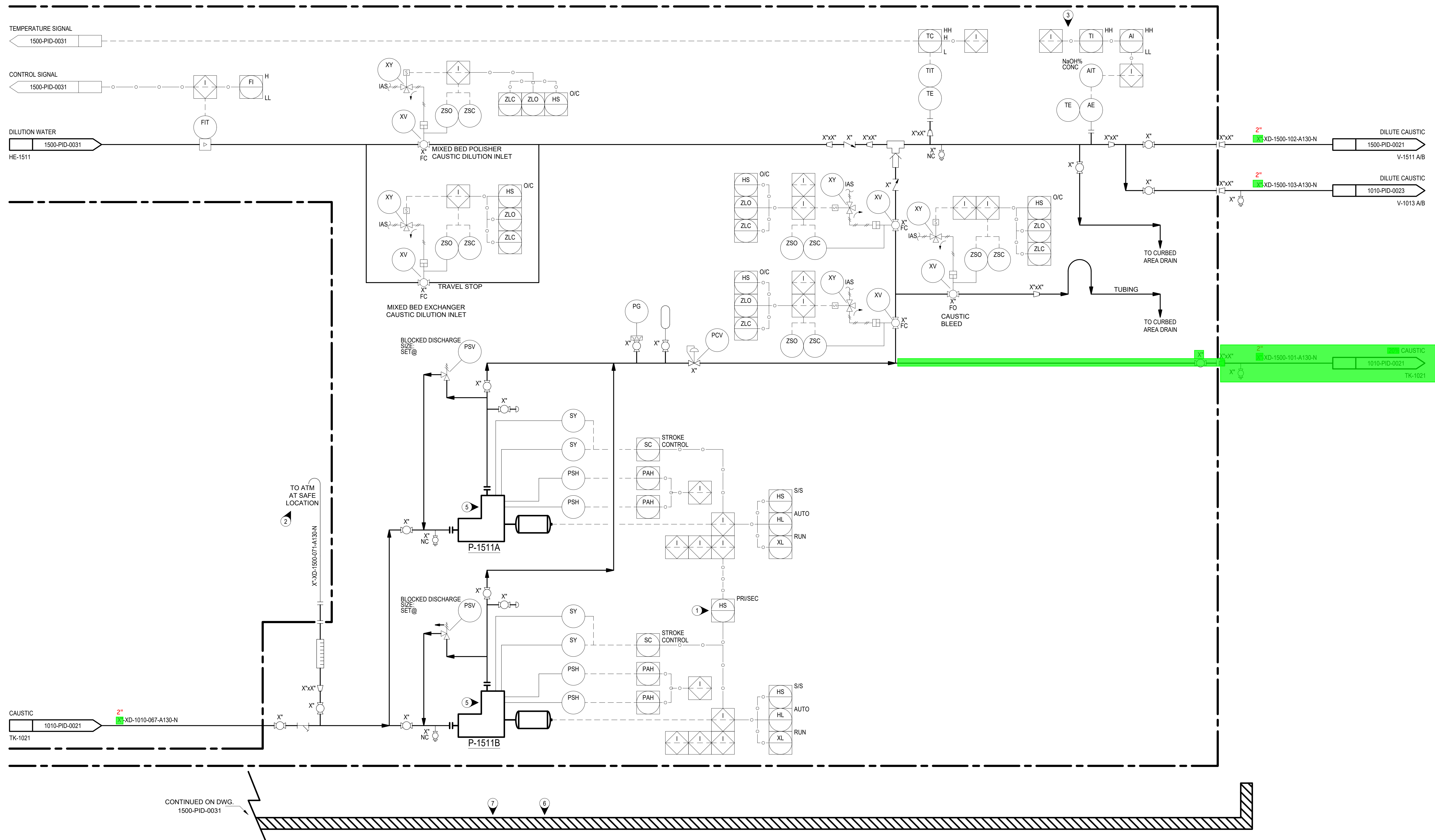
PROCESS FLOW DIAGRAM	
PIPING & INSTRUMENTATION DIAGRAM	
CONDENSATE POLISHING PACKAGE	
CONDENSATE CAUSTIC DILUTION HEATER	
PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW	

SSP ENGINEERING & CONSULTING		FIRM NUMBER #13619	
13100 NORTHWEST FREEWAY SUITE 625 HOUSTON, TEXAS 77040			
PROJECT NO: 25MA02		SCALE: NONE	
DRAWN BY: GS		CHECKED: APB	
DESIGN BY:		ENG APVL: PDN	
DRAWING NO:		1500-PID-0031	

REV

A

DRAWING LOCATION: c:\users\jason\one-drive - eng\ssps files - 25ma02 - aromatics utility scope old eyes\revin\04 - design\04 - process\04s - (Border Rev 1 5 18 25). PLOT DATE: ---. PLOT BY: jason. DRAWING: 1500-pid-0033.dwg.



- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.
- NOTES
1. DOSING PUMPS ALTERNATE DUTY AFTER EACH USE.
2. CALIBRATION COLUMN VENT IS EXTENDED TO A HEIGHT ABOVE THE MAXIMUM TANK ELEVATION.
3. INTEGRAL TEMPERATURE ANALOG SIGNAL IN ANALYZER.
4. ALL MOTOR DRIVEN EQUIPMENT PROVIDED WITH A 3 POSITION START/ RUN/ STOP SWITCH. SPRING RETURN FROM START TO RUN, MAINTAIN STOP.
5. CHECK VALVES INTERNAL TO THE PUMP.
6. COMMON CURBED AREA FOR DEMIN CAUSTIC DILUTION SYSTEM AND CONDENSATE POLISHING PACKAGE CAUSTIC DOSING PUMPS AND CONDENSATE CAUSTIC DILUTION HEATER.
7. CAUSTIC RESISTANT LINING.

NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT, EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER. CONTRACTOR, SUB-CONTRACTORS & SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

MOTIVA

CLIENT: MOTIVA ENTERPRISES LLC

PLANT: PORT ARTHUR REFINERY

JOB NO:

PIPING & INSTRUMENTATION DIAGRAM

CONDENSATE POLISHING PACKAGE

CAUSTIC DOSING PUMPS

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

SSP ENGINEERING & CONSULTING

13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

PROJECT NO: 25MA02

DRAWN BY: GS

DESIGN BY:

DRAWING NO:

FIRM NUMBER #13619

SCALE: NONE

CHECKED: APB

ENG APVL: PDN

REV

1500-PID-0033

A

CHECK PRINT

SULFURIC ACID DOSING PUMPS
 RATED CAPACITY :
 DIFFERENTIAL PRESSURE :
 RATED POWER :
 MOC (CASING / INTERNAL) :
 INSULATION / TRACING :
 TRIM : VT-1500-069-A520-N
 SEAL PLAN :

- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.

1. DOSING PUMPS ALTERNATE DUTY AFTER EACH USE.
2. CHECK VALVES INTERNAL TO THE PUMP.
3. ALL MOTOR DRIVEN EQUIPMENT PROVIDED WITH A 3 POSITION START/ RUN/ STOP SWITCH. SPRING RETURN FROM START TO RUN, MAINTAIN STOP.
4. COMMON CURBED AREA FOR DEMIN ACID DILUTION SKID AND CONDENSATE POLISHING PACKAGE SULFURIC ACID DOSING PUMPS.
5. CALIBRATION COLUMN VENT IS EXTENDED TO A HEIGHT ABOVE THE MAXIMUM TANK ELEVATION.
6. ACID RESISTANT LINING.

[illegible]

JOB NO.

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW



1500 BID 0005

REV

AMMONIA TANK

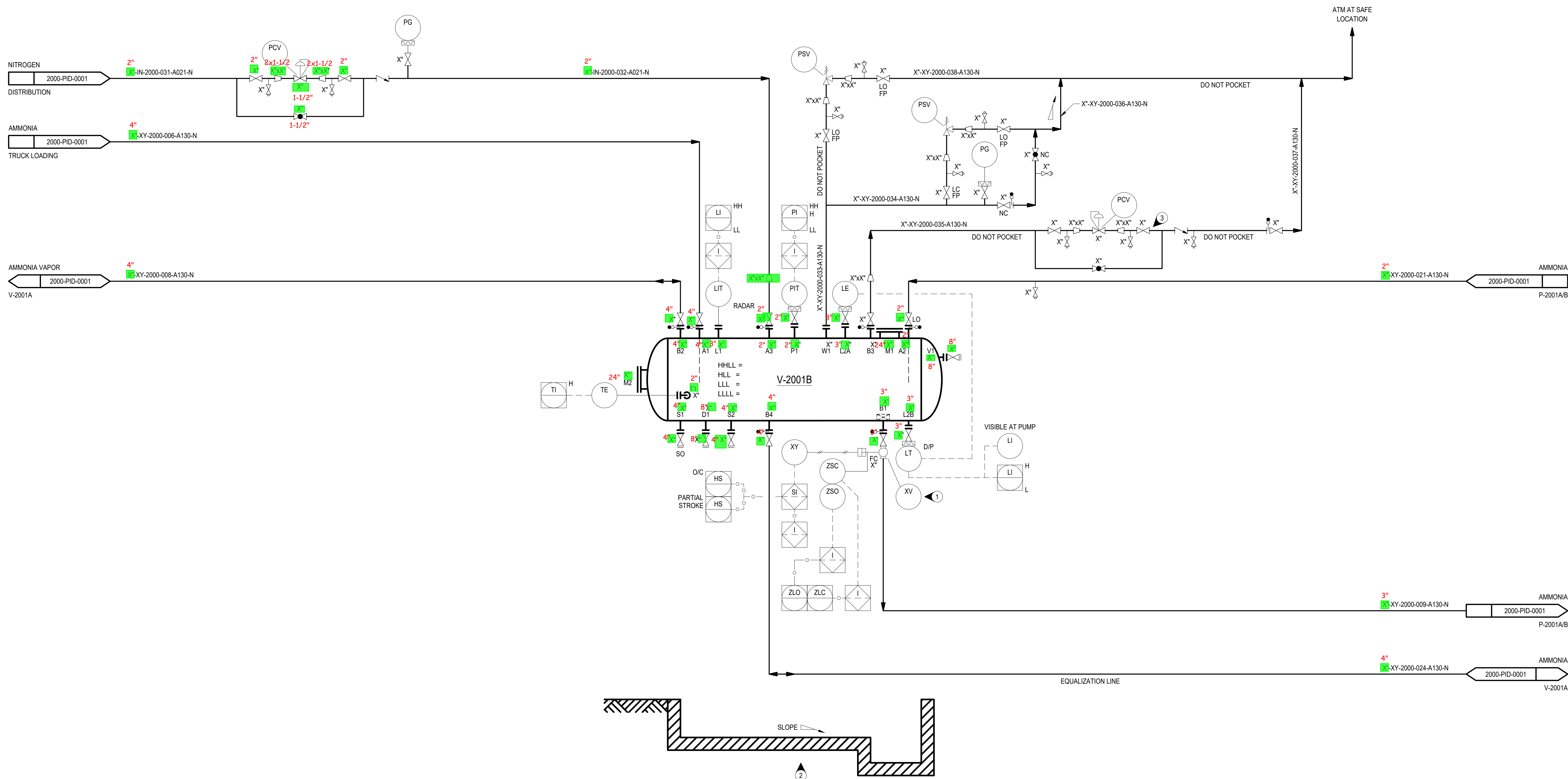
SIZE :
TOTAL VOLUME :
DESIGN PRESSURE :
DESIGN TEMP :
MOC :
TRIM : VT-2000-039-A130-N
INSULATION / TRACING :

A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS
SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.

3. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.

EMERGENCY BLOCK VALVE (EBV) - FIRE SAFE

1. EMERGENCY BLOCK VALVE (EBV) - FIRE SAFE.
2. COMMON IMPOUND AREA FOR AMMONIA DRUMS
3. LOCATED AT HIGH POINT.



CHECK PRINT

NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER, DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR FOR EXTENSIONS TO THIS PROJECT, EXCEPT BY AGREEMENT IN WRITING AND NOT BE USED TO REPRESENT THE ENGINEER'S DESIGN OR TO BE USED TO HOLD ANY PARTY RESPONSIBLE FOR DEFENDING TO CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE, AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT, SHALL NOT INDICATE ANY DETAIL OR SYSTEM FOR CONSTRUCTION. THE ENGINEER'S DESIGN INTENT SHALL NOT BE USED TO CONSTRUCT OR PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK, CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR OBTAINING ALL NECESSARY PERMITS, ERECTION, INSTALLATION, AND COORDINATION. MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, OR SEQUENCES. THE ENGINEER'S DESIGN INTENT SHALL NOT BE USED TO CONSTRUCT OR PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS

[illegible]

MOTIVA

PIPING & INSTRUMENTATION DIAGRAM

AMMONIA TANK E



**SSP ENGINEERING
& CONSULTING**
13100 NORTHWEST FREEWAY
SUITE 625,
HOUSTON, TEXAS 77040

FIRM
NUMBER
#13619

PROJECT NO: 25MA02

SCALE: NONE

DRAWN BY: GS

CHECKED: APB

DESIGN BY:
DRAWING NO:

2000-PID-0003

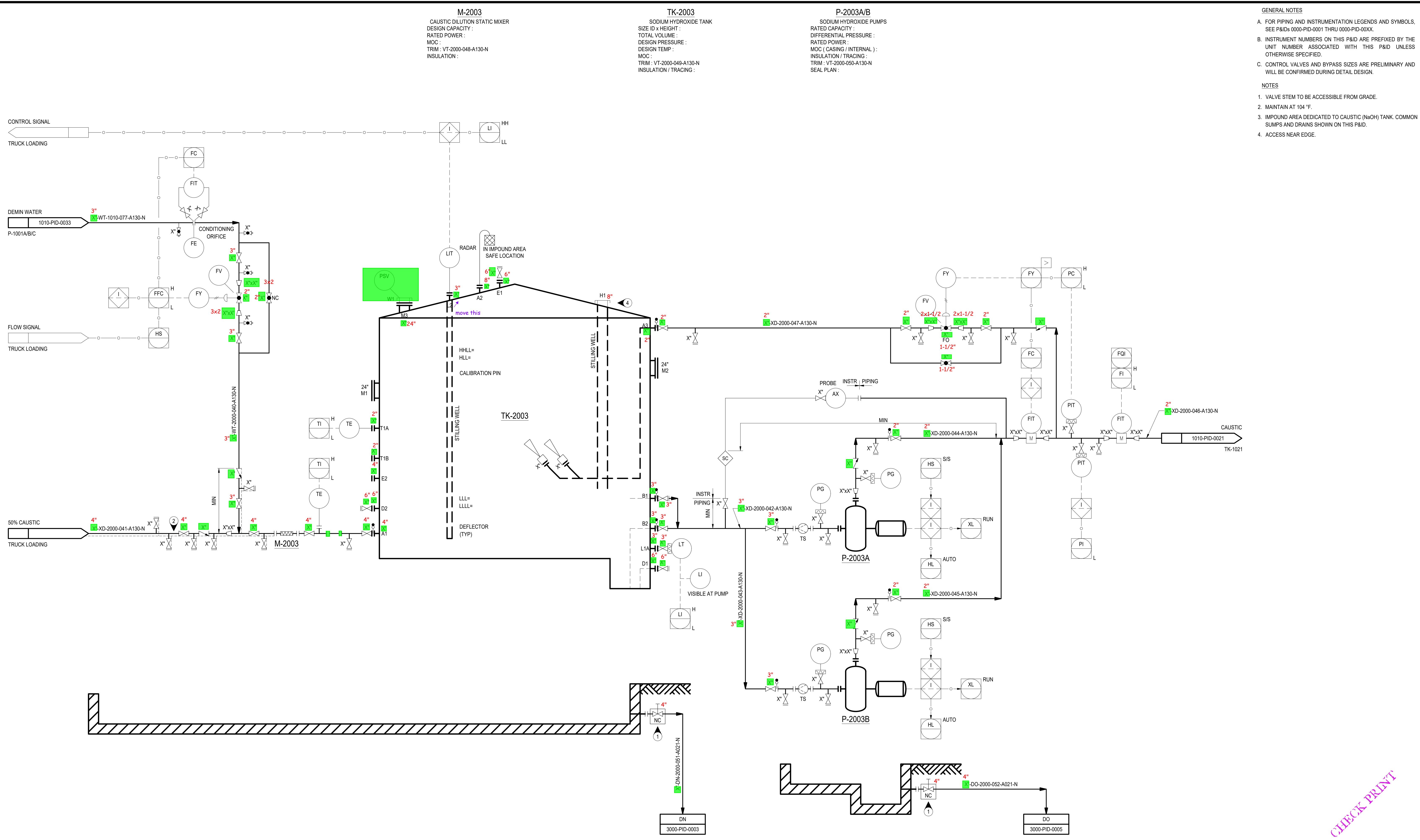
**REV
A**

CLIENT: MOTIVA ENTERPRISES LLC

PLANT: PORT ARTHUR REFINERY

JOB NC

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW



- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDS 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.
- NOTES
1. VALVE STEM TO BE ACCESSIBLE FROM GRADE.
2. MAINTAIN AT 104 °F.
3. IMPOUND AREA DEDICATED TO CAUSTIC (NaOH) TANK. COMMON SUMPS AND DRAINS SHOWN ON THIS P&ID.
4. ACCESS NEAR EDGE.

NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT, EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT, SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

MOTIVA

CLIENT: MOTIVA ENTERPRISES LLC

PLANT: PORT ARTHUR REFINERY

JOB NO:

PIPING & INSTRUMENTATION DIAGRAM

SODIUM HYDROXIDE STORAGE

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

SSP ENGINEERING & CONSULTING

13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

FIRM NUMBER #13619

PROJECT NO: 25MA02
SCALE: NONE
DRAWN BY: GS
CHECKED: APB
DESIGN BY:
ENG APVL: PDN
DRAWING NO: **2000-PID-0005** **REV A**

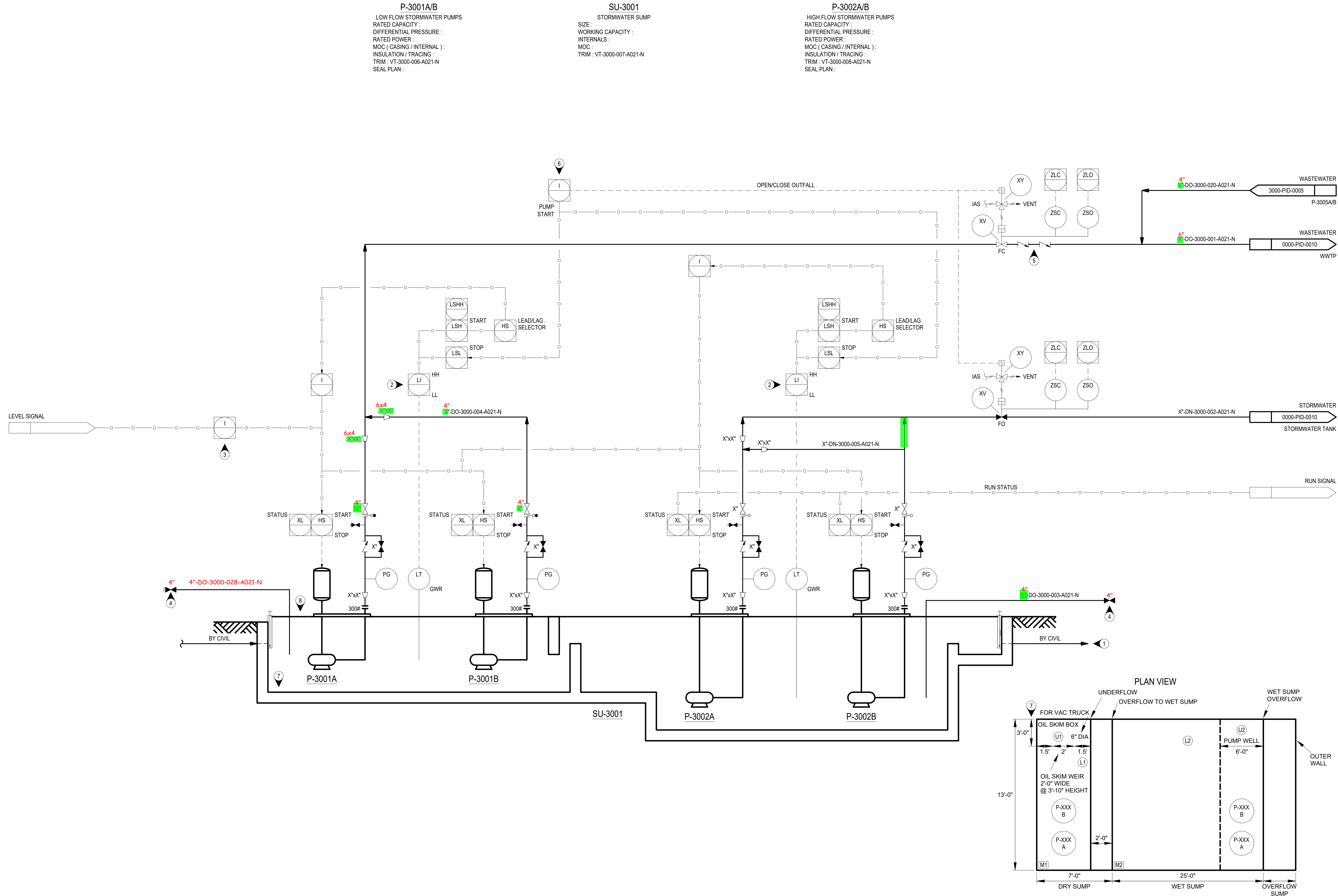
1. VALVE STEM TO BE ACCESSIBLE FROM GRADE.
2. IMPOUND AREA DEDICATED TO ACID TANK. COMMON SUMPS AND DRAINS SHOWN ON THIS P&ID.
3. ACCESS NEAR EDGE.



**FIRM
NUMBER
#13619**

007	REV A
------------	------------------

DRAWING LOCATION: c:\users\jason\one\drive - eng\ssps\files - 25ma02 - aromatics utility scope\old eyes\revin\04 - process\pids - process\pids, (Border Rev 1 5 18 25), PLOT DATE: ---, PLOT BY: jason, DRAWING: 3000-pid-0001.dwg



CHECK PRINT

NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT. EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER, CONTRACTOR, SUB-CONTRACTORS, & SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

PIPING & INSTRUMENTATION DIAGRAM
STORMWATER SUMP

CLIENT: MOTIVA ENTERPRISES LLC
PLANT: PORT ARTHUR REFINERY
JOB NO:

SSP ENGINEERING & CONSULTING
13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

FIRM NUMBER #13619

PROJECT NO: 25MA02
DRAWN BY: GS
DESIGN NO:
SCALE: NONE
CHECKED: APB
ENG APVL: PDN
PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW
3000-PID-0001 REV **A**

DRAWING LOCATION: c:\users\jason\one\drive - eng\ssps files - 25ma02 - aromatics utility scope old eyes\rev\04 - design\04 - process\04a, (Border Rev 1 5/18/25), PLOT DATE: ---, PLOTTED BY: jason, DRAWING: 3000-pid-0003.dwg

SU-3003

NONOILY WATER SUMP

SIZE :
WORKING CAPACITY :
INTERNALS :
MOC :
TRIM : VT-3000-014-A021-N

P-3003A/B

NONOILY WATER PUMPS

RATED CAPACITY :
DIFFERENTIAL PRESSURE :
RATED POWER :
MOC (CASING / INTERNAL) :
INSULATION / TRACING :
TRIM : VT-3000-015-A021-N
SEAL PLAN :

GENERAL NOTES

A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.

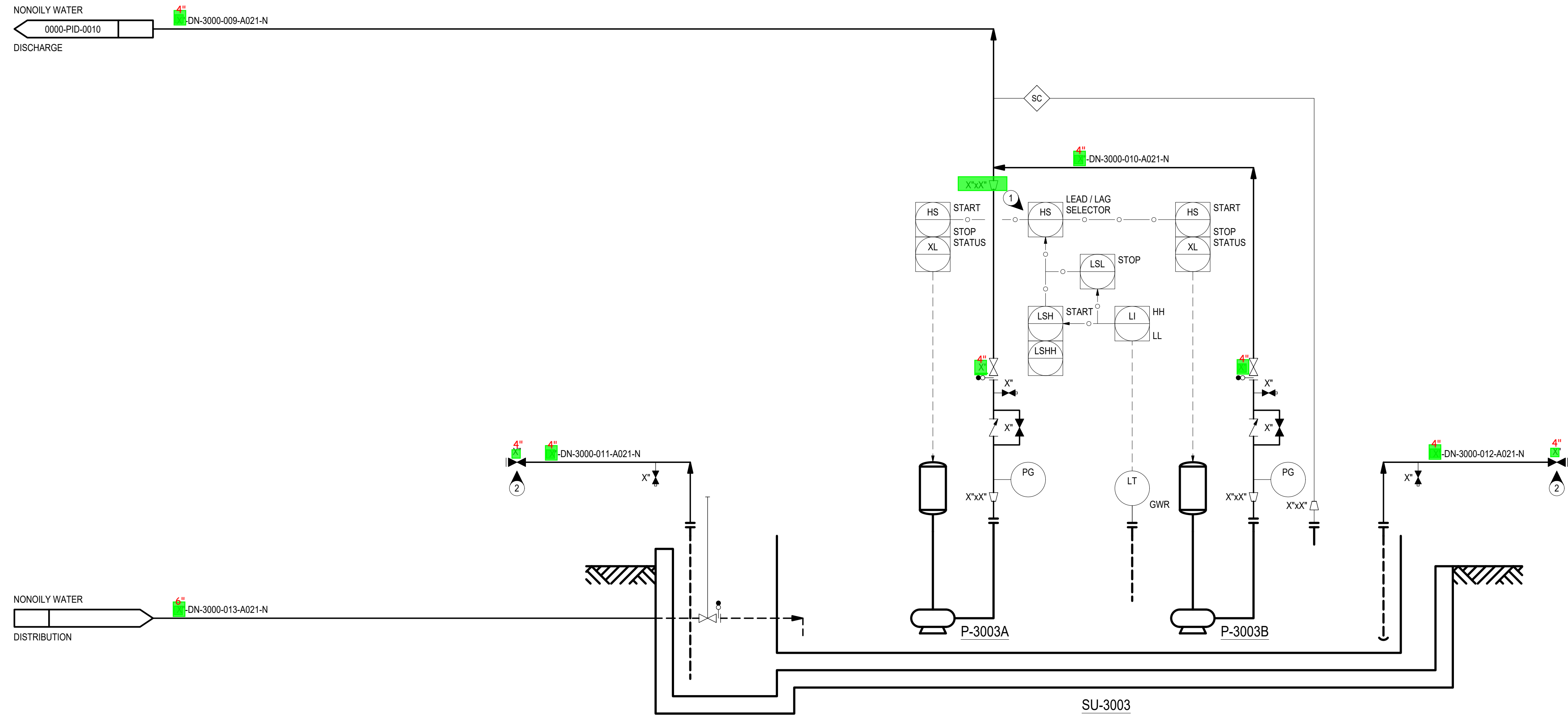
B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.

C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.

NOTES

1. EACH PUMP SET IS IN A LEAD/ LAG CONFIGURATION. THE LEAD PUMP IN THE PUMP SET WILL START ON HIGH LEVEL AND THE LAG PUMP WILL START ON HIGH HIGH LEVEL. BOTH PUMPS IN THE PUMP SET WILL STOP ON LOW LEVEL.

2. TRUCK LOADING CONNECTION.



CHECK PRINT

NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT. EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER, CONTRACTOR, SUB-CONTRACTORS, & SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT, SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUE FOR APPROVAL	GS	APB	PDN						

MOTIVA

CLIENT: MOTIVA ENTERPRISES LLC
PLANT: PORT ARTHUR REFINERY
JOB NO:

PIPING & INSTRUMENTATION DIAGRAM

WASTEWATER
NONOILY WATER COLLECTION

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

SSP ENGINEERING & CONSULTING

13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

PROJECT NO: 25MA02
DRAWN BY: GS
DESIGN BY:
DRAWING NO:

SCALE: NONE
CHECKED: APB
ENG APVL: PDN
3000-PID-0003

FIRM NUMBER #13619
REV A

2. TRUCK LOADING CONNECTION.

CHECK PRINT

NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT. EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER, CONTRACTOR, SUB-CONTRACTORS, & SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE, AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT, SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND CO-ORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SPECIFIED BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

MOTIVA

ENTERPRISES LLC

CLIENT: MOTIVA ENTERPRISES LLC

PLANT: PORT ARTHUR REFINERY

JOB NO:

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

PROJECT NO: 25MA02

DRAWN BY: GS

DESIGN BY:

DRAWING NO:

SCALE: NONE

CHECKED: APB

ENG APVL: PDN

3000-PID-0005

SSP

ENGINEERING & CONSULTING

13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

FIRM NUMBER

#13619

P-3007A/B
SANITARY LIFT STATION PUMPS
RATED CAPACITY :
DIFFERENTIAL PRESSURE :
RATED POWER :
MOC (CASING / INTERNAL) :
INSULATION / TRACING :
TRIM : VT-3000-029-A021-N
SEAL PLAN :

GENERAL NOTES

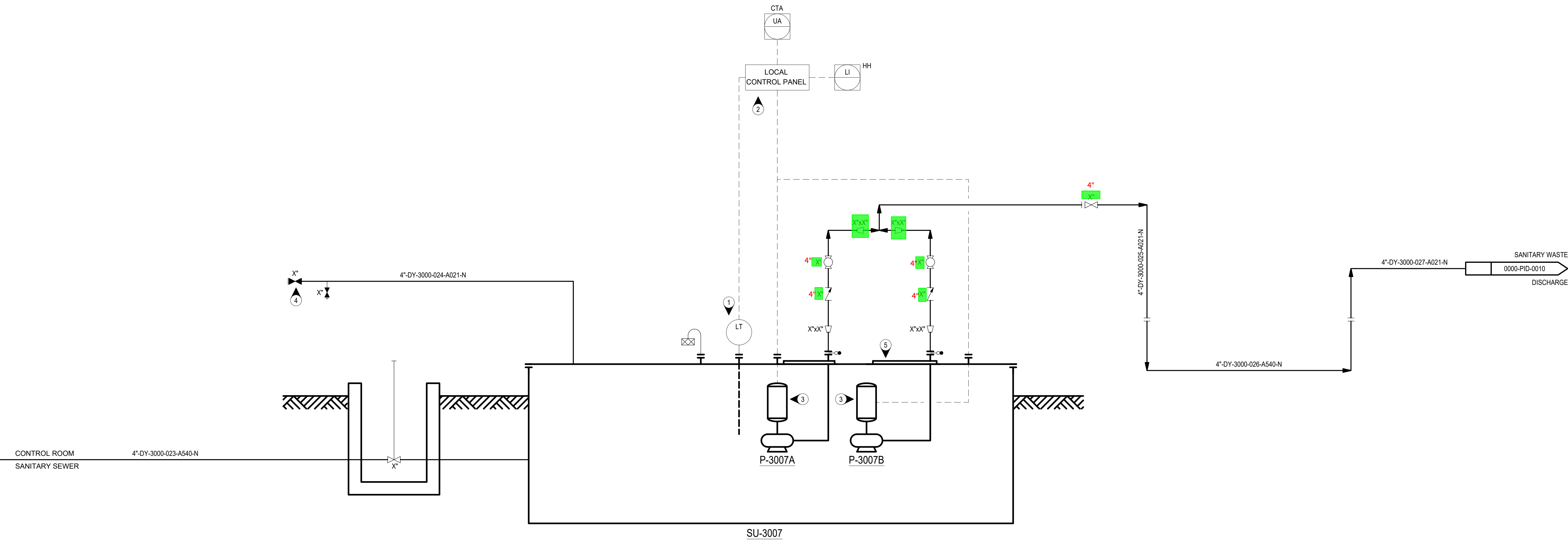
A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.

B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.

C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.

NOTES

1. CONTROLS START ONE PUMP AT HIGH LEVEL AND THE SECOND AT HIGH HIGH LEVEL.
2. PRIMARY PUMP OPERATION ALTERNATIVES BETWEEN PUMPS.
3. LOW LIQUID LEVEL STOPS BOTH PUMPS.
4. TRUCK LOADING.
5. 38"x20" PUMP ACCESS DOOR.



CHECK PRINT

NOT PUBLISHED. ALL RIS REQUIRED BY THE ENGINEER, DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT. EXCEPT BY AGREEMENT IN WRITING AND IN THE PRESENCE OF THE ENGINEER, THE CONTRACTOR SHALL BE RESPONSIBLE FOR THE PROPER USE OF INSTRUMENTS OF SERVICE. THE CONTRACTOR SHALL BE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE, AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH INFORMATION. THE CONTRACTOR SHALL BE RESPONSIBLE FOR OBTAINING ALL NECESSARY PERMITS, CONSTRUCTION CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND CO-ORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR OBTAINING ALL NECESSARY PERMITS, AND FOR THE PROPER STORAGE AND HANDLING OF MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEAD, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES; THE CONTRACTOR SHALL BE RESPONSIBLE FOR THE PROTECTION OF EXISTING UTILITIES, STRUCTURES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:

[illegible]

CLIENT: MOTIVA ENTERPRISES LLC
PLANT: PORT ARTHUR REFINERY
JOB NO:

PIPING & INSTRUMENTATION DIAGRAM

WASTEWATER
SANITARY LIFT STATION

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW



**SSP ENGINEERING
& CONSULTING**
13100 NORTHWEST FREEWAY
SUITE 625,
HOUSTON, TEXAS 77040

PROJECT NO: 25MA02

DRAWN BY: GS

DESIGN BY:

DRAINING NO:

**ENGINEERING
CONSULTING**
BEST FREEWAY
E 625,
#13619

SCALE: NONE

CHECKED: APB

ENG APVL: PDN

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200	201	202	203	204	205	206	207	208	209	210	211	212	213	214	215	216	217	218	219	220	221	222	223	224	225	226	227	228	229	230	231	232	233	234	235	236	237	238	239	240	241	242	243	244	245	246	247	248	249	250	251	252	253	254	255	256	257	258	259	260	261	262	263	264	265	266	267	268	269	270	271	272	273	274	275	276	277	278	279	280	281	282	283	284	285	286	287	288	289	290	291	292	293	294	295	296	297	298	299	300	301	302	303	304	305	306	307	308	309	310	311	312	313	314	315	316	317	318	319	320	321	322	323	324	325	326	327	328	329	330	331	332	333	334	335	336	337	338	339	340	341	342	343	344	345	346	347	348	349	350	351	352	353	354	355	356	357	358	359	360	361	362	363	364	365	366	367	368	369	370	371	372	373	374	375	376	377	378	379	380	381	382	383	384	385	386	387	388	389	390	391	392	393	394	395	396	397	398	399	400	401	402	403	404	405	406	407	408	409	410	411	412	413	414	415	416	417	418	419	420	421	422	423	424	425	426	427	428	429	430	431	432	433	434	435	436	437	438	439	440	441	442	443	444	445	446	447	448	449	450	451	452	453	454	455	456	457	458	459	460	461	462	463	464	465	466
---	---	---	---	---	---	---	---	---	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----

RE